Design and Optimization Model of Floating Offshore Wind Platforms

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DTU Wind

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DTU Wind and Energy Systems is a department of the Technical University of Denmark with a unique integration of research, education, innovation and public/private sector consulting in the field of wind and energy. Our activities develop new opportunities and technology for the global and Danish exploitation of wind and energy. Research focuses on key technical-scientific fields, which are central for the development, innovation and use of wind energy and provides the basis for advanced education.

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Abstract

In the decarbonization of the energy sector, the Floating Offshore Wind Turbine (FOWT) technology offers great potential as it allows deployment at deeper seas with stronger, more stable winds. The acceleration of the deployment of floating wind is especially dependant on decreasing in Levelized Cost Of Energy (LCOE) and this can be achieved through upscaling and reducing platform costs.

The 5 MW OC4 semi-submersible floating platform is upscaled to fit the 15 MW IEA reference wind turbine. This is based on a literature study investigating theoretical and practical scaling trends. The upscaling is done based on two key paremeters; the radius from center to offset column r_{MO} and the diameter of the upper offset column d_{UC}.

The upscaled 15 MW floater was optimized in order to reduce the steel mass of the substructure with the same key parameters (r_{MO} and d_{UC}) as design variables while ensuring hydrostatic stability and structural integrity.

The model is developed in Python, using OpenMDAO, an open source multi-disciplinary design analysis and optimization tool. The hydrostatic and -dynamic properties of the optimized design are analyzed in the frequency domain using RAFT. The structural constraints were based on assumptions of static equilibrium.

The design was optimized by increasing r_{MO} and decreasing d_{UC}. The dynamic analysis showed that the design was also optimized in regards to the system motion in surge and pitch and the maximum nacelle acceleration, while it worsened in heave motion.

A sensitivity analysis was carried out in order to investigate how the draught, ballast distribution, pontoon wall thickness and maximum allowable pitch angle respectively influenced the optimized design.

The optimization model demonstrated the potential of serving as a valuable tool in the design process for future semi-submersible platforms for FOWTs by providing fast and valuable insight into the relationships between variables and the impact of certain design parameters on both mass and dynamic behavior. The study highlights the importance of continuing to develop and refine optimization models for FOWT systems.

Keywords: Offshore Code Comparison Collaboration Continuation (OC4), semisubmersible, WT, Response Amplitudes of Floating Turbines (RAFT), NREL, Hydrostatic stiffness, Foundation

Preface

This Master Thesis is the finalization of two Master's degrees. One in Mechanical Engineering and one in Sustainable Energy Engineering both at the *Technical University of Denmark* under the *Department of Wind Energy* with supervision from Katherine Dykes. The thesis has been done in close cooperation with *National Renewable Energy Laboratory (NREL)* with Matthew Hall as supervisor and *PEAK Wind ApS* with Ilmas Bayati and Thor Snedker as supervisors.

It is assumed that the reader has a basic understanding of maritime engineering and wind energy.

The thesis has been made possible by the help, advice and guidance from our supervisors.

Thank you.

DTU Lyngby Campus, February 5, 2023

Hans Frederik Greve Lehmann

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Contents

Nomenclature

Acronyms

Symbol	Description	Units
BC	“Base Column”.	[-]
BEM	“Boundary Element Method”.	[-]
CB	“Cross Brace”.	[-]
CoB	“Center of Buoyancy”.	[-]
CoF	“Center of Floatation”.	[-]
CoG	“Center of Gravity”.	[-]
DL	“Delta Lower pontoon”.	[-]
DoF	“Degree of Freedom”.	[-]
DU	“Delta Upper pontoon”.	[-]
FEA	“Finite Element Analysis”.	[-]
FEM	“Finite Element Method”.	[-]
FOWP	“Floating Offshore Wind Platform”.	[-]
FOWT	“Floating Offshore Wind Turbine”.	[-]
IEA	“International Energy Agency”.	[-]
LCoE	“Levelized Cost of Energy”.	[-]
MC	“Main Column”.	[-]
MDAO	“Multi-Disciplinary Analysis and Optimization”.	[-]
NREL	“National Renewable Energy Laboratory”.	[-]
O&G	“Oil and Gas”.	[-]
O&M	“Operations and Maintenance”.	[-]
OC	“Offset Column”.	[-]
OC4	“Offshore Code Comparison Collaboration Continuation”.	[-]
PSD	“Power Spectral Density”.	[-]
RAFT	“Response Amplitudes of Floating Turbines”.	[-]
RAO	“Response Amplitude Operator”.	[-]
RNA	“Rotor Nacelle Assembly”.	[-]
RPM	“Revolutions Per Minute”.	[r/min]
SWL	“Still Water Line”.	[-]

TLP	"Tension-Leg Platform".	[-]
TRL	"Technology Readiness Level".	[-]
UC	"Upper Column".	[-]
YL	"Lower pontoon".	[-]
YU	"Upper pontoon".	[-]

Greek Symbols

Symbol	**Description**	**Units**
$\hat{\xi}$	Complex response amplitude	[m]
∇	Water displacement	[m^3]
ω_n	Natural frequency	[rad/s]
ρ	Density	[kg/m^3]
σ	Standard deviation of distribution	[]
σ_{cr}	Critical uniaxial stress	[Pa]
σ_y	Yielding strength of steel	[Pa]
θ	Static inclination angle in pitch	[°]
θ_{max}	Maximum static inclination angle in pitch	[°]
ζ_0	Wave amplitude	[m]

Indices

Symbol	**Description**	**Units**
1	1^{st} degree of freedom: Surge	[-]
2	2^{nd} degree of freedom: Sway	[-]
3	3^{rd} degree of freedom: Heave	[-]
4	4^{th} degree of freedom: Roll	[-]
5	5^{th} degree of freedom: Pitch	[-]
6	6^{th} degree of freedom: Yaw	[-]

Latin Symbols

Symbol	**Description**	**Units**
$\hat{f}$	Excitation force from wind and waves	[N]
$\overline{BM_C}$	Distance from CoB to metacenter	[m]
$\overline{GB}$	Distance from CoG to CoB	[m]
$\overline{GM_C}$	Distance from CoG to M_C	[m]
$\overline{GZ}$	Righting lever	[m]
A	Cross-sectional area of bar	[m^2]
a_{nac}	Nacelle acceleration	[m/s^2]
A_{sub}	6x6 added mass matrix of substructure	[kg]
A_{wp}	Waterplane area	[m^2]
B	6x6 damping matrix	[Ns/m]

B_ϕ	Roll damping	[Nm]
C_{moor}	6x6 stiffness matrix of mooring system	[N/m]
C_{struc}	6x6 stiffness matrix of structure	[N/m]
D	Draught	[m]
D_R	WT rotor diameter	[m]
d_{BC}	Diameter of base column	[m]
d_{MC}	Diameter of main column	[m]
d_{OC}	Diameter of offset column	[m]
d_p	Diameter of pontoons	[m]
E	Young's modulus	[Pa]
F_T^{max}	Maximum thrust force on WT rotor	[N]
F_B	Buoyancy force	[N]
f_b	Freeboard height	[m]
f_p	Probability correction factor	[-]
F_{RES}	Resulting force	[N]
g	Gravitational force	[m/s^2]
H_s	Significant wave height	[m]
h_{BC}	Height of base column	[m]
h_{CB}	Height of cross brace	[m]
h_{UC}	Height of upper column	[m]
I	Area moment of inertia of waterplane	[m^4
i	Number of degrees of freedom	[-]
I_J	Second moment of area about the J^{th} axis	[m^4]
I_{an}	Annular moment of inertia	[m^4]
I_{jk}	Moment of inertia about the jk^{th} axis	[m^4]
I_{xx}	Waterplane moment of inertia about x-direction	[m^2]
I_{yy}	Waterplane moment of inertia about y-direction	[m^4]
k_l	Effective length factor (buckling)	[-]
K_{55}	Hydrostatic stiffness in pitch	[N/m]
K_{hydro}	6x6 hydrostatic stiffness matrix	[N/m]
K_{moor}	Mooring system hydrostatic stiffness matrix	[N/m]
l_{CB}	Length of cross brace	[m]
l_{YU}	Length of upper pontoon	[m]
M	Mass	[kg]
m	Number of constraints	[-]
M_C	Metacenter	[-]
M_H	Heeling moment	[Nm]

m_n	n^{th} order spectral moment	$[-]$
M_R	Righting moment	[Nm]
M_{bal}	Mass of ballast	[kg]
m_{MC}	Mass of main column	[kg]
m_{RNA}	Mass of rotor-nacelle assembly	[kg]
M_{struc}	6x6 mass matrix of structure	[kg]
M_{sub}	Mass of substructure	[kg]
m_{tower}	Mass of turbine tower	[kg]
n	Number of decision variables	[-]
$p(x)$	Probability distribution function	[-]
P_{crit}	Critical load	[N]
P_{des}	Design load	[N]
r_{MO}	Distance from main column center to offset column center	[m]
Re	Reynolds number	$[-]$
S_ζ	Wave energy spectrum	[J/m^2]
S_J	JONSWAP spectrum	[J/m^2]
S_R	Motion response spectrum	[J/m^2]
s_{be}	Safety factor for bending	[-]
s_{bu}	Safety factor for buckling	[-]
s_y	Safety factor for y	[-]
T_0	Modal period	[s]
T_m	Mean roll period	[s]
T_n	Natural period	[s]
T_n	Natural period	[s]
T_p	Mean wave period	[s]
T_z	Mean zero-crossing period	[s]
t_{MC}	Thickness of main column	[m]
t_{OC}	Thickness of offset columns	[m]
t_p	Thickness of pontoons	[m]
V	Displaced volume	[m^3]
v_w	Wind speed	[m/s]
x_{wp}	x-coordinate for weighted center of waterplane area	[m]
y_{wp}	y-coordinate for weighted center of waterplane area	[m]
z_B	z-coordinate for CoB	[m]
z_G	z-coordinate for CoG	[m]
z_{FL}	Vertical distance from waterplane to fairlead	[m]
z_{hub}	Vertical hub height of WT rotor	[m]

1 Introduction

Future advancements in offshore wind power capacity are estimated to increase faster than in the last two decades. At the beginning of 2022 the total installed offshore wind power capacity was at 35 GW. Already by 2030 this is predicted to reach approximately 382 GW, and to further increase to approximately 2002 GW by 2050. This is driven mainly by the commissioning of floating wind units, recent technological advancements, significant economic growth, and improvements in turbines [1].

The majority of foundations currently in use are bottom fixed foundations as opposed to their floating counterpart due to the lower technology readiness level of the floaters. However, bottom fixed foundations are limited to shallower waters and nearly 80 % of the world's offshore wind resource is placed at water depths above 60 m, where bottom fixed wind turbines are not feasible [2].

The improvements of Floating Offshore Wind Turbines (FOWTs) will increase the number of sites suitable for wind farms. By extending the scope of sites to deeper waters, more powerful and consistent winds can be utilized while reducing visual pollution and interference with fishing and shipping lanes. In order to accelerate the implementation of floating wind parks, better early-stage project design and financial evaluation tools must be developed. By creating a model of costs relative to site and technical inputs, one can support optimization of floating offshore wind farm development in the preliminary phase and support the commercialization of the industry.

1.1 State of Floating Wind

Floating wind as an industry is relatively immature but rapidly evolving. The technology timeline in Table 1.1 gives an overview of some of the most important past, present, and future FOWT projects:

Table 1.1 Technology timeline

2008	**Blue H Technologies** is the first to deploy a major FOWT prototype. The TLP with an 80 kW turbine is installed in the Adriatic Sea at a water depth of 113 m, 22 km from the shore of SE Italy [3].
2009	The **Hywind Demo** project by Equinor is commissioned as the first full scale FOWT. The 2.3 MW spar buoy FOWT located in Norway produced more than 40 GWh before being sold in 2019.
2011	The first semi-submersible FOWT is deployed off the Portuguese shore. The **WindFloat 1 Project** featured a 2 MW Vestas V-80 wind turbine. Enduring extreme Atlantic Ocean weather conditions, it proved the robustness of the technology [4].

Table 1.1 Technology timeline

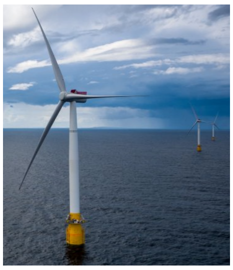
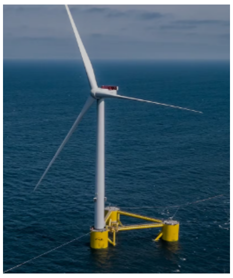
2013	The **VolturnUS 1:8** FOWT is the first grid-connected FOWT in the US. The 20 kW semi-sub was launched by the University of Main as part of the DeepCwind consortium [5].
2013	**Fukushima Mirai**, a compact semi-sub supporting a 2 MW downwind turbine, is launched in Japan as a part of the Fukushima FORWARD project [6].
2015	**Fukushima Shimpuu**, also part of Fukushima FORWARD, breaks the record as the largest FOWT at 7 MW with a semi-sub foundation. It was decommissioned in 2020 due to low profitability [7], [8].
2017	The 30 MW **Hywind Scotland Equinor** project park is the world's first floating wind farm, consisting of five FOWTs of 6 MW each. A 60-70 % cost reduction has been achieved compared to the predecessor, Hywind Demo, demonstrating the feasibility of floating wind farms that could be ten times larger. Each year since Hywind Scotland started production, the floating wind farm has achieved the highest average capacity factor of all UK offshore wind farms [9]. 
2018	The **FloatGen** project launches a 2 MW FOWT on a barge foundation 22 km off shore in the Biscay Bay by a consortium of Europe-wide partners for assessing feasibility [10].
2019	**WindFloat Atlantic** is the first full-scale project to use semi-submersible technology and operated by Principle Power. It consists of three 8.4 MW Vestas V164 turbines supported by WindFloat units located 20 km off shore from Portugal [11].
2021	The 50 MW **Kincardine Offshore Windfarm** by Principle Power located 15 km off the east coast of Scotland goes into operation. The farm consists of five 9.5 MW Vestas V164 turbines supported by WindFloat units joined by the 2 MW WindFloat 1. It is currently the world's largest operational wind farm [12], [13]. 
2023	**Hywind Tampen** is the world's first renewable power for offshore oil and gas. With its 11 wind turbines of 8.6 MW supported by spar buoys and with a system capacity of 88 MW, it will meet about 35 % of the annual electricity demand of the five drilling rigs in the north sea [14].
2024	**Maine Aqua Ventus** is expected to go into operation. This demonstration project will deploy a 11 MW turbine on the VolturnUS concrete semi-submersible hull in Maine [15].
2026	**Pentland** Floating Offshore Windfarm located 6 km off the North coast of Caithness, Scotland will be the largest one once completed with a power generation of up to 100 MW [16].
2026	South Korea is estimated to have power from its first floating windfarm, with an initial capacity of 500 MW. The **Korea Floating Wind** project will eventually grow into five projects with a capacity of 2 GW in Korea's East Sea [17].

TABLE 1.1 Technology timeline

2026	The **Redwood Coast Offshore Wind Project** is expected to deliver its first power from the 150 MW floating wind farm 40 km off the coast of California in waters up to 900 m deep [18].
2028	The 3 **GW MarramWind FOWP Project**, which is a partnership between Shell and ScottishPower, located NW from Aberdeen is expected to be operational. This is the largest single site from the ScotWind Leasing for offshore wind tender which include 20 developments that total up to 27.6 GW in capacity with the majority being at water depths requiring floating foundations.

According to [19], the wind capacity from FOWTs will increase to 14 GW by 2030 and 289 GW by 2050. Currently, the world wind capacity installed is at 837 GW with only 60 GW coming from offshore wind and 0.1 GW from floating wind [20]. This exponential growth can be explained by the expected drop in Levelized Cost Of Energy (LCOE) as the technology matures. Baseline cost analysis does not indicate economic viability without further optimization, innovation, and up-scaling to commercial plant sizes [21].

In Figure 1.1, it is illustrated how the drivers of changes for the global average LCOE of wind is expected to evolve towards 2050. Due to the immaturity of the technology, floating wind will experience a drastically higher reduction compared to its fixed counterparts. The reductions in the levelized costs for fixed and floating offshore wind will be 39 % and 84 % respectively. The largest change for FOWT is seen from "non-turbine investment cost" adding to the argument of how important the advance in substructure optimization is in order to drop the expenses. Economy of scale will also add to the cost reduction as a movement from prototype to mass production will reduce the cost of manufacturing and installation. Furthermore, the expected increase in capacity factor for FOWTs is a considerable contributor to the LCOE reduction. The large impact compared to their fixed counterpart can be explained by the ability to reach stronger and more stable winds further off shore, and the substructures becoming more stable, allowing for a higher turbine efficiency.

In general, research and industry show promising signs towards the ambitions of floating wind becoming a competitive tool for renewable energy. Equinor states: "Our goal is that floating wind be competitive with other forms of energy by the year 2030. We expect floating offshore wind to be the next big breakthrough in renewables." [22]

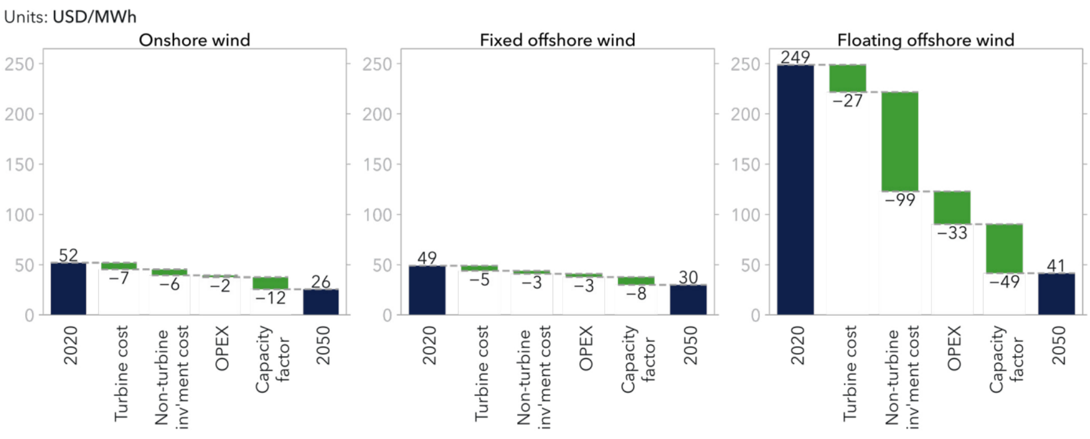


Figure 1.1: Drivers of change for the global average levelized cost of wind between 2020 and 2050 [19].

1.2 Scope of the Project

The feasibility of deep-water wind turbines is not questioned as long-term serviceability of floating structures has already been successfully demonstrated by the marine and offshore Oil and Gas (OG) industries over many decades. However, the economics that allowed the deployment of thousands of offshore oil-rigs have yet to be demonstrated for floating wind turbine platforms [23].

The capital costs for the wind turbine itself will not be significantly higher than current marinized turbine expenses in shallow water. Therefore, the economics of deep-water wind turbines will be determined primarily by the additional expenses of the floating structure and power distribution system (e.g., longer transmission runs) [23]. As a result, tools to estimate and lower the cost of the floating structure is crucially important to accelerate the investment in floating wind.

The key to cost-effective energy production is having the largest possible turbine size [24]. By doubling the rotor diameter, the swept area and hence the power output is quadrupled. On top of this, there are economies of scale in wind turbines, i.e., larger machines are usually able to deliver electricity at a lower cost than smaller machines. The reason for this is that the cost of foundations, road building, electrical grid connection, as well as a number of components in the turbine (the electronic control system, etc.) are somewhat independent of the size of the machine. In addition to this, maintenance costs are partly independent of the size of the machine [25]. For this reason, wind turbines are increasing in size. Thus, this thesis will investigate the upscaling of an existing platform and turbine from 5 MW to 15 MW.

2 Theory

In this section, the general concept of a FOWT is covered by introducing the different types of floating substructures, Sec. 2.1, and their station keeping systems, Sec. 2.2. Relevant upscaling trends and methods are explained in Sec. 2.1. Stability theory of a FOWT is explained in Sec. 2.4, and general FOWT design considerations is discussed in Sec. 2.5. In Sec. 2.6, relevant theory of wave excitation and responses in the frequency domain is presented. Some structural analysis is introduced in Sec. 2.7 and finally, the theory behind design optimization is explained in Sec. 2.8.

2.1 Floating Substructures

Several floating substructures for wind turbines have been designed, investigated, and developed in recent years. Floating substructures can generally be categorized into three classical designs: the spar, semi-submersible, and Tension-Leg Platform (TLP), all of which originates from the O&G industry [3]. Each class utilizes a different stabilizing effect as their primary method to achieve static stability, and can be seen in figure 2.1.

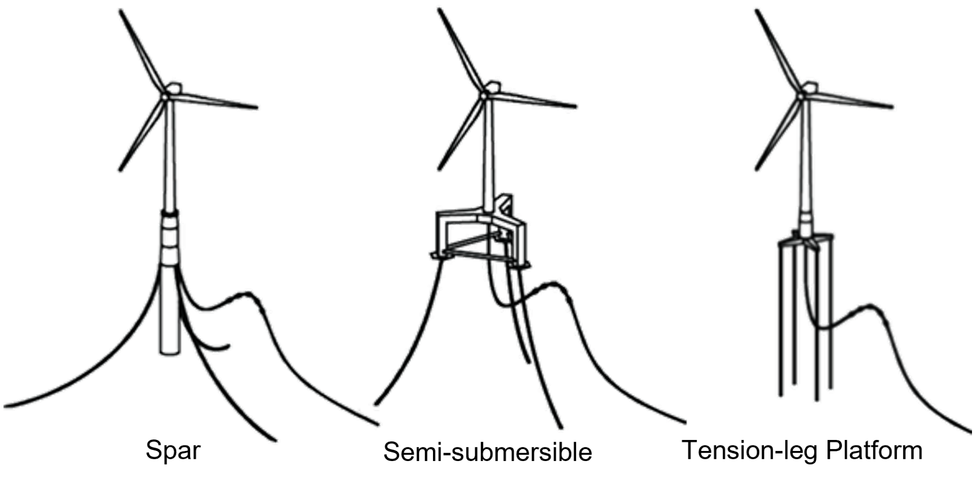


Figure 2.1: The three most common floater designs: spar, semi-submersible, and tension-leg platform [26].

2.1.1 Spar

The spar design is generally a deep-draughted cylinder made from either steel or concrete. The deep draught is ballasted at the bottom in order to ensure a low center of gravity for optimal ballast stabilisation. Production is simple and cheap compared to the other designs [21]. However, the deep draught (100 m draught for the Norwegian 2.3 MW Hywind FOWT, and estimated to be approximately 110 m for a 15 MW turbine [27]) limits where they can be used, built, and towed, making assembly and service more expensive and complicated, and limits the sites where this design is applicable for use.

2.1.2 Semi-Submersible

The semi-submersible (and barge) design uses a large water-plane area and distributed surface buoyancy to gain stability. This inherently makes them more sensitive to wave forces, and hence more exposed to fatigue loading. However, it benefits from a shallower draught resulting in easier quayside assembly, commissioning, and tow-outs. During operation, it also enables the FOWT to be towed to a service port for larger Operations and Maintenance (O&M) tasks such as main component replacements [28]. The design of the semi-submersible typically consists of three or four buoyant columns made from steel or concrete connected by a truss system. The disadvantage can be additional joints and an increased amount of structure near the waterline, leading to higher corrosion risk and generally increased cost due to increased overall mass [29].

2.1.3 Tension-Leg Platform

The TLP uses mooring line tension and a buoyant platform in order to achieve stability. This gives it the lowest platform motion under operation while still being the smallest and lightest of the three floater designs. The disadvantage is instability until the mooring lines have been connected and tensioned, making the installation more complicated and expensive. As the mooring lines must be in tension, the mooring is subject to higher loads and horizontal forces leading to higher complexity and cost (see Sec. 2.2). TLPs are also more exposed to natural disasters and failures due to their high dependency on the mooring lines [3],[21].

2.1.4 Choosing a Substructure

As each design class has its pros and cons, the choice of substructure largely depends on external conditions such as metocean conditions, water depth, sea and soil conditions, and local O&M facilities. In general, the semi-submersible platform and the spar platform are more viable options in economic terms [30]. Spar and TLP designs have a reduced sensitivity to waves due to their smaller cross-sectional area at the water level. In terms of Technology Readiness Level (TRL), only spar and semi-submersible (including barge) designs have been proven using

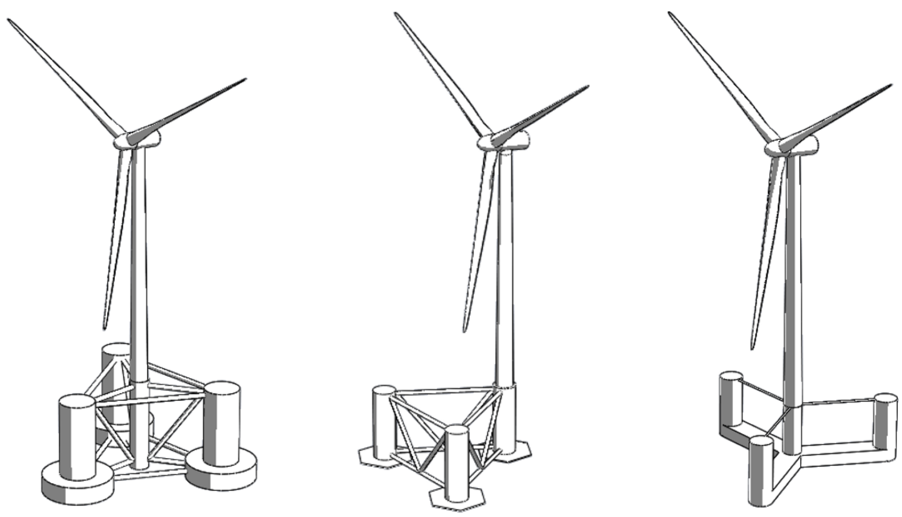


Figure 2.2: Three different semi-submersible designs: OC4 (l), WindFloat(m), Volturn US-S (r).

a full-scale prototype [30]. Furthermore, their self-stability is beneficial as it allows for the use of cheaper, less complex vessels for installation [28].

Since the semi-submersible structure has been favoured by many platform developers due to its relatively low draught resulting in ease of assembly, towing, and commissioning [21], this paper focuses on this platform topology.

A variety of different semi-submersible design concepts exist. A few of these can be seen in Fig. 2.2. On the left is the OC4 developed for the DeepCwind project for 5 MW [31]. In the middle is the WindFloat three-column floating platform compatible with any standard offshore wind turbine and suitable for deployment in waters deeper than 40 m. [32]. On the right is the VolturnUS-S developed for the IEA 15 MW Offshore WT [33]. The WindFloat is operating in Portugal and Scotland with a market leading 75 MW installed, the VolturnUS has been tested in 1:8 scale, while the OC4 is purely a reference platform at the time of writing.

In [34], a wide range of offshore wind turbine platforms and mooring systems are generated from the parameterization of relevant design variables. The semi-submersible design results for multi-body platforms showed that semi-submersible platforms with four floats demonstrated better stability and were more cost-effective than other semi-submersible designs. Designs that bear close resemblance to the OC4 performed well in this multi-objective design optimization. The study also suggested that taut mooring systems would combine well with the semi-submersible to further increase the stability.

2.1.5 Platform Material Selection

When designing the platform, a parameter to consider is its material. Steel and concrete structures are the two options utilized in the industry. Deciding factors when designing a floater are material properties, such as strength, durability, and density, as well as the material cost, availability, and carbon footprint.

The availability of steel is generally low in Europe while concrete is more widely available. In 2021, Asia contributed 72 % of the world's steel production while European countries only accounted for 8 % [35]. In a comparative study between steel and concrete substructures for FOWT by DNV for WindWorks Jelsa AS (WWJ) [36], the carbon footprint, cost, and availability of the two materials are considered. When transport from Asia is excluded from the comparison, the concrete semi-submersible was estimated to be more expensive than the steel alternative by 11 %. It should be noted that the steel substructure cost considered in the study is representative of manufacturing in Asia. Manufacturing in e.g. Europe will increase steel substructure cost by approximately 25 – 30 % (local European transport costs not included).

With regard to their carbon footprint, the concrete floaters had a 2.5-3 times lower carbon footprint compared to their steel equivalents, depending on end-of-life strategy. Introducing recycled steel to the raw material input can reduce the carbon footprint of the steel concepts significantly. However, with 70 % recycled steel content, the study still favors the concrete concepts [36].

2.2 Mooring and Anchoring Systems

Mooring systems are a well tested technology within the O&G industry but for FOWT applications, the requirements differ primarily by having higher pitch stability requirements due to the added torque from the wind.

The mooring methods can be divided into two categories; catenary and taut or tension leg mooring, utilizing two different principles to keep the floater stationary [37].

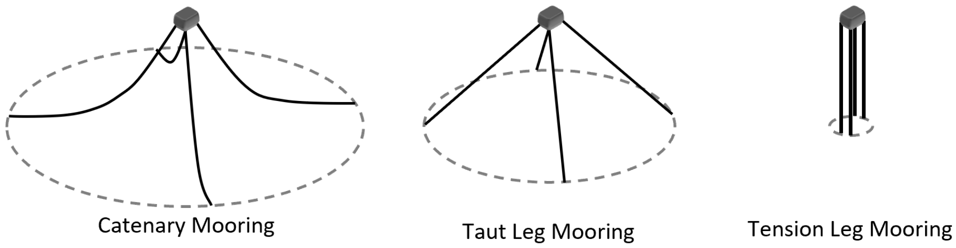


Figure 2.3: Different mooring systems.

2.2.1 Catenary Mooring

With catenary mooring, the motions of the floater is limited by the weight of the lower section of mooring chain that rests on the seafloor. Typically, this means that the anchor radius is about 4-8 times the water depth, which will ensure a horizontal and reduced load on the anchors. [38] As water depths increase, the required chain (and price thereof) increase many-fold. For this reason, catenary lines are known to be favored for applications with water depths lower than 100 m in the O&G industry [39]. The FOWT is less restricted in its position both vertically and horizontally.

2.2.2 Taut and Tension Leg Mooring

For taut and Tension Leg (TL) mooring, stability is maintained by the high tension in the mooring lines. The literature is divided on whether to use the two terms interchangeably [3], [37], or as two different concepts [39], [40], [41]. The distinction can be made based on the angle they have to the sea bed. For taut, the angle between the line and the seabed is typically between 30 and 40 degrees, while it is vertical for TL. Common for them both is that the anchor radius will be significantly lower and so will the footprint on the seabed. Since the mooring lines go in an almost straight line from the seabed to the fair-lead, the length of the mooring lines scales linearly with water depth. For this reason, a taut leg (or semi-taut) is usually adopted in waters deeper than 200 m. The requirements for the anchors are much greater as they must also be able to withstand vertical loads. With taut or TL mooring, a better station keeping performance can be obtained and the mooring system greatly benefits the stability of the floater.

For taut mooring, the material of the cables are typically synthetic fiber compositions as they have potential to deliver cost savings compared to conventional steel components due to lower weight, great fatigue performance, and less dynamic tension from their low stiffness. Available materials of synthetic fibers are HMPE (e.g., Dyneema) and polyester. Both have a long and proven track record in the oil and gas industry. For TLPs, the legs commonly consist of multiple tubular steel members called tendons.

2.2.3 Anchoring

The choice of anchor types mainly depends on the soil conditions of the seabed, the mooring system, and holding capacity required. In Fig. 2.4, a variety of different anchor types are illustrated. The drag embedded anchor is best suited to cohesive sediments and horizontal loads with the benefit of a very simple installation process and it being easily recoverable. The three pile anchors (3.-5.) can all accommodate vertical loads, and work by the same principle of soil friction. What differentiate them is their method of installation. The dead weight is the most simple anchor and can require vertical loads but due to its large size and weight, costs can increase. The vertical load anchor is installed like a conventional drag embedment anchor but penetrates much deeper. When the anchor mode is changed from the installation mode to the vertical loading mode, the anchor can withstand both horizontal and vertical loads. Although designed to suit deep water mooring applications, its unidirectional load capacity allows mooring objects in confined sub-sea infrastructures such as in vicinity of pipeline and cables [37].

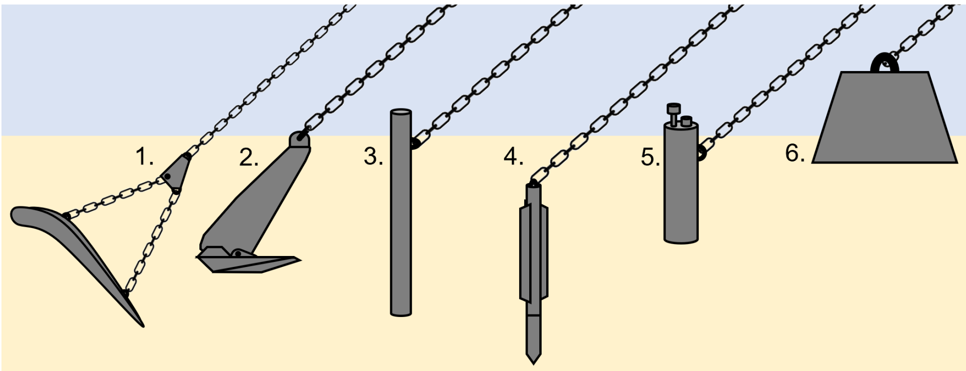


Figure 2.4: Anchor types: (1) Vertical load anchor, (2) Drag embedment anchor, (3) Drive pile, (4) Torpedo pile , (5) Suction pile , (6) Dead weight.

2.3 Upscaling of Semi-Submersible Floating Platforms

As mentioned, the size of offshore wind turbines have increased significantly over the last decade to reach higher energy production per turbine and bring down the overall cost of energy. For FOWTs, the design of the floating substructure is directly related to the supported wind turbine and tower, and the dynamics of the whole system changes at it scales.

This section explores the theoretical scaling laws for semi-submersible floating platforms to support the increasing sizes of the wind turbines and reach commercial plant-sizes for FOWT. Furthermore, an analysis for eight existing semi-submersible platforms, designed to support wind turbines with power rating between 5 to 15 MW, have identified trends related to real platform design characteristics [42]. From this, an initial upscaled design of the NREL 5 MW OC4 semi-submersible to 15 MW could be identified.

2.3.1 Theoretical Scaling Laws

As a FOWT experiences loads from both wind and waves, the scaling of the platform is closely related to the scaling of the rotor [42]. The scaling factor of the wind turbine, p_1, is defined by the increased size in rotor diameter, D, while the scaling factor of the floating support structure, p_2, is defined by the increased size of the characteristic lengths of the platform l. This could be diameters of columns, lengths of pontoons, the draught d_z, or other dimensions. Both scaling factors can be seen in Eq. 2.1.

$$p_1 = \frac{D_{scaled}}{D_{base}} \qquad , \qquad p_2 = \frac{l_{scaled}}{l_{base}} \tag{2.1}$$

Three different scaling laws, explaining the ratio between s and p for the platform design, have been proposed in literature [42]. The **power** scaling law suggests that the linear dimensions of the rotor and platform increase proportionally, as defined in Eq. 2.2. The **mass** scaling law propose that the ratio between turbine and platform follows the square-cube law, as defined in Eq. 2.3. The **fixed draught** scaling law preserves the mass properties, while keeping the draught constant, as defined in Eq. 2.4.

$$p_2 = p_1 \approx \sqrt{\frac{P_{scaled}}{P_{base}}} \quad (2.2) \qquad p_2 = p_1^{2/3} = \sqrt[3]{\frac{m_{scaled}}{m_{base}}} \quad (2.3) \qquad \begin{cases} p_{1,xy} & = \sqrt{\frac{m_{scaled}}{m_{base}}} \\ p_{1,z} & = 1 \end{cases} \quad (2.4)$$

The study showed that linear dimensions should not increase proportionally to the rotor diameter to maintain platform stability. Also, for existing reference FOWT, a draught around 20 m was maintained for all turbine sizes, suggesting that the fixed-draught law apply best with existing trends on this point. The mass and fixed-draught scaling laws both provide a satisfactory hydrostatic stiffness, and the product of r_{MO} and d_{UC} scales closer to the mass scaling law procedure. The analysis of the dynamic response showed that platforms following the mass scaling law will not experience increase in motion amplitudes, while the fixed draught procedure might lead to higher motion amplitudes.

A different study, [43], performed two theoretical upscaling procedures of the NREL 5 MW OC4 to a 15 MW system. One scaled on a combination of r_{MO} and d_{UC} and can be seen in Eq. 2.5, while another solely scaled on r_{MO} and can be seen in Eq. 2.6.

$$p_{r\&d} = 1.42 \approx \sqrt[3]{\frac{P_{scaled}}{P_{base}}} \quad (2.5) \qquad p_d = 2.07 \approx \sqrt[2/3]{\frac{P_{scaled}}{P_{base}}} \quad (2.6)$$

The metal mass of the substructure mainly increase with the combined scaling, while the solely r_{MO} scaling might pose challenges in the design of the connecting pontoons and braces, as these bear significantly increased bending moment due to their increased lengths. The combined scaling gives an increased displaced volume, resulting in a higher buoyancy force, and increased necessary ballast mass. The combined scaling also resulted in heave and pitch natural periods increasing and that way moving further away from the general wave period of ocean waves, while scaling only on distance resulted in decreasing natural periods in heave and pitch, bringing them further into the general range of ocean waves. The study [43] concluded that the combined scaling is the most reasonable of the two, for the OC4 semi-submersible.

In [44], an upscaling rule of a semi-submersible floater is investigated by surveying the design in the Fukushima FORWARD project. Here it is suggested that several parameters are customized for the explicit site. For example, the draught should be decided from the port depth, while the freeboard should be decided based on the maximum wave height at the site. Additionally, the diameter of the main column is determined from the diameter of the turbine tower base. The floater mass including ballast is scaled according to the square-cube law, as specified in Eq. 2.3. Here, the ratio of the turbine mass is used instead of the power ratings considering technology development. The diameter of the offset column and the distance between the columns is a variable and should be determined by balancing the ratio of floater restoring moment in the pitch direction as the same ratio of maximum overturning moment. The distance between the columns are increased with 5 % from 2 MW to 5 MW and from 5 MW to 10 MW.

2.3.2 Practical Scaling Trends

An analysis of eight existing reference semi-submersible floating platforms showed that none of the above-mentioned scaling laws were followed in practice [42]. The review of the existing semi-submersible floaters showed a large variety of design solutions, and although a direct comparison is difficult because the floater types are different, the study showed that the key geometric parameters that drive the design of semi-submersible platforms are the product of the radius from center to offset column, r_{MO}, and the diameter of the upper column, d_{UC}, (see Fig. 2.5, showing top view of the OC4 floater). The best-fit curve of the relation between $r_{MO} \cdot d_{UC}$ of the eight platforms and the rotor diameter, D_R, of the corresponding wind turbines is given in Eq. 2.7, and showed a high goodness-of-fit value of $R^2 = 0.911$, indicating that it is a good relation to scale by.

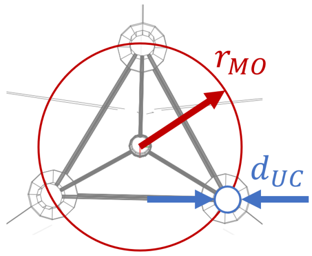


Figure 2.5: r_{MO} & d_{UC}.

$$r_{MO} \cdot d_{UC} = 1.63 \cdot D_R^{1.09} \tag{2.7}$$

2.4 Hydrostatics and Stability of FOWT

In this section, basic concepts of intact stability will be presented in order to better understand what elements go into creating a stable floater. The motion of a FOWT is defined with 6 degrees of freedom (DoF), as seen in Fig. 2.6. The oscillatory translatory motions refer to *surge*, *sway*, and *heave*, while the oscillatory angular motions refer to *roll*, *pitch*, and *yaw*. Stability is defined by the FOWT ability to maintain its position and orientation for all 6 DoF while being subject to external forces from wind and waves.

All of the theory derived in this section applies to a body floating freely in still water. That is, there are no waves, and the body is at rest relative to the water surface, vertically as well as horizontally. This section is largely based on [45].

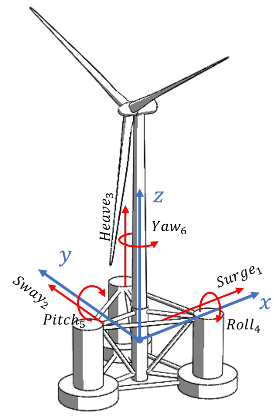


Figure 2.6: Six DoF

2.4.1 Flotation

Floating is met when the total weight of all components of the FOWT system is in equilibrium with the buoyancy force. This is Archimedes' principle. The weight of the system, gm, is adjusted by changing the ballast, while the buoyancy force is defined from the volume of the displaced water, ∇, multiplied with the water density, ρ, and the gravitational force, g. The law for floating is described in Eq. 2.8.

$$gm = \rho_w g \nabla \tag{2.8}$$

As any object has a **Center of Gravity** (CoG) for where the total weight of the body may be thought to be concentrated, so does any floating body have a **Center of Buoyancy** (CoB), for where the buoyancy force may be thought to always act upon. The CoB is defined as the geometric center of the submerged body. Under normal floating conditions with no external forces acting on the body, the CoG and CoB will be on the same vertical line. When a floating body is subject to displacement, the CoG will stay fixed to the body while the CoB follows the new geometric center of the submerged body, as illustrated in Fig. 2.7. The point about which a floating body is rotating when subject to a torque is called the **Center of Flotation** (CoF) and is the centroid of the area of the waterplane [46]. This point is also the origin of the coordinate system in RAFT as illustrated in Fig. 2.6.

2.4.2 The Metacenter

The metacenter, M_C, is another geometric point which is of importance to a floating body in upright equilibrium. It indicates how stable the floating body is in its upright position. For a stable floating body, which is given a small angular movement away from its equilibrium position, the metacenter can be assumed to be located in a fixed point defined as the point of intersection between the center line of the cross section containing the CoG (and the old CoB) and the vertical line through the new CoB due to any smaller angular displacement of the body as seen in Fig. 2.7.

The assumption of a fixed metacenter is only reasonable for small angles. For most conventional hull forms, the metacenter can be taken as fixed up to angles of heel of about 10° [47].

2.4.3 Initial Stability

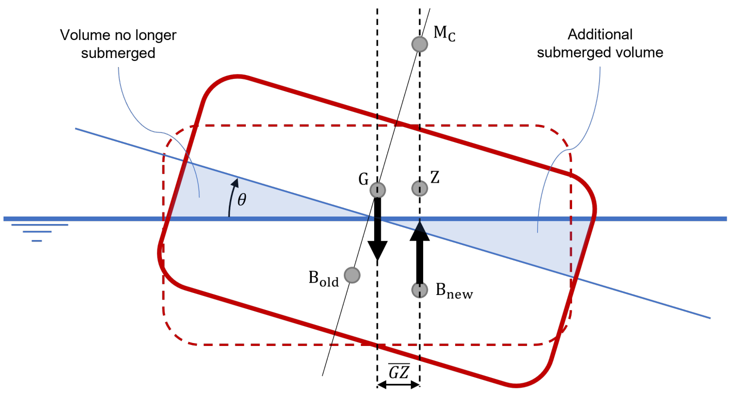


Figure 2.7: Depending on the position of CoG, G, relative to the metacenter, the force couple formed by buoyancy and weight creates a restoring moment.

The fundamental concept of the initial stability of a floating body, which is when the body is upright or very nearly so, can be explained by considering the body to be inclined from the upright by an external force (moment) which is then removed. See Fig. 2.7. As the body inclines, the CoG stays fixed to the body while the CoB changes as the displaced volume changes. The two points are no longer vertically aligned, resulting in a righting moment defined by the buoyancy and righting lever being the horizontal distance between the two centers, $\overline{GZ}$.

$$M_R = \rho g \nabla \cdot \overline{GZ} \tag{2.9}$$

The righting lever $\overline{GZ}$ can be found from the inclination angle θ and the distance between the CoG and metacenter M_C:

$$\overline{GZ} = \overline{GM_C} \sin(\theta) \tag{2.10}$$

From this, it can be seen why the metacenter is important. In the Fig. 2.7, it is seen how M_C is above G, giving positive stability. Had the new CoB been directly beneath the CoG, then M_C and G had been coincident resulting in neutral stability. Finally, if instead the inclination had caused M_C to be below the CoG, the body would be unstable as the moment would rotate the body further from vertical and possibly capsizing. For this reason, the sign of $\overline{GM_C}$ can be taken as a criterion for initial stability:

$$\overline{GM_C} > 0 \Leftrightarrow \text{Stable} \qquad \overline{GM_C} = 0 \Leftrightarrow \text{Neutral} \qquad \overline{GM_C} < 0 \Leftrightarrow \text{Unstable} \tag{2.11}$$

As seen from the figure, the distance $\overline{GM_C}$ can be expressed as in Eq.2.12:

$$\overline{GM_C} = \overline{BM_C} - \overline{GB} \tag{2.12}$$

Where $\overline{GB}$ is the distance between CoB and CoG, while BM_C is the distance between CoB and the metacenter, which for small angles can be express by Eq. 2.13 [45]:

$$\overline{BM_C} = \frac{I_x}{\nabla} \tag{2.13}$$

where I_x is the area moment of inertia of the waterplane. For $\overline{GB}$, this can be defined from the z-coordinates (see Fig. 2.6) of the center of buoyancy and gravity respectively as seen in Eq. 2.14:

$$\overline{GB} = z_B - z_G \qquad \Leftrightarrow \qquad -\overline{GB} = z_G - z_B \tag{2.14}$$

In combination, Eq. 2.9 - 2.14 give a new expression for the righting moment [48]:

$$M_R = \rho g \nabla \Big(\frac{I_x}{\nabla} + (z_B - z_G) \Big) \sin(\theta) = \rho g I_x \sin(\theta) + \rho g \nabla (z_B - z_G) \sin(\theta) \tag{2.15}$$

To see the initial stability analysis applied to the OC4 wind turbine, go to App. A.2.

2.4.4 Hydrostatic Stiffness

The hydrostatic stiffness, K_{hydro}, of a floating structure describes the variations in mass and buoyancy when the structure is exposed to changes from its base position. It is hence a frequently used parameter for creating stability criteria for floating structures [42], [49]. In matrix form, the relation between the forces acting on the structure and its movement in each DoF can be described by Eq. 2.16:

$$[F] = [K_{hydro}][x] \tag{2.16}$$

where $[F]$ is the vector of forces (or moments for rotational DoF) acting on the structure and $[x]$ is the translation (or angle for rotational DoF), while $[K_{hydro}]$ is the hydrostatic stiffness matrix.

Only the heave, roll, and pitch components are contributing to the hydrostatic stiffness as motion of these DoF will result in changes in the platform position relative to the waterplane. They are given with respect to the reference coordinate system axes [50].

Three effects contribute to the hydrostatic of a FOWT; the change in load due to change in submerged volume, the change in moment due to change in CoG and CoB, and finally the effects from the mooring system. As mentioned in Sec. 2.1, the spar, semi-submersible, and TLP substructure all respectively utilizes each of these three effects.

In RAFT, the hydrostatic stiffness is calculated for each member and summed to get the hydrostatic properties of the structure [51]. The global hydrostatic stiffness matrix, K_{hydro}, excluding the mooring stiffness, is given in Eq. 2.17:

$$K_{hydro} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho g A_{wp} & \rho g A_{wp} y_{wp} & -\rho g A_{wp} x_{wp} & 0 \\ 0 & 0 & \rho g A_{wp} y_{wp} & \rho g I_{xx} + \rho g \nabla z_{cb} - m_{tot} g z_{cg} & -\rho g A_{wp} x_{wp} y_{wp} & 0 \\ 0 & 0 & -\rho g A_{wp} x_{wp} & -\rho g A_{wp} x_{wp} y_{wp} & \rho g I_{yy} + \rho g \nabla z_{cb} - m_{tot} g z_{cg} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{2.17}$$

A_{wp} is the waterplane area and x_{wp} and y_{wp} is the coordinates for its weighted center. ∇ is the displaced volume, and z_{cb} is the vertical center of buoyancy location. I_{xx} and I_{yy} is the waterplane moments of inertia about the x and y directions, respectively.

As seen from Eq. 2.17, the hydrostatic stiffness in pitch and roll is similar except for the axis of the moment of inertia. As most substructures are symmetrical, this term will be the same, resulting in stability in pitch and roll being equal. The full hydrostatic stiffness for pitch can be seen in Eq. 2.18, with the three effects making up the hydrostatic stability contribution:

$$K_{55} = \underbrace{\rho g I_{yy}}_{\text{waterplane}} + \underbrace{\rho g \nabla z_{cb} - m_{tot} g z_{cg}}_{\text{B-G relative position}} + \underbrace{K_{moor,55}}_{\text{mooring}} \tag{2.18}$$

Note that K_{55} from Eq. 2.18 bears similarity to the hydrostatic stiffness element from Eq. 2.15. Since Eq. 2.15 describes a free-floating body, there is no need to distinguish between the buoyancy force $\rho g \nabla$ and the weight mg of the system as they are identical. For Eq. 2.18, this distinction is necessary as the weight of the catenary mooring is not included in the total weight of the structure mg resulting in $\rho g \nabla > mg$. This is instead accounted for in the mooring stiffness contribution, K_{moor}. With the new definition of the hydrostatic stiffness, the righting moment in pitch can be simplified as shown in Eq. 2.19:

$$M_{R,pitch} = K_{55}\theta \tag{2.19}$$

The buoyancy stiffness contribution is the relation between the buoyancy force and weight, so for free-floating structures, its contribution is dependent on the horizontal placement of the two centers. A spar buoy gets its stability by having z_{cg} being far below z_{cb} by the use of a heavy deeply placed ballast. If z_{cg} is instead above z_{cb}, the buoyancy stiffness contribution will be negative. This is the case for the OC4 structure as seen in App. A.4.

The contribution to the total stiffness from the mooring system, K_{moor} is best represented by a 6×6 matrix, since it can generate counteracting forces in all DoF with possibly coupled terms [48].

In order to understand which geometric parameters play a key role in the contribution to the waterplane stiffness, the global second moment of area for a three offset column semi-submersible, can be approximated using Eq. 2.20 [42]:

$$I_{yy} \approx \frac{3\pi}{8}\left(r_{MO}^2 d_{UC}^2 + \frac{d_{UC}^4}{8}\right) \tag{2.20}$$

For the semi-submersible 5 MW OC4, the effect from the waterplane is the driving factor, while the B-G relation is a small negative contribution. In App. A.4, the hydrostatic stiffness in pitch of the 5 MW OC4 is found analytically and compared with computational results from RAFT.

2.5 Design of FOWT

In this section, the overall considerations driving the design of a semi-submersible platform for a FOWT are investigated, including what parameters are important to account for in regards to the hydrostatics and stability of a FOWT. As the industry is still relatively immature, no detailed guidelines are available when it comes to design of FOWTs.

2.5.1 Maximum Pitch Angle

It is desired that wind turbines operate in a close to upright position, as a larger heeling angle will change the projected rotor plane area to the wind, and the blade angle of attack resulting in a reduction in the power output. Although it is necessary to impose a maximum pitch angle, the exact value is still open for discussion. Literature refers to a value of 10° for the maximum pitch angle [42], [48].

2.5.2 Freeboard Height and Draught Limits

Freeboard and draught limits can be set due to both stability and logistic considerations. One of the reasons for semi-submersible platforms being favored by the industry is its low draught, which allows for easier assembly, towing, and commissioning [21]. Generally, platform developers try to keep the draught close to 20 m regardless of the size of the wind turbine [42].

In heavy storms, an incoming wave which significantly exceeds the freeboard can impact and wet the deck leading to *green water loads*. Green water loading may cause extensive structural damages to floating offshore units and is of considerable concern to the stability and survivability of the platform [52]. In order to avoid these loads, a minimum vertical distance between sea water level and the top of the support structure is set, also called the minimum freeboard height. This minimum freeboard height depends on the local metocean conditions, but an initial estimate for preliminary design is 10-12 m [42], [48].

2.5.3 Standard for Hydrostatic Stability

The standard DNV-ST-0119 for floating offshore wind turbine structures [53] specifies general principles and requirements for the design of FOWTs. For intact floating stability, it is stated that the floating structure shall be capable of maintaining stability during operation of the wind turbine at the wind speed that produces the largest rotor thrust. The floating structure shall also be capable of maintaining stability during standstill of the wind turbine in severe storm conditions. The largest rotor thrust force is reached when the turbine is operating at rated wind speed, while during storm conditions, it is common to either pitch the wind turbine blades to feather, while maintaining the rotor plane perpendicular to the direction of the wind, or to yaw the rotor 90°out of the wind.

The standard further specifies that stability is maintained for semi-submersibles when the following two requirements are met:

"The area under the righting moment curve to the second intercept or downflooding angle, whichever is less, shall be equal to or greater than 130% of the area under the wind heeling moment curve to the same limiting angle".

"The righting moment curve shall be positive over the entire range of angles from upright to the second intercept".

The static stability of a floating structure can be presented from the righting moment, or just by the lever (GZ distance from Fig. 2.7) as the forces are constant. This is visualized in a static stability curve or GZ-curve in Fig. 2.8.

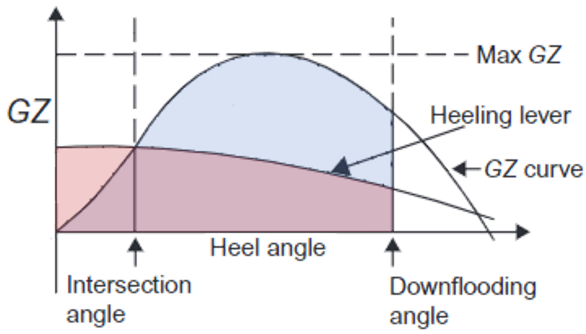


Figure 2.8: Arbitrary GZ-curve for visualisation of the concept. The blue area must be 130 % the size of the red area dictated by the DNV-ST-0119 standard.

The range of angles over which the lever arm GZ is positive, is called the *range of stability*. Freeboard height and reserve of buoyancy are important for determining this range. The area under the curve represents the ability of the floating body to absorb energy imparted to it by winds, waves or other external loads. The GZ-curve apply to the full range of inclinations, meaning that it reflects both initial intact stability (small angles) and stability at larger angles [45].

Deriving the entirety of the curve is complex and requires specifically developed software and detailed information on the geometry and mass distribution of the floating body [45].

Horizontal axis wind turbines give the best power output when operating at upright position. For this reason as well as stability considerations, FOWTs are designed to never heel beyond small angles for operation at maximum rotor thrust. The theory for initial intact stability (small angles) is explained in details in Sec. 2.4.3, where the equation for the righting moment is also derived. A calculation of the stability at larger heeling angles is not performed, as the FOWT will be designed to never go beyond small angles of heel. This is done by designing the structure so that the hydrostatic restoring moment in pitch is equal to or greater than the overturning moment resulting from the maximum thrust force acting on the wind turbine.

2.5.4 Natural Frequencies and Wave Forces

Wind and wave forces impose oscillatory motion to the FOWT, and it is important that these motions are minimized to ensure an appropriate system performance. Ocean waves contain substantial energy in certain periods, and ideally, the system should be designed so that the natural frequencies of the floating structure is outside the range of wave excitation, which has a period of 5-25 s, to avoid resonance. For an undamped system, the natural frequency of the FOWT system is estimated from the restoring and inertial properties of the system according to Eq. 2.21, where C refers to the hydrostatic stiffness matrix in the i-th mode of motion, M and A are respectively the mass and zero-frequency added mass matrix in that mode [54] [55].

$$f_{n,i} = \frac{1}{2\pi}\sqrt{\frac{C_{ii}}{M_{ii} + A_{ii}(\omega_{n,i})}} \tag{2.21}$$

For semi-submersible floaters, the heave, pitch, and roll motions might have natural frequencies within the general spectral range of ocean waves, while natural periods in surge and sway are governed by the mooring system and usually have higher natural frequencies [55]. Due to the symmetry of the geometry of the floater, natural frequencies in pitch and roll are almost similar, but small variations can occur due to the contribution from the Rotor Nacelle Assembly (RNA) which is non-symmetrical.

The FOWT system response to wind and wave forces can be analyzed with a frequency domain approach. In a frequency domain analysis, the relative wave spectrum is modeled, and initial and geometric characteristics of the system are derived.

2.5.5 Coupling of Natural Frequencies and Rotor Speed

Coupling of the rotational frequency of the wind turbine and the motion of the whole FOWT system should be avoided. This means that the rotor speed, 1P, and the blade passing frequency, 3P, of the wind turbine should be out of the range of the natural frequency of the semi-submersible platform [42], [55].

2.6 Wave Excitations and Response

2.6.1 Wave Spectrum

In general, the wave energy spectrum, $S_\zeta(\omega)$, derived from an analysis of an irregular wave record obtained at a particular time and location in the ocean will be a unique result that will never be exactly repeated.

In coastal waters such as the North Sea, the wave spectrum is rarely fully developed and for this reason the Joint North Sea Wave Project (JONSWAP) modified the Bretschneider spectrum to consider a limited fetch. The result is the JONSWAP spectrum, $S_J(\omega)$ from Eq. 2.22, which will be used in this report [56]. The distribution can be seen in Fig. 2.9 for different sea states.

$$S_J(\omega) = 0.658 \cdot 3.3^J \frac{A}{\omega^5} \exp\left(\frac{-B}{\omega^4}\right) \tag{2.22}$$

where

$$J = \exp\left[\frac{-1}{2\lambda^2}\left(\frac{\omega T_0}{2\pi} - 1\right)^2\right] \quad , \quad A = \frac{H_s^2}{4\pi}\left(\frac{2\pi}{T_z}\right)^4 \quad , \quad B = \frac{1}{\pi}\left(\frac{2\pi}{T_z}\right)^4 \tag{2.23}$$

H_s is the significant wave height, T_z is the mean zero-crossing wave period, and $\lambda = 0.07$ for $\omega \leq 2\pi/T_0$ or $\lambda = 0.09$ for $\omega > 2\pi/T_0$ where T_0 is the modal period. For a Bretschneider spectrum like this, $T_0 = 1.296T_p = 1.41T_z$ [56].

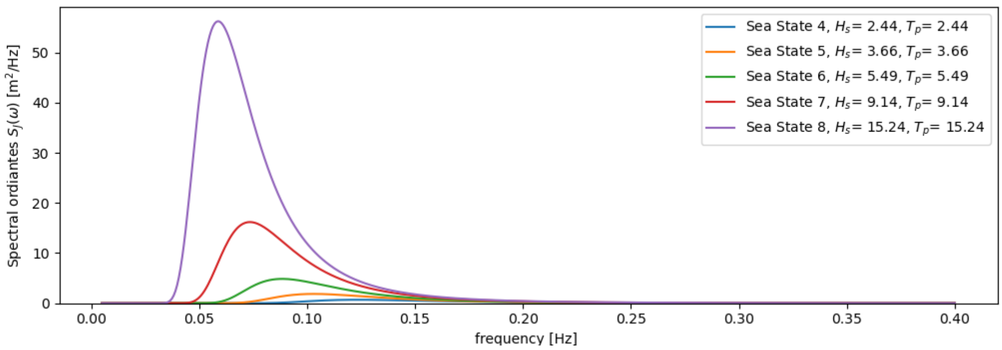


Figure 2.9: JONSWAP wave spectra for different sea states defined in [33].

2.6.2 Response Amplitude Operators

In the field of design of floating structures, a Response Amplitude Operator (RAO) is effectively a transfer function used to determine the effect that the environment will have upon the motion of a floating body. ROAs are usually obtained from scale models tested in a model basin, or from running specialized CFD, often both. They are graphs of the resulting non-dimensional amplitudes plotted as a function of encountered frequency and give the proportion of wave amplitude transferred by the ship system into the ship motions [57], [56].

The complex-valued RAOs of a floating structure are found by solving the equations of motions, Eq. 3.11, for the specific structure.

As mentioned, the amplitudes are assumed to be proportional to the wave amplitude, ζ_0, and it is usual to express them in non-dimensional form: linear displacement amplitudes are non-dimensionalized by dividing by the wave amplitude, and angular motion amplitudes are divided by the wave slope amplitude $k\zeta_0$ in radians [54], [56]:

$$RAO_i = \left|\frac{\hat{\xi}_i}{\zeta_0}\right| \quad \text{for} \quad i = 1,2,3 \qquad \text{and} \qquad RAO_i = \left|\frac{\hat{\xi}_i}{k\zeta_0}\right| \quad \text{for} \quad i = 4,5,6 \tag{2.24}$$

The use of RAOs to describe the behaviour of floating structures assumes a linear relationship between the waves and motion of the ship. As mentioned in Sec. 3.5, the linear assumption is not always valid, but through linearization and iterations it provides a reasonable approximation to the motion of the FOWT.

In RAFT, the linearization inherent in the viscous damping is dependent on the response amplitude, see Eq. 3.11, resulting in a coupling between the excitation forces and the RAOs [58]. As the effect of rotor aerodynamics on the dynamic response is modeled using the derivative of thrust with respect to wind speed, especially the wind will have a significant impact on the damping of the FOWT [51].

By definition, the frequency-domain model assumes that the platform motions are at the same frequency as the incident waves and that the incident waves are regular. While this means that the transient response of the system cannot be modeled, the assumption of linearity implies that the responses at different wave frequencies can be superimposed according to a wave spectrum to predict the system behavior in irregular sea states [58].

2.6.3 Response Spectrum

The response spectrum - also called the motion energy spectrum, power spectrum, or power spectral density (PSD) - for the individual motion component is calculated by filtering the encountered wave energy spectrum with the appropriate RAO. Assuming that the RAO has been properly normalised,the response spectrum, $S_R(\omega)$, is obtained in the wave frequency domain from Eq. 2.25 [56]:

$$S_R(\omega) = S_\zeta(\omega)|RAO(\omega)|^2 \tag{2.25}$$

The spectral analysis (Eq. 2.25) is an important tool in the assessment of the sea-keeping performance of an offshore structure. High RAO ordinates occurring at frequencies where there is relatively high levels of wave energy will give large contributions to the response spectrum. The motions which are most sensitive to modal period are those that have distinct RAO peaks, like relative motion and roll.

The response spectrum explains the power content of a signal as a function of frequency per unit of frequency. This means that in the response graph, the amplitudes are normalized by the frequency resolution [59].

In Fig. 2.10, the wave spectrum, transfer functions (RAO), and response spectra in surge, heave, and pitch have been generated in RAFT for the original OC4 5 MW semi-submersible. It can be seen how the RAO acts as a filter on the wave spectrum to yield the response spectrum. To keep the response as low as possible, it is important to have the peaks of the weave spectrum and RAO not being overlayed.

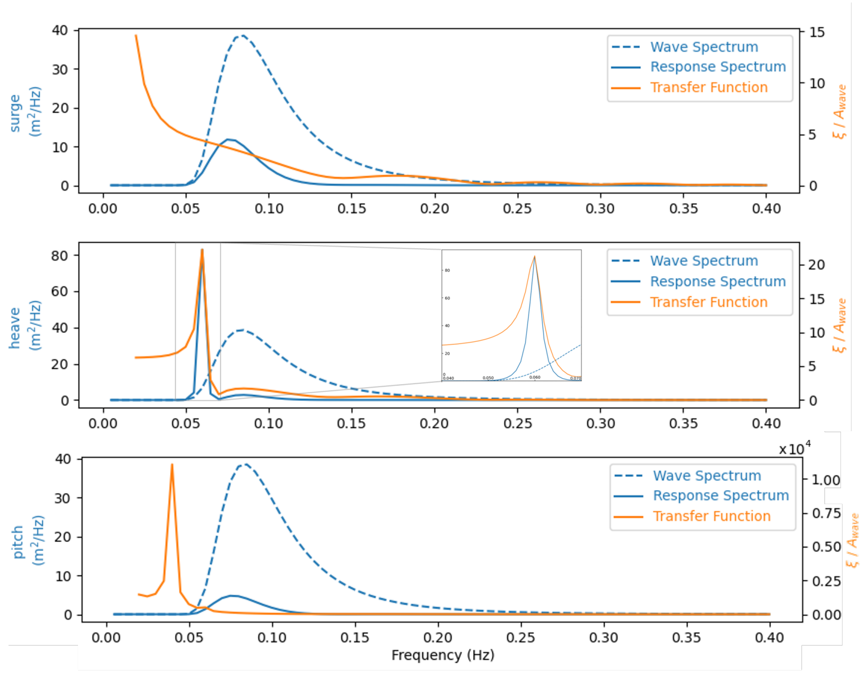


Figure 2.10: Wave and response spectra along with transfer functions (RAOs) of the original 5 MW OC4 semi-submersible at a wind speed of 8 m/s, wave height of 6 m and wave period of 12 s.

2.6.4 Statistic in Frequency Domain

Very often, the analysis of ocean statistics can, advantageously, be performed in the frequency domain. In the frequency domain, the n-th order spectral moments are defined by Eq. 2.26:

$$m_n = \int_0^\infty \omega^n S_\zeta(\omega) d\omega \qquad , \qquad (n = 0, 1, 2, ...) \tag{2.26}$$

A useful statistic is that the variance, σ^2, is found as the 0-th order moment (for an ergodic process with zero mean). In Eq. 2.27, this is shown for the wave spectrum but applies for the response spectrum as well [56].

$$\sigma^2 = m_0 = \int_0^\infty S_\zeta(\omega) d\omega \tag{2.27}$$

The Root Mean Square (RMS) value is defined as the square root of the variance, i.e., $RMS = \sigma = \sqrt{m_0}$. For an input signal with zero mean, RMS is identical to the Standard Deviation (SD).

It is often of relevance to estimate the **maximum** value from a given spectrum.

One approach to estimate this is with the *three-sigma rule* in which a requirement is defined as three standard deviations above the mean value, see Eq. 2.28. This has e.g. been used in similar studies like [60] and [34]. For a normal distribution, this accounts for 99.7 % of the probability, and it is thus empirically useful to treat it as near certainty [61].

$$\max = \mu + 3\sigma \tag{2.28}$$

Another way to estimate a maximum is with the assumption of a Rayleigh distribution rather than a normal distribution. In many engineering cases, the Rayleigh distribution is assumed to be a good approximation of the distribution of the individual wave heights in measured time histories of sea surface elevation. The Rayleigh distribution is a special case of the Weibull distribution and given from Eq. 2.29 [56]:

$$F(x) = 1 - \exp - \Big(\frac{x}{2m_0}\Big) \tag{2.29}$$

Its probability distribution function $p(x)$ is given by:

$$p(x) = \frac{dF}{dx} = \frac{2x}{2m_0} \exp - \Big(\frac{x}{2m_0}\Big)^2 \qquad , \qquad x \geq 0 \tag{2.30}$$

When m_0 is known, a maximum value for a spectrum can then be estimated from a determined probability value [62].

2.7 Structural Analysis

Structural integrity is very important when designing a substructure, as failure can result in severe damage, leading to significant monetary losses and human injuries or death. The structural integrity refers to its ability to withstand loads and maintain its designed function for its intended lifetime, including extreme cases. Structural failure refers to the loss of a load-carrying component or the structure itself, and happens when the material is stressed beyond its strength limit.

It can be difficult to accurately evaluate the strength of a complex structure like a substructure. Finite Element Analysis (FEA) is a valuable tool as it provides a reasonable estimate using a detailed computational analysis while being much cheaper, faster and more practical compared to comprehensive testing. FEA is the use of Finite Element Method (FEM) which works by subdividing a larger structure into a mesh of smaller elements which can be described using differential equations and solved numerically by approximating a solution by minimizing an associated error function. While FEA is a valuable tool, it is beyond the scope of this thesis.

When designing the structure, it should be included in the calculations that the structure should be designed stronger than needed for normal usage to allow for emergency situations, or unexpected loads such as unintended collisions by drifting service vessels, unintended change in ballast distribution, dropped objects during maintenance operations, loss of mooring lines, fire, or explosions [53]. For structures which are to be designed for use in areas exposed to rare tropical storms, these loads should also be considered. It should also account for general additional environmental loads, such as the presence of waves.

This margin is often described using a safety factor, s, which can be defined as the fraction of the total structural capability over what is needed. From this, a structural design constraint can be formulated from the criteria that a given design load, P_{des}, multiplied by a safety factor must not exceed the critical load, P_{cr}, as shown in Eq. 2.31

$$P_{cr} \geq P_{des} \cdot s \tag{2.31}$$

Different elements in the structure will be subject to different loads such as tension, compression, bending, torsion, and shear. Some relevant structural failure mechanisms are described below.

2.7.1 Yielding

In order to secure against failure, it is important to ensure that the internal forces in the structure does not surpass the yield strength of the material. The yield strength, σ_y, is a material property and is the stress corresponding to the yield point at which the material begins to deform plastically. Typical steel of the type ASTM A36 has a yield strength of approximately 250 MPa.

The uniaxial normal stress, σ, throughout a bar, across any cross sectional area, can be expressed simply by the magnitude of the internal compression/tension forces, N, and cross sectional area of the bar, A, as seen in Eq. 2.32:

$$\sigma_{des} = \frac{N}{A} \tag{2.32}$$

From this, a criteria against yielding can be set up so that the maximum allowable yielding, σ_{des}, is within the boundaries of yielding with a safety factor, s_y, as seen in Eq. 2.33:

$$\sigma_{des} \cdot s_y \leq \sigma_y \tag{2.33}$$

2.7.2 Buckling

When a beam is experiencing axial tension, its load capabilities are defined by the yield strength before it begins to deform plastically. When subject to axial compression however, this equilibrium state can become unstable long before the yield strength of the material is reached. This phenomena of stability loss is called buckling, and is a very relevant structural criteria to investigate. It can be analytically investigated using Euler's critical load given by the formula in Eq. 2.34 [63].

$$P_{b,cr} = \frac{\pi^2 EI}{(k_l l)^2} \tag{2.34}$$

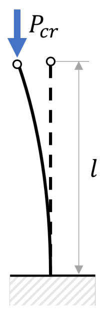


Figure 2.11

k_l is the effective length factor which varies for different elementary cases depending on how the beam is suspended at each end. One of the most critical conditions is a fixed-free condition as seen in Fig. 2.11, where $k_l = 2$. E is Young's modulus and I is the second moment of inertia.

For a beam with a cross section of an annular, the second moment of inertia is given by Eq. 2.35.

$$I_{an} = \frac{\pi}{4}(r^4 - (r-t)^4) \tag{2.35}$$

where r is the outer radius and t is the wall-thickness as seen in Fig. 2.12 (c).

Buckling has been used as the sole structural criteria in [34] for optimization of a similar semi-submersible structure to the OC4, as it was estimated to be the most critical failure point.

2.7.3 Bending

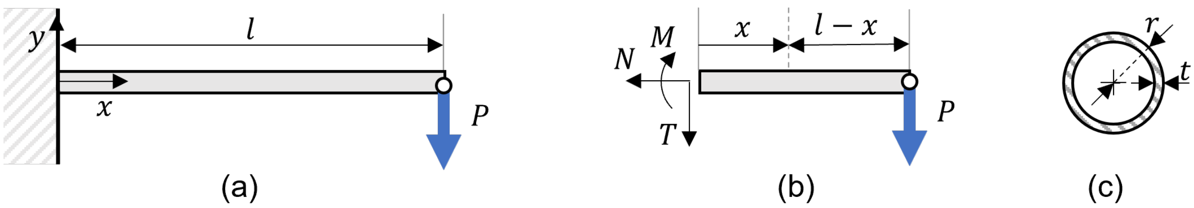


Figure 2.12: Bending

Another relevant failure cause to investigate is bending. Bending typically happens when a beam is subject to an external load applied with a force component perpendicularly to the longitudinal axis. In Fig. 2.12 (a), this is shown for a cantilever beam with a point load at its end. As the beam is fully defined, the external forces can be found from the force equilibrium in all DoF as shown in Eq. 2.36 and from the subsystem in Fig. 2.12 (b).

$$\rightarrow: N = 0 \qquad \uparrow: T = -P \qquad \curvearrowright: M = P(x-l) \tag{2.36}$$

As there are no axial forces, the beam is subject to pure bending with the internal stresses defined by Eq. 2.38:

$$\sigma(x,y) = \frac{-M(x)}{I} \tag{2.37}$$

The numerical greatest stress is found for $x = 0$, $y = \pm r$. Applying this to a beam with a cross section of an annular as seen from Eq. 2.35, the maximum internal stress is:

$$|\sigma|_{\max} = \frac{-M(0)}{I} r = \frac{Pl}{\frac{\pi}{4}(r^4 - (r-t)^4)} r \tag{2.38}$$

2.7.4 Fatigue

Since FOWTs are expected to operate for 20–25 years, the fatigue strength is one of the key factors in securing the safety as for all offshore structures. Generally, the fatigue damage analysis of an offshore structure is to be performed in the time domain, but a few frequency domain fatigue analyses for semi-submersible platform have also been proposed [64]. First, the stress range and the number of cycles are calculated from the time history data, and subsequently, the fatigue damage is calculated by using an S-N curve based on Palmgren Miner's law. The stress is found by dividing the bending moments by the sectional modulus [65]. For estimating the total fatigue damage, a large number of load conditions must be included in the simulation since the fatigue accumulates over the service life. Since FOWTs are subjected to various types of uncertain loads, such as wind, wave, current, and mooring force, the coupled dynamics need to be considered to analyze the whole FOWT structure as a system [66]. For a complex structure like a semi-submersible, the procedure would require FEA and is outside the scope of this project.

2.8 Design Optimization

Design optimization is a tool to replace an iterative design process and accelerate the design cycle.Design optimization require a formulation of the optimization problem, including the objective function (the goal of the optimization), the design variables (parameters that can be changed), and the constraints that must be satisfied (the limits on the design variables that defines the feasible space). When the problem has been formulated, the optimization algorithm evaluates the design and seeks the best possible solution from a set of possible solutions by automatically changing the design variables. A final design is identified when the optimal conditions of the objective function are satisfied and no other designs close by are better. Although the optimization is automated, human decisions are still needed in the design optimization process, and especially the formulation of the optimization problem is crucial [67].

A general optimization problem formulation is given in Eq. 2.39, with the objective function, f, number of decision variables, n, constraints, g, number of constraints, m, and lb and ub as the lower and upper bounds of the decision variables:

$$\begin{array}{ll} given & \vec{x} = \langle x_1, x_2, .., x_n \rangle \\ minimize & f(\vec{x}) \\ s.t. & g_1(\vec{x}) \leq b_1 \\ & g_2(\vec{x}) \leq b_2 \\ & \ldots \\ & g_m(\vec{x}) \leq b_m \\ & lb_i \leq x_i \leq ub_i \end{array} \tag{2.39}$$

After running the optimization and an optimal design has been identified, the design must be assessed and evaluated by engineers. At this stage, they might decide on reformulating the optimization problem. Post-optimality studies are done to interpret the optimal design which has been identified, and the general design trends. This can be done by varying the design variables or other parameters and quantifying their effect on the objective function or the constraints. An overview of the whole design optimization problem is seen in Fig. 2.13.

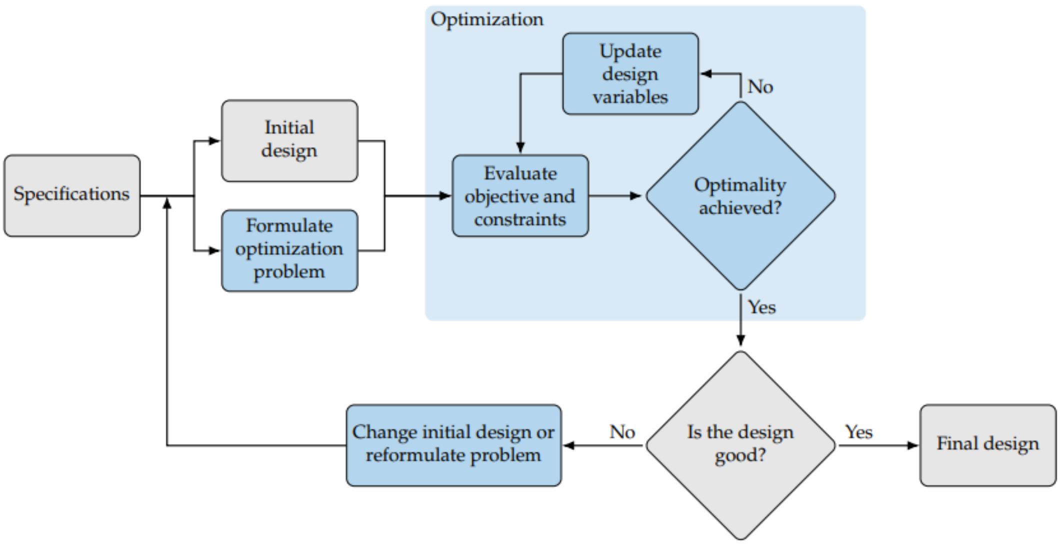


Figure 2.13: Overview of the design optimization process [67].

2.8.1 Optimization of Floaters

Floating offshore wind turbines are complex systems, consisting of many elements, and they can be optimized with respect to several different aspects, such as cost, reliability, performance, manufacturability, and structural integrity. Furthermore, the structures must withstand challenging metocean conditions, as it will often be preferable to place them in locations of deep water with strong winds followed by large waves. This means that FOWTs are exposed to a complicated loading (aero- and hydrodynamic), and they experience motion in all six degrees of freedom.

Due to interdependencies among the different subsystems, the optimization of only one subsystem will not result in an overall optimal design. It is difficult to find an all-in-one solution for an optimized design, and for this reason, one must select certain aspects to focus on in the optimization. Essentially, an integrated design approach in the optimization framework is necessary to achieve significant cost reductions in floating wind energy systems [21]. If the system is solely evaluated on reducing mass with the constraint of not capsizing, the turbine might pitch and move too much resulting in a reduced power output.

2.8.2 Cost Optimization of FOWT Systems

A detailed cost function of a FOWT is a combination of the cost of the wind turbine, floating platform, mooring system, and anchor cost.

For the floating platform, the driving factor determining the cost is its mass. Although the cost per mass differs for cylinders and members respectively, a constant steel cost per kg is usually used for the platform cost. The cost of the mooring system is modeled using the total length of the lines and the maximum steady-state mooring line tension which is implemented on the mooring system. The cost model for mooring lines is defined based on a factor with the unit [cost/m kN] which is multiplied by the line length and the maximum steady-state line tension. The anchor cost is defined by combining the anchor installation and technology costs. Anchor cost is modeled based on the maximum steady-state load on the anchors, and a fixed per-anchor installation cost is also included in a cost model [34].

Installation and manufacturing cost is highly dependent on the location at which the FOWT will be installed, and the size of the wind farm (economies of scale). A detailed cost analysis, covering location, transportation, and manufacturability, could be carried out when a specific project and site is specified.

3 Methodology

This chapter outlines the technical processes and methods employed in the development of an optimized 15 MW OC4 FOWT.

First, relevant specifications are introduced for the wind turbine, floater, and environmental conditions in Sec. 3.1. It is explained what method is used for upscaling the platform in Sec. 3.2, and the method used for evaluating the structural integrity of the platform is described in Sec. 3.4. The workflow of RAFT is introduced in order to understand how the FOWT is analyzed in Sec. 3.5. The optimization process is described in Sec. 3.6, and important performance parameters are described in Sec. 3.7.

3.1 Specifications

3.1.1 The OC4 Semi-Submersible Floater

The floater design which has been upscaled and optimized in this project is the OC4 semi-submersible structure, which has been developed in the Offshore Code Comparison Collaboration Continuation (OC4) as a part of the DeepCwind project. The turbine specification for the OC4 was the National Renewable Laboratory (NREL) offshore 5 MW baseline wind turbine, which is a representative utility-scale, multi-megawatt turbine [33].

The OC4 semi-submersible consists of a main column (MC) attached to the turbine tower, and three offset columns (OC) that are connected to the main column through a series of pontoons and cross members. The offset column consists of an upper column (UC) and a larger diameter base column (BC), which contribute to suppressing motion. The connecting pontoons consist of five sets with three of each one. Connecting each main- to offset column is an Upper Pontoon (YU), Lower Pontoon (YL), and Cross Brace (CB). Connecting the offset columns to each other is also an Upper Delta pontoon (DU) and Lower Delta pontoon (DL). All pontoons are of the same diameter.

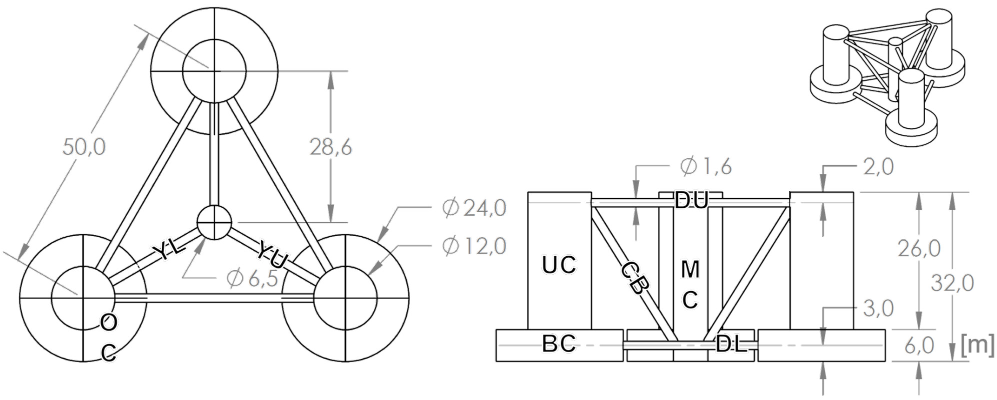


Figure 3.1: Plan and side view of the OC4 semi-submersible platform.

The wall thickness is 6 cm for the offset columns, 3 cm for the main column, and 1.75 cm for the pontoons. It should be noted that this thickness is not an expression for the actual desired thickness of a finished semi-submersible. In reality, the thickness might be lower, but in return have added internal structure with stiffeners adding to the strength of the structure. This is too complex to model and define, and as a result, the pontoons are assumed to be simple cylinders with a thickness that represents the strength of the structure [33].

To ensure a mean heave offset around 0, the OC4 semi-submersible is ballasted with water in the BCs and the UCs. The levels are 5.06 m in the BCs and 7.77 m in UCs respectively.

3.1.2 The IEA 15 MW Reference Wind Turbine

One of the important roles of reference wind turbines is that they serve as publicly available design parameters to be used for studies of new technologies or methodologies. The IEA 15 MW offshore wind turbine reflects a cooperation between NREL and the DTU, to ensure a reference turbine that matches the current generation of wind turbines used in the industry. The turbine is chosen for this project as it is appropriate in size for the future of FOWTs, and as the necessary data for this turbine is available for usage as part of the RAFT program. The IEA 15 MW reference wind turbine is a Class IB direct-drive machine, with a rotor diameter of 240 m and a hub height of 150 m [68].

3.1.3 Site Conditions

For the hydrostatic stability analysis, two load cases must be considered as stated in the DNV-ST-00119 Standard for Floating Offshore Wind Turbine Structures [53]. The two cases are; operation at the wind speed that produces the largest rotor thrust, and standstill in severe storm conditions.

In accordance with the standard DNV-OS-J101 for design of offshore wind turbine structures, the spectral density of the sea elevation may be presented by the JONSWAP spectrum, where wave height and wave period must be specified. According to the standard, simultaneous observations of wind and wave data should be obtained for a given site to estimate wave height and wave period.

First, the wind speed is identified for the case of operation at maximum rotor thrust, which corresponds to the rated wind speed. Next, a conditional wave height for the given wind speed must be identified, and the conditional wave peak period for given significant wave height and wind speed must be selected. The wave conditions corresponding to extreme conditions must be selected as well. For a site-specific analysis, the wave height and period of the specific site at these two conditions is estimated and used in the dynamic analysis.

For the scope of this project, general base case conditions for operation at maximum rotor thrust is used for the dynamic analysis in the model. A wind speed of 10.6 m/s is used, as it corresponds to the rated wind speed of the IEA 15 MW wind turbine, and the significant wave height and wave period are wave conditions that represent an expected case. For the extreme 50-y case, reasonable wave specification are defined from [69], which represent a site in Norway. The load values for the two cases are summarized in Tab. 3.1. A water depth of 200 m was chosen because it is the baseline water depth of the RAFT examples. For both cases, the turbulence intensity is set to 0.1. As default, the wind and wave heading is set to 0° meaning, that it is in the direction of the x-axis (see fig. 2.6).

It should be noted that this is not a complete set of environmental conditions, but should represent reasonable conditions to evaluate the floater stability.

Table 3.1: Load case.

Case	Condition	Wind speed v_w [m/s]	Sig. wave height H_s [m]	Avg. wave period T_P [s]
1	Max thrust, operating	10.6	8.5	13.1
2	Extreme storm, standstill	40	15.6	14.5

3.2 Upscaling

The first part of this project consists of an upscaled model of the 5 MW OC4 semi-submersible floating structure to a 15 MW system. From Sec. 2.3, different theoretical upscaling procedures for semi-submersible floating platforms were explained. The study concluded that there is not one way to upscale a semi-submersible, and that theoretical procedures are not always strictly adhered to in practice, although an important take from the analysis was, that scaling of the platform is related to scaling of the rotor, as the ratio between these parameters affects loads, mass and stability of the whole system. With this in mind, it was decided to use the practical scaling trends for the initial upscaling. When analyzing the linear dimensions on existing semi-submersible reference structures, the main criteria was the product of the radius from the center column to the center of the OC, r_{MO}, and the diameter of the UC, d_{UC}. Additionally, [43] supports the importance of scaling both on radius and diameter. In the initial upscaled design, these two dimensions have been controlled through defined scaled values based on Eq. 2.7. The platform is designed to fit with the IEA 15 MW turbine. As the wind turbine has a rotor diameter of 240 m, it gives the following $r \times d$ value:

$$r_{MO} d_{UC} = 1.63 \cdot 240\text{m}^{1.09} = 640\text{m}^2$$

The draught was unchanged due to logistics considerations, although this decision is also backed up by the fixed draught scaling law. The remaining vertical dimensions; freeboard, height of BC and UC are fixed for simplicity. This is based on conclusions from Sec. 2.3 and 2.5.2. These dimensions of the floater may be redefined with respect to transportation, installation, as well as manufacturing considerations. An overview of the initial upscaling approach can be seen in Tab. 3.2.

Table 3.2: Floater upscaling geometry.

Parameter	Symbol	Description
Radius from center to OC	r_{MO}	decided from WT rotor diameter (Eq. 2.7)
Diameter of UCs	d_{UC}	decided from WT rotor diameter (Eq. 2.7)
Draught	D	fixed (local port facilities)
Freeboard height	fb	fixed (local wave profile)
Diameter of BCs	d_{BC}	double that of d_{UC}
Height of BCs	h_{BC}	fixed
Thickness of OCs	t_{OC}	fixed
Diameter of main column	d_{MC}	equivalent to WT tower
Thickness of main column	t_{MC}	equivalent to WT tower
Diameter of pontoons	d_p	fixed
Thickness of pontoons	t_p	changed to maintain same stress (App. A.7)

The thickness of pontoons and cross-braces are also fixed to ensure the same internal stress as of App. A.7. For the MC, the diameter and thickness have been chosen to fit that of the bottom of the turbine tower. The diameter of the BC has been set to maintain the same ratio with the

UC of 2:1. All remaining dimensions have been defined to be driven by r_{MO} and d_{UC}, see App. A.1. This also enables the whole OC4 to be re-defined from just the two design variables, which must happen for each iteration in the optimization.

3.2.1 Mooring

It should be noted that as the dimension of the floater changes, so will the distance between the fairleads and anchoring points, resulting in a change in mooring line tension. This will affect the hydrostatic stiffness, making the results difficult to compare. This change is accounted for by adjusting the length of the mooring line, in such a way that the change in tension is limited. The full explanation to how this is achieved can be found in App. A.3.1. For situations where the mooring stiffness is wished to be as non-intrusive as possible, the mooring line is made as slag as possible (see App. A.3.1).

3.3 Hydrostatic Stability

The maximum wind heeling moment is identified for the maximum rotor thrust force, with the distance from the rotor location to the point of rotation. For a free-floating body this is the CoF, but for a FOWT positioned by centenary mooring, this point of rotation is better assumed to be around the vertical position of the fairleads [70]. The heeling moment, M_H, can be expressed as seen in Eq. 3.1.

$$M_H = F_T^{max} \cdot (z_{hub} - z_{FL}) \tag{3.1}$$

where z_{hub} is the vertical position of the middle of the turbine's rotor from the waterplane and z_{FL} that of the fairleads. $F_T^{\max}$ is the maximum total thrust force assumed to act on the center of the rotor.

The moment counteracting the heeling moment is the righting moment, and so by setting the maximum heeling moment to never surpass the righting moment, Eq. 2.19, a hydrostatic stability criteria can be defined.

$$M_H \leq M_{R,pitch,max} \tag{3.2}$$

By defining a maximum allowable pitch, $\theta_{\max}$, and solving for the structure-dependent hydrostatic stiffness in pitch, a design criteria can be found as shown in Eq. 3.3, imposing a minimum stiffness:

$$K_{55} \geq \frac{M_H}{\theta_{max}} \tag{3.3}$$

The situation where this inequality becomes an equality is when the heeling moment is balanced by the righting moment. This is equivalent to the intersection angle in the GZ-curve of Fig. 2.8, and the value of this angle corresponds to the maximum allowable pitch angle.

3.4 Structural Analysis

The evaluation of the structural integrity of the OC4 semi-submersible must be done for each iteration in the optimization loop. It is hence important that this is possible somewhat automatically. As RAFT is not using the same format as OpenFAST which is a wind turbine simulation tool with FEM capabilities from NREL, it was deemed unfeasible to incorporate FEA of the model for each iteration within the scope of this project [71]. Instead, the OC4 structure will be analytically evaluated under certain assumptions and based on this, several structural design criteria can be set.

3.4.1 Hydrostatic Loading

The structure is first analyzed statically by looking at an isolated offset to main column link. This is done in three different cases all shown in Fig. 3.2. The first case is a true static case analyzing the forces in the pontoons in still water. The two other cases are extreme cases, which are not truly static but relevant for investigating the extreme forces the pontoons may be subject to. In [34], a very similar case to case (c) is used with buckling as the sole structural criteria. The full calculations are shown in App. A.5.

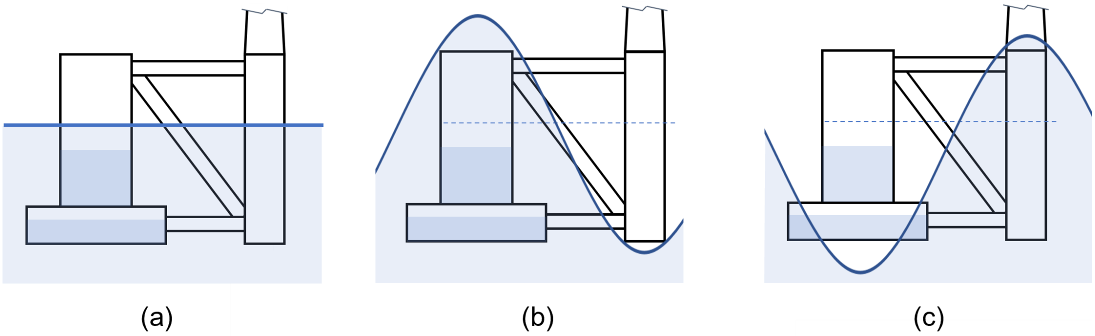


Figure 3.2: **(a)** is the still water case, **(b)** is an extreme case with wave top at offset column, and **(c)** for main column.

The following **assumptions** will be made:

- No external forces are acting on the isolated OC-MC link other than buoyancy.
- The delta pontoons will be disregarded.
- The pontoons and cross braces will be assumed to act like a truss structure meaning that they only carry axial loads (no torque or transverse loads).
- All three offset columns contribute equally to carry the weight on the main column.
- The weight and buoyancy of the pontoons can be disregarding.
- The force in the upper and lower pontoons carries the same horizontal load, $F_{YU} = F_{YL}$, as both columns are regarded as rigid structures.

From the assumptions, the axial forces in the pontoons can be expressed as shown in Eq. 3.4.

$$F_{CB} = F_{RES}\frac{l_{CB}}{h_{CB}} \qquad F_{YU} = F_{YL} = -F_{RES}\frac{l_{CB}}{2l_{YU}} \tag{3.4}$$

where h_i is the height and l_i is the length of the respective pontoon i. F_{RES} is the resulting vertical force acting on each side of the truss structure. For the three cases, the resulting force is given by:

$$F_{RES,a} = \frac{1}{3}\Big[(m_{RNA} + m_{tower} + m_{MC})g - D_{MC}\pi r_{MC}^2 g\rho_w\Big] \tag{3.5}$$

$$F_{RES,b} = \frac{1}{3}\Big[(m_{RNA} + m_{tower} + m_{MC})g\Big] \tag{3.6}$$

$$F_{RES,c} = \Big[\pi\left(r_{UC}^2 - (r_{UC} - t)^2\right)h_{UC}\rho_s + \pi\left(4r_{UC}^2 - (2r_{UC} - t)^2\right)h_{BC}\rho_s + 9t\rho_s{r_{UC}}^2 + m_{bal,OC}\Big]g \tag{3.7}$$

where ρ_s and ρ_w are the densities of steel and water respectively, and $m_{bal,OC}$ is the ballast mass of the offset column, defined in Eq. A.26.

From Eq. 3.4, it can be seen how the force will be greater in the CB, as long as $h_{CB} < 2l_{YU}$. This is the case for the original 5 MW OC4 and as the structure is upscaled and becomes wider while it maintains the same draught and freeboard, this inequality will only become greater. The model still investigates both pontoons.

It can also be seen how F_{CB} and F_{YU} have opposite signs relative to F_{RES}, meaning that if F_{RES} is positive, CB is in tension, while YU and YL are in compression. As buckling only happens for compression and can happen before yielding, the model determine what pontoon is subject to compression and investigate buckling here.

For the original 5 MW OC4 design, yielding will always occur before buckling. This is due to the abnormally low wall-thickness to radius ratio of the pontoons of 1.1%. By setting the critical buckling force (Eq. 2.34) equal to that of yielding (Eq. 2.32), and solving for the length, it is found that only for pontoon lengths greater than 51.2 meters will buckling happen before yielding. For comparison, a very similar semi-submersible structure from [34] has a fixed wall thickness to radius ratio of 5%.

From Eq. 2.32 and 2.33, the **yielding** criteria can be set up with respect to the normal forces rather than the internal stress, as seen in Eq. 3.8:

$$P_{y,cr} - P_{y,des} \cdot s_y \geq 0 \tag{3.8}$$

where the critical load is defined as $P_{y,cr} = \sigma_y A$ and $P_{y,des}$ is the design load, which includes F_{CB}, F_{YU}, and F_{YL}.

Likewise, the **buckling** criteria can be expressed as seen in Eq. 3.10:

$$P_{bu,cr} - P_{bu,des} \cdot s_y \geq 0 \tag{3.9}$$

where $P_{bu,cr}$ is given from Eq. 2.34, and $P_{bu,des}$ is the design load, which includes F_{CB}, F_{YU}, and F_{YL} if they are negative, meaning that the given beam is in compression.

Bending

The internal stress during bending is proportional to the length of the beam as seen in Eq. 2.38. It is hence a relevant failure mechanism to evaluate when upscaling and optimizing the structure. From the assumptions of the pontoons acting as a truss structure, it follows that no bending is present as they can carry no torque or transverse loads. Reality is not as simple and the large radius of the pontoons speaks against this assumption. An attempt to accommodate for this was made by incorporating a separate structural analysis specifically for investigating bending, by assuming that the forces are carried explicitly by the upper and lower pontoons for the still water case. It was decided not to use failure against bending as a constraint in the main optimization model, but the effect will still be investigated and is optional in the model. The constraint is implemented with the following constraint:

$$P_{be,cr} - P_{be,des} \cdot s_{be} \cdot c_{be} \geq 0 \tag{3.10}$$

where $P_{be,cr}$ is the critical bending load given from Eq. A.28, $P_{be,des}$ is design load equal to $F_{RES,a}/2$, s_{be} the safety factor for bending, and c_{be} is the correction factor to accommodate for the corrections used to increase the bending stiffness. For further clarification, see App. A.5.2.

3.4.2 Dynamics

In reality, the structure is not in static equilibrium but moves when influenced by the external forces. One way to analyze the dynamic loads on the system would be to investigate the difference in acceleration between the different structures. This has been done in App. A.6 and is optional to include in the optimization code. However, as RAFT analyses the structure as a single rigid body, it was decided not to include this analysis in the main optimization model at it was deemed too pseudo. Furthermore, using the dynamic RAFT results to evaluate the design criteria makes the optimization much slower computationally and more case specific as the dynamic response must be found for each iteration.

3.4.3 Safety Factor

The safety factor can be estimated as a product of individual considerations of multiple different measures such as: material properties, load stress, geometry, failure analysis, and desired reliability [72]. As an example, SAE 316L steel, typically used in offshore structures, is a well-tested material, and so a material-safety factor of 1 would be sufficient. On the other hand, the failure of the pontoons would be catastrophic, leading to a high reliability-safety factor.

The DNVGL-ST-0119 standard for FOWT structures, [53], provides load factors (safety factors) to be used when characteristic loads from different load categories are combined to form the design load for use in design. These are for the Ultimate Limit States (ULS) which cases (b) and (c) are examples of. Here, a safety factor of 1.35 is required.

It is not within the scope of this project to estimate a reliable safety factor for a final design. However, determining a safety factor is still important in order to define the constraints of the structure.

Deciding on a Main Structural Setting

For the original OC4 5 MW design, the physical dimensions already has questionable structural integrity. In the Definition of the OC4, [33], it is stated that "For a realistic full-scale system, the size of the cross braces and pontoons would need to be increased and internal stiffeners would need to be added to the columns."

This speaks to the fact that the OC4 is a reference floater and its dimensions being a starting point rather than a finished working design.

It was hence either necessary to strengthen the OC4 by changing the geometry of the members, use over-tuned material properties, or simply lower the structural requirements.

It was decided rather to have a less critical safety factor and structural constraint than to deviate too much from the original OC4 5 MW design, as it would make the mass parameter difficult to compare.

The structural constraints are also already simplifications and should be replaced by more complex structural analysis with the help of FEA before a final design is implemented.

For the bending constraint, over-tuned material properties was still incorporated with the correction factor c_{be} in order to investigate this possibility.

In the default optimization model, is was decided to use case (a) with a safety factor for buckling, s_b, and yielding, s_y, of 3. This was deemed a good compromise between reasonable structural integrity while still allowing a sufficiently large solution space.

Case (b) and (c) are still optional and were investigated as well as the bending constraint. These were all investigated with a safety factor of 1 in order to see their effects clearly.

3.5 RAFT

Response Amplitudes of Floating Turbines (RAFT) is an open-source Python software for frequency-domain analysis of floating wind turbines, developed by the National Renewable Laboratory (NREL). RAFT incorporates quasi-static mooring reactions, strip theory, and potential-flow hydrodynamics, blade-element-momentum aerodynamics, and linear turbine control in order to compute the response spectra of a floating wind system. It can be used to calculate RAOs, response spectra, mean/static properties, and response metrics based on mean and root-mean-square values of each output [73], [51].

Frequency-domain models are an important tool for designing floating structures because they can calculate the coupled response of a system significantly faster than time-domain simulations. This is done by constructing a linear, frequency-dependent representation of the system and then solving for the harmonic, steady-state system response at each excitation frequency [51].

A thorough guide on how to get started with using RAFT can be found in App. B.1.

3.5.1 Model Structure

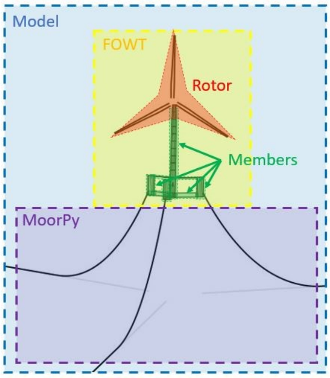


Figure 3.3: RAFT model objects [74].

RAFT represents a floating wind turbine system as a collection of different object types as illustrated in Fig. 3.3. Each object corresponds to a class in Python. The **Model** class contains the full system. Within it, the **FOWT** class describes the floating platform and turbine. The mooring system is handled by a separate model, with **MoorPy**. Within the FOWT class, the floating platform and turbine tower are represented as a single rigid body composed of multiple members in the **Member** object. Here, parametrization of substructure members of either cylindrical or rectangular geometries and the tower as a tapered cylindrical member is handled. The turbine Rotor-Nacelle Assembly (RNA) is represented by a **Rotor** class. The nacelle is represented as a lumped mass, and the rotor blades as a series of blade elements with aerodynamic and control properties.

3.5.2 Modelling Approach

An overview of the solution process is shown in steps below [51]:

1. Evaluate structure mass and hydrostatic characteristics.

In the FOWT class the static properties is computed by summing the contributions of all the members of the floater, the tower, and the RNA to fill in its 6x6 mass and hydrostatic stiffness matrices. The final global hydrostatic stiffness matrix can be seen in Eq. 2.17.

2. Model mooring system and solve unloaded equilibrium position.

The mooring system is modeled using MoorPy (another open-source tool-set from NREL) to solve the quasi-static equilibrium of the system returning the offset, reaction forces, mooring stiffness, and line tension.

3. Evaluate linear hydrodynamic coefficients, including excitation at select sea states.

With a strip-theory approach, RAFT applies the relative form of the Morison equation for transverse flow across a strip of a cylinder to all submerged members. It accounts for both wave and body velocity by discretization of each member into strips to which the equations are applied based on the local geometry and wave kinematics. The local damping matrices are transformed to the platform reference point and summed to get the overall substructure damping matrix, $\mathbf{B}_{sub}$.

4. Evaluate aerodynamic coefficients at select wind speeds.

These are calculated for each mean wind speed with the steady-state Blade-Element-Momentum (BEM) theory solver CCBlade, and used to model the rotor aerodynamic contributions to aerodynamic added mass, damping and excitation.

5. Apply mean loads and solve for mean offset position.
6. Reevaluate aerodynamic coefficients to account for tilt.

The hub-height dependent quantities are then transformed to be about the platform reference point, accounting for their coupled effect on surge and pitch for the final aerodynamic added mass, damping, and excitation.

7. Compute rotor response coefficients (including effect of control).
8. Linearize viscous drag excitation and damping at given sea state.
9. Solve for system response.

RAFT iterate over steps 7-9 until the response converges, which is necessary due to the linearization used on many inherently nonlinear phenomena. The dynamic response of the system as a function of frequency is solved from the frequency-dependent equation of motion, given by Eq. 3.11:

$$(-\omega^2[\mathbf{M}_{struc}+\mathbf{A}_{sub}(\omega)+\mathbf{A}_{aero}(\omega)]+i\omega[\mathbf{B}_{sub}(\omega)+\mathbf{B}_{aero}(\omega)]+\mathbf{C}_{struc}+\mathbf{C}_{moor})\hat{\xi}(\omega) = \hat{\mathbf{f}}(\omega) \tag{3.11}$$

where $\mathbf{M}$ is the mass and inertia of the floating structure, $\mathbf{A}$ added mass, $\mathbf{B}$ damping, $\mathbf{C}$ stiffness, $\hat{\mathbf{f}}$ excitation force from wind and waves, and $\hat{\xi}$ response. Both $\hat{\mathbf{f}}$ and $\hat{\xi}$ are represented as a mean plus a Fourier series of complex amplitudes.

The frequency dynamics (Eq. 3.11) are assumed to act about an operating point defined as the mean state of the system for a given load case, found by solving the static equilibrium given by Eq. 3.12.

$$\mathbf{C}\bar{\xi} = \hat{\mathbf{f}}_{aero} + \hat{\mathbf{f}}_{hydro} + \hat{\mathbf{f}}_{moor}(\bar{\xi}) \tag{3.12}$$

where $\mathbf{C}$ is the total hydrostatic stiffness matrix, $\hat{\mathbf{f}}_{aero}+\hat{\mathbf{f}}_{hydro}$ the mean wind and wave load, and $\hat{\mathbf{f}}_{moor}(\bar{\xi})$ is the nonlinear mooring system reaction force (which includes the effective mooring stiffness).

In an eigenvalue analysis, an undamped system with no excitation loads is assumed, giving a simplified Eq. 3.11 where $\mathbf{B}(\omega)$ and $\hat{\mathbf{f}}(\omega)$ is set to zero. By setting all but one element in the response vector $\hat{\xi}$ to zero, the angular frequency, ω is solved for in each DoF [55].

3.6 Optimization

Optimization is a complex discipline involving a wide variety of problems, where the type of problem will determine the types of methods for solving it. This section begins with a presentation of the problem formulation, followed by a classification of the optimization method employed. At last, an overview of the model workflow is presented.

3.6.1 Problem Formulation

The objective of the optimization is to minimize the cost by minimizing the mass of steel for the floating structure. This is done by regulating the geometry of the structure through the design variables.

The design variables are the diameter of the UCs, d_{UC} and the radius from the center of the main column to the center of the OC, r_{MO}. These design variables has been chosen as they, in combination, has a defining influence on the whole geometry of the substructure as described in App. A.1. They have a direct impact on the hydrostatic stiffness of the platform by influencing the waterplane area as expressed in Eq. 2.20, but will also drastically influence the critical buckling and bending force as the length of the pontoons also increase as seen from Eq. 2.34 and 2.38 respectively.

The constraints are set in order to maintain hydrostatic stability and structural integrity. The hydrostatic stability is defined from the maximum heeling moment, 3.1, where the maximum total thrust force is 2300 kN for the IEA 15 MW reference WT [68] and the maximum static pitch angle is set to 10° as suggested by literature [42], [48]. The safety factors for buckling, s_b, and yielding, s_y are both set to 3 as covered in Sec. 3.4.3. All the constraints are inequality constraints, defined to be less or equal to zero and has been normalized in order to make the optimization process more rigid and uniform for all parameters.

The upper and lower bounds of the design variables are determined with consideration to avoiding them reaching their bounds, and allowing the optimization model a great level of flexibility when searching for an optimal design. The upper bounds could also be set for site specific conditions if e.g. the FOWTs were to be towed inside a harbour for O&M service.

Manufacturing and location specific constraints which could be relevant for the final cost of a FOWT, are not regulated in the optimization problem formulation. Instead, these can be considered when defining the fixed parameters in the initial design specifications.

The optimization can be summarized by the problem formulation described in Eq. 3.13:

$$
\begin{aligned}
\underset{r_{MO},d_{UC}}{\text{minimize}} \quad & M_{sub} \\
\text{subject to} \quad & K_{55} - \frac{F_T^{max} \cdot z}{\theta_{max}} \geq 0 \\
& P_{b,cr} - P_{b,des} \cdot s_b \geq 0 \\
& P_{y,cr} - P_{y,des} \cdot s_y \geq 0 \\
\\
& 20 \leq r_{MO} \leq 80 \\
& 3 \leq d_{UC} \leq 25
\end{aligned}
\tag{3.13}
$$

After an optimized design has been found, the eigenfrequencies and dynamic behaviour are investigated.

3.6.2 Optimization Driver

Multi-disciplinary Design Analysis and Optimization (MDAO) is a field of engineering that uses optimization methods to solve design problems incorporating a number of disciplines. FOWTs are inherently complex and multi-disciplinary systems that require MDAO approaches for optimization of design and operational strategies [75].

OpenMDAO is an open-source high-performance computing platform for MDAO, written in Python [76]. It was first conceived to solve aircraft design problems by NASA in 2010, and the framework is now widely used in other sectors such as the wind industry [77].

OpenMDAO has been used to create the optimization model for this thesis. Specifically, the optimization workflow has been set up with the build-in *ScipyOptimizeDriver COBYLA* (Constrained Optimization By Linear Approximation optimizer). It is a numerical optimization method for constrained problems where the derivative of the objective function is not known [78].

The full optimization code can be seen in App. B.2.

3.6.3 Problem Classification

Generally, optimization problems can be classified according to some categories, which have been listed below. The classification of the problem in question is underlined.

- <u>Continuous</u> or discrete design variables: This problem is continuous as the variables in the model are allowed to take on any values within the defined range. Discrete design variables are those that have only certain levels or quantities that are possible, such as the choice between materials or standard sizes.
- Linear or <u>non-linear</u>: The problem is non-linear as both constraints and objective functions has a non-linear relation to the design variables. However, linear approximation is used during optimization for COBYLA.
- <u>Constrained</u> or unconstrained: In a constrained problem, the decision variables can only take on certain values with respect to some defined constraints.
- <u>Deterministic</u> or stochastic: The model is deterministic meaning that it produces the same solution every time, for a particular set of inputs, and gives a certain theoretical guarantee that the solution is the global one.
- <u>Single decision</u> or dynamic: Dynamic optimization allows the decision variables to depend on such factors as experience, and these variables may in turn depend on previous decisions, and will have many solutions in the feasible region while static models find one optimal point.
- <u>Single-</u> or multi-objectives: As the optimal solution is solved for one objective function, substructure mass, it is a single objective model. Other performance indicators will still be analyzed and discussed, but are not included as an objective in the problem.
- Single- or <u>multi-disciplinary</u>: In a multi-disciplinary problem, each sub-system has its own design variables, objective function and constraints. Some design variables can be common to several sub-systems. The disciplinary outputs from one discipline can be needed to evaluate another sub-system. In this case there is a coupling between two disciplines.

3.6.4 Model Workflow

In Fig. 3.4, an overview of the design optimization process for the work presented in this thesis can be seen. From specifications, the design is upscaled to achieve a preliminary design. In practice, the preliminary design consists of an YAML file defining the geometric, structural, aero- and hydrodynamic parameters of the whole FOWT system. The optimization is an iterative process where the design is evaluated based on the problem formulation, and updated until optimality is achieved within a certain tolerance. The validity of the calculated results is limited by the accuracy of the underlying assumptions. RAFT is then used to evaluate the hydrodynamics of the optimized design. The design process is iterative and an optimized design may reveal limitations and opportunities in the design which would lead to conceptual changes that can be incorporated into a new preliminary design concept.

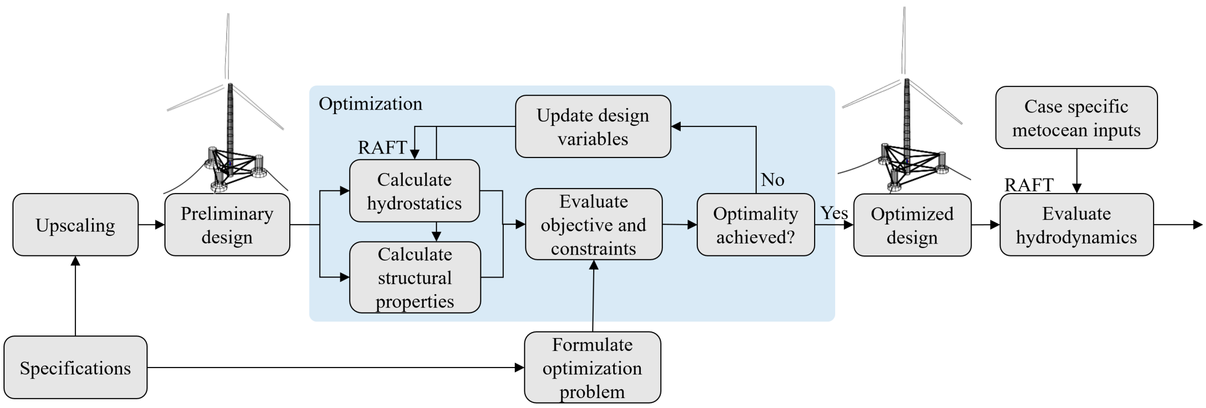


Figure 3.4: Workflow with the design optimization process.

3.7 Performance

While the target of the optimization is to minimize the mass of the structure, it is relevant to identify and track operational performance indicators for the whole FOWT system. Some specifications concerning the performance envelope of a coupled FOWT system are; maximum displacements, maximum velocities, maximum accelerations, bandwidths of motions, and maximum forces and moments [60].

3.7.1 Maximum Nacelle Acceleration

A way to increase the performance of a floating offshore wind turbine is to minimize the nacelle acceleration. This motion creates extra load on the wind turbine blades, and may cause fatigue in the drive train. Generally, large motion of the wind turbine nacelle may damage the equipment and decrease the lifetime of the system [34]. This motion is defined from the response of nacelle acceleration at specific sea state conditions.

This performance metric is defined with the maximum nacelle acceleration, specifically in the direction of the wind. The maximum nacelle acceleration is defined as three times the standard deviation of the nacelle acceleration (as of Eq. 2.28). Here the mean is zero, resulting in Eq. 3.14:

$$\max(a_{nac}) = 3 \cdot \sigma_{a_{nac}} \tag{3.14}$$

The standard deviation of the acceleration is calculated from Eq. 2.27 with the RAO of the nacelle acceleration. This is given from the RAO in surge (translation along the longitudinal axis) and pitch (rotation about the lateral axis) multiplied by the distance from the center of rotation to the CoG of the nacelle, z_{nac}. This is specified in Eq. 3.15. For further clarification on the relevance of the DoF in use, see Fig. 2.6.

$$RAO_{a_{nac}}(\omega) = -\omega^2(RAO_1(\omega) + RAO_5(\omega)z_{nac}) \tag{3.15}$$

3.7.2 System Motions

A common way to analyze the motions of the FOWT system is by comparing the standard deviations of the motion response in various sea states and wind speeds [54], [60], [79]. Minimizing the motion of the FOWT system in pitch, also important for the control of the turbine, as oscillations will make the turbine move in and out of the wind, resulting in fluctuating wind speed. This can be difficult for the controller to adjust for which will essentially impact the power production. Furthermore, the load on the rotor will be increased, and the consequence will be seen in the O&M cost. Heave motion will mainly impact the loads on the platform, and may also be a determining factor for O&M as higher heave motion will restrict the accessibility of the FOWTs for O&M endeavours.

The motion response spectrum is derived from Eq. 2.25, and the standard deviation can be found for each DoF with Eq. 2.27. Combined they give the standard deviations of the response over a given frequency range as seen in Eq. 3.16.

$$\sigma_{R,i} = \sqrt{\int_{\omega_{\min}}^{\omega_{\max}} S_\zeta(\omega)|RAO_i|^2 d\omega} \tag{3.16}$$

The standard deviations of the response spectrum are expected to be largest for heave, surge, and pitch DoF, as these are the directions of environmental loads.

4 Results and Discussion

In this chapter, the optimization results for the upscaled 15 MW OC4 is presented. First, the results of the upscaling to a preliminary 15 MW design and RAFT analysis is covered in Sec. 4.1. This design is then optimized using the previously covered optimization model and a comparison is made between the two in 4.2 and in 4.3 it is also compared to the Volturn US-S semi-submersible. Finally a sensitivity analysis is carried out in order to investigate the factors that impact the optimization and the extent of their influence in Sec. 4.4.

4.1 Preliminary Design

The geometry of the initial upscaled design of the OC4 semi-submersible floating platform to support the IEA 15 MW wind turbine is shown in Tab. 4.1. The pontoon thickness is 2.5 cm as of App. A.7. The ballast levels have been adjusted to 3.79 m in the BCs, and 7.65 m in the UCs.

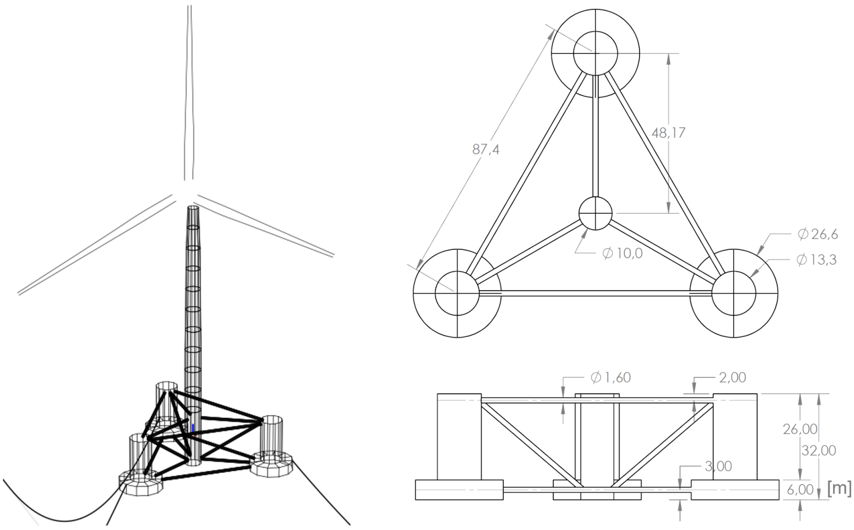


Figure 4.1: OC4 upscaled to carry 15 MW WT.

Table 4.1: 15 MW OC4 semi-submersible floater geometry.

Parameter	Symbol	Value	Unit
Radius from center to OC	r_{MO}	48.17	[m]
Diameter of UCs	d_{UC}	13.29	[m]
Mass of steel structure	M_{sub}	5274	[ton]
Ballast mass	M_{bal}	10814	[ton]
Hydrostatic stiffness in pitch	K_{55}	2522	[MN/m]

The upscaling methodology determined that the product of r_{MO} and d_{UC} should be scaled according to Eq. 2.7, while the exact value of each of these two parameters can be decided by the designer. The upscaled diameter of the UC was decided by observing the dimensions of other semi-submersible reference platforms [42]. It was noted that the OC4 had a significantly larger offset column diameter than the remaining 5 MW designs, and the dimensions of other larger reference platforms (10-15 MW) was in the range of 10.5-15 m. The final diameter was scaled with a factor 1.11 to be inside this range at a value of 13.29 m. r_{MO} was then defined from the $r \times d$ value, resulting in a scaling factor of 1.67. This scaling followed somewhere in between the power and mass scaling laws. In [43] a distance and radius scaling approach was followed for the OC4, resulting in a metal mass scaling exponent of 1.41. In this thesis, the mass of the platform metal mass was scaled with a factor 1.37, which is lower due to the low scaling factor of the UC that has the most significant impact on the metal mass.

4.1.1 Natural Frequencies

From an eigenvalue analysis, the natural frequencies of the floater in all six DoF have been identified. The values are shown in Tab. 4.2. The eigenvalue analysis revealed that the heave natural frequency is inside the general spectral range of ocean waves which as a period range of 5-25 s [55]. Due to the geometric symmetry of the floater, it has pitch and roll values with a similarity that extends down to the fourth decimal place. The natural period in roll is slightly larger than in pitch due to the contribution from the RNA's CoG which is slightly off-center, meaning that the moment of inertia in roll is slightly larger than in pitch. The surge and sway motions are equal and governed by the mooring lines.

Table 4.2: Natural periods of preliminary 15 MW OC4 semi-submersible floater design.

DoF i	Surge 1	Sway 2	Heave 3	Roll 4	Pitch 5	Yaw 6
T_{n_i} [s]	256.4	256.4	16.75	33.56	33.56	198.4

When looking at the coupling between natural frequencies and rotational frequency, 1P, and blade passing frequency, 3P, no coupling was found. The IEA 15 MW WT has an operating rotor speed interval of 5-7.5 RPM [42]. This corresponds to a 1P ranging from 7.9-12 s, and a 3P ranging from 2.6-4 s. These are all below the natural periods of the FOWT.

4.1.2 Dynamic Analysis

A hydrodynamic analysis of the FOWT in the frequency domain has been performed in RAFT. The FOWT has been analyzed for two cases; operation at maximum rotor thrust, denoted as case 1, and standstill in extreme storm conditions, denoted as case 2. Specifications for the two cases can be found in Tab. 3.1. The maximum nacelle acceleration and the standard deviations of system motions in surge, heave, and pitch directions, are specified in Tab. 4.3. The case specific wave period will impact the motion in heave, which is also where the largest standard deviation is seen.

Table 4.3: 15MW OC4 semi-submersible preliminary design system motions

Parameter	Symbol	Case 1	Case 2	Unit
Maximum nacelle acceleration	a_{nac}	0.0234	1.65	[m/s^2]
Standard deviation in surge	σ_1	0.940	1.98	[m]
Standard deviation in heave	σ_3	1.03	2.24	[m]
Standard deviation in pitch	σ_5	0.358	0.817	[°]

4.2 Optimized Design

An optimization of the semi-submersible floating structure was performed with the objective of reducing the mass of the substructure while maintaining the required hydrostatic stability for steady-state performance during operating condition while also assuring structural integrity in the pontoons and cross-braces which are considered the most critical members. It was found that a smaller structure than the initial up-scaled design was able to achieve the desired hydrostatic stability and structural integrity. This was accomplished by increasing r_{MO} and decreasing d_{UC}. A summary of the optimized floater geometry is found in Tab. 4.4 as well as the percentage deviation from the preliminary design. The adjusted ballast levels of the optimized design are 3.79 m in the BCs and 7.65 m in the UCs.

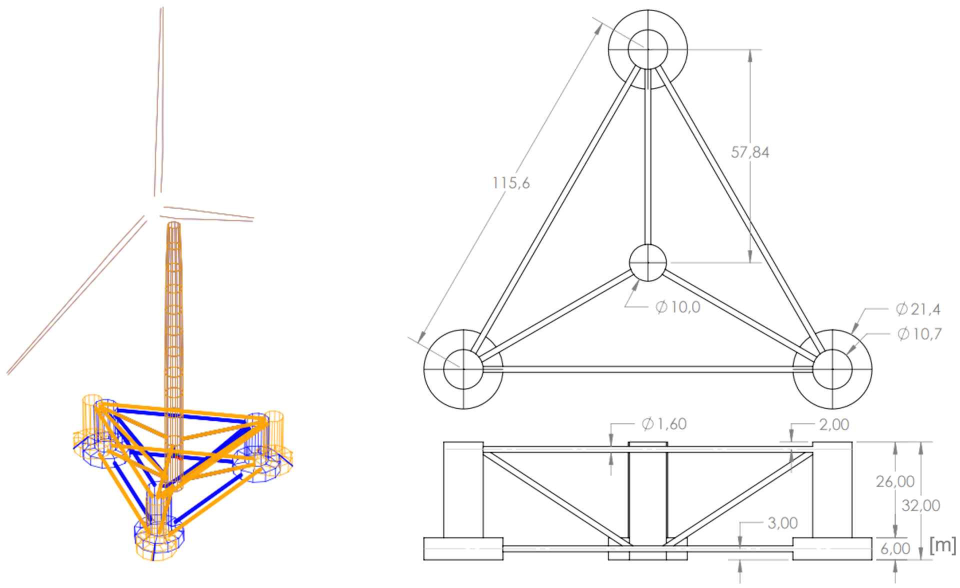


Figure 4.2: Optimization of upscaled OC4 15 MW. Blue is initial and orange is optimized.

Table 4.4: Optimized 15 MW OC4 semi-submersible floater design variables, mass and hydrostatic stiffness.

Parameter	Symbol	Value	Unit	Dev.[%]
Radius from center to OC	r_{MO}	57.84	[m]	+20.1
Diameter of upper columns	d_{UC}	10.72	[m]	-19.3
Mass of steel structure	M_{sub}	4374	[ton]	-18.6
Ballast mass	M_{bal}	6224	[ton]	-41.4
Hydrostatic stiffness in pitch	K_{55}	2172	[MN/m]	-13.9

4.2.1 Natural Frequencies

The natural frequencies of the optimized floater in all six DoF have been identified and the periods are shown in Tab. 4.5. The natural periods in surge, sway, and yaw have decreased significantly compared to the initial design, but are still far above the range of ocean waves. The natural periods in pitch and roll have slightly increased, while the natural period in heave is still inside the range of ocean waves and has decreased slightly compared to the initial design (see Tab. 4.2). The decrease in natural frequency in heave can be explained by Eq. 2.21. The stiffness component in heave is dependent on the waterplane area, and has decreased due to the reduced diameter of the offset columns. The mass and added mass is reduced relatively more which explains the decrease in the natural period.

Table 4.5: Natural periods of optimized 15 MW OC4 semi-submersible floater design.

DoF i	Surge 1	Sway 2	Heave 3	Roll 4	Pitch 5	Yaw 6
$T_{n,i}$ [s]	119,6	119,6	15.71	34.04	34.04	113.8

4.2.2 Optimization Workflow

The workflow of the optimization algorithm has been tracked and is plotted with normalized values in Fig. 4.3. The mass has been normalized with respect to the original value, while the constraints are normalized with respect to their critical value. The first graph shows the objective function, the second graph shows the design variables, and the last graph shows the three inequality constraints which represent conditions that must be satisfied. A constraint is tight if the inequality becomes an equality. The iteration tracker reveals the sensitivity of the objective function to changes in the design variables. Most notably is it that even small variations in d_{UC} gives large variations in both the mass and the hydrostatic stiffness constraint.

As seen from Eq. 2.20, a variation in d_{UC} has a greater impact on the waterplane area contribution to hydrostatic stiffness than r_{MO}, but by increasing r_{MO} relatively more than what d_{UC} is reduced, this is compensated for. As the general mass and volume of the semi-submersible decrease, the CoB-CoG relative position contribution to K_{55} will also become even more negative.

It can be seen how buckling is far more sensitive than yielding. r_{MO} is related to l_{CB} with an approximately linear relationship. As an increase in l_{CB} will both reduce the critical load squared and increase the design load linearly, this will have a major impact on the buckling constraint. For yielding, the design load is equally dependent, but the critical load is constant.

The iteration tracker reveals that for the optimized design, the hydrostatic stiffness and buckling are tight constraints, while yielding is a loose constraint.

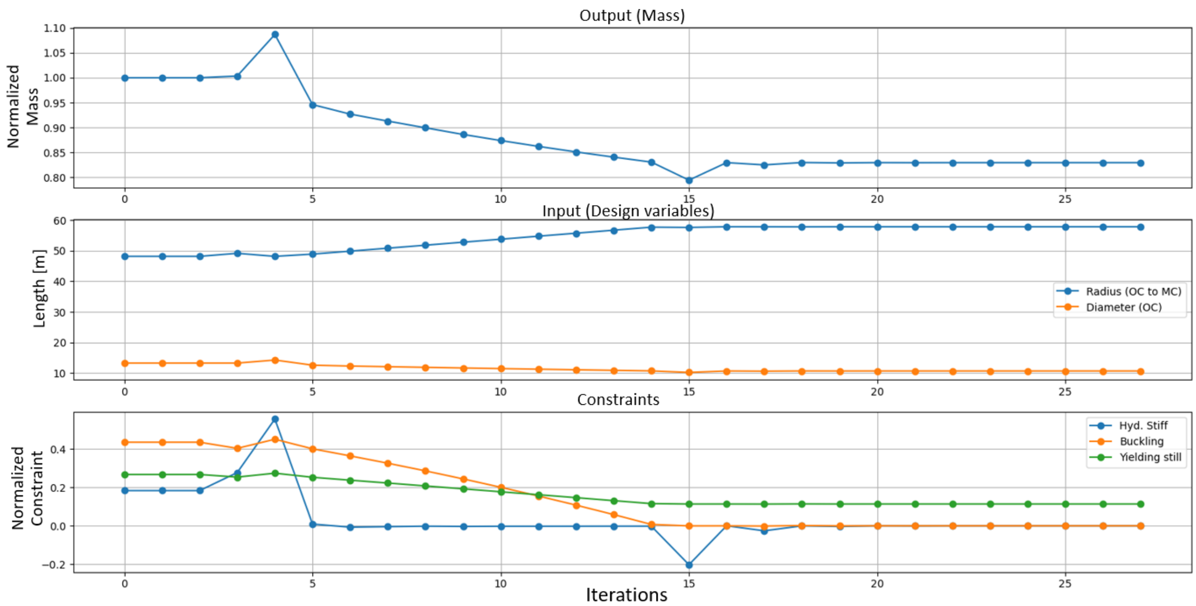


Figure 4.3: Iteration tracker for the optimization of the 15 MW semi-submersible floater.

4.2.3 Dynamic Analysis

A dynamic analysis of the optimized OC4 semi-submersible floater has been performed in RAFT, and the results of the analysis is presented in this section.

System Motions

The system motions standard deviations and maximum nacelle acceleration are shown in Tab. 4.6 for the two load cases specified in Tab. 3.1. Additionally, the percentage deviation to the initial design is shown for the two load cases respectively.

Table 4.6: Optimized 15 MW OC4 semi-submersible system motions.

Parameter	Symbol	Case 1		Case 2		Unit
		Value	Dev.[%]	Value	Dev.[%]	
Max. nacelle acc.	a_{nac}	0.0152	−34.9	1.19	−28.0	[m/s^2]
SD in surge	σ_1	0.915	−2.7	1.93	−2.5	[m]
SD in heave	σ_3	1.14	10.7	2.30	2.5	[m]
SD in pitch	σ_5	0.349	−2.5	0.707	−13.5	[°]

In the optimized design, the system motion SD decrease in surge and pitch directions while it increases in the heave direction. One explanation could be that the natural period in heave has decreased from 16.75 s to 15.31 s in the optimized design, making it closer to the mean wave period of both cases. A more significant cause for the decrease is that as the diameter of the UC decreases so does the diameter of the BC which function as heave plates. Heave plates are attached to the columns of semi-submersible platforms to improve the vertical plane stability by providing increased added mass in this plane [80]. During the optimization, the added mass in heave, A_{33}, is reduced by 44 %.

The maximum nacelle acceleration decreases to 1.19 m/s^2 for the extreme 50-y condition (case 2), and also decreases further for operating condition (case 1). The nacelle acceleration in question is dependent on the movement in surge and pitch. According to [81], the two most important design requirements in regards to the floater motions during operation are that the pitch angle should stay below 10°, and that the horizontal nacelle acceleration should not exceed 3 m/s^2. Another study [34] suggests that the nacelle acceleration should not exceed 1 m/s^2. Although this indicates uncertainty about an exact limit, it should be noted that the optimized OC4 semi-submersible FOWT only reaches close to these values in standstill during extreme wave loads, while they are well within tolerances during operation.

The mean displacements and the dynamic variations of the platform motion response in surge, heave, and pitch have been investigated under varying wind and constant wave loads and the results are shown in Fig. 4.4. The wave conditions are maintained the same as case 1, while the wind speed has been varied from 5-20 m/s, with approximately 3 m/s intervals. All runs represent an operating case.

The mean surge motion reaches up to 17 m, which is due to the FOWT being moved downwind to its new mean state position. The standard deviations express the movement of the FOWT during operation. The standard deviations are almost unaffected by changing wind speeds, but slightly increased at the rated wind speed. The pitch peaks at rated wind speed with a mean of 9.4°, as this is where the thrust force is greatest. It is still below 10° which corresponds well to the design constraint. For a similar study for the original OC4 5 MW FOWT and an optimized design, the results showed strong resemblance [79].

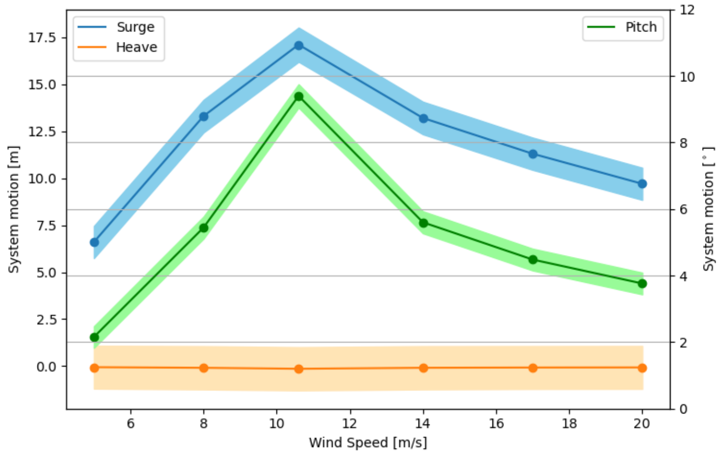


Figure 4.4: Mean system motions (line) and SD (shade span) of optimized design during operation at various wind speeds (5-20 m/s).

Response Amplitude Operators

The RAOs of the system for operational load case 1 are compared for the initial upscaled and the optimized 15 MW design in Fig. 4.5. RAO is highest in surge due to the wind heading. Heave shows a peak around the natural frequency for both designs respectively. The RAO in pitch also shows a peak around the natural frequency of the system, but the peak is significantly lower for the optimized design.

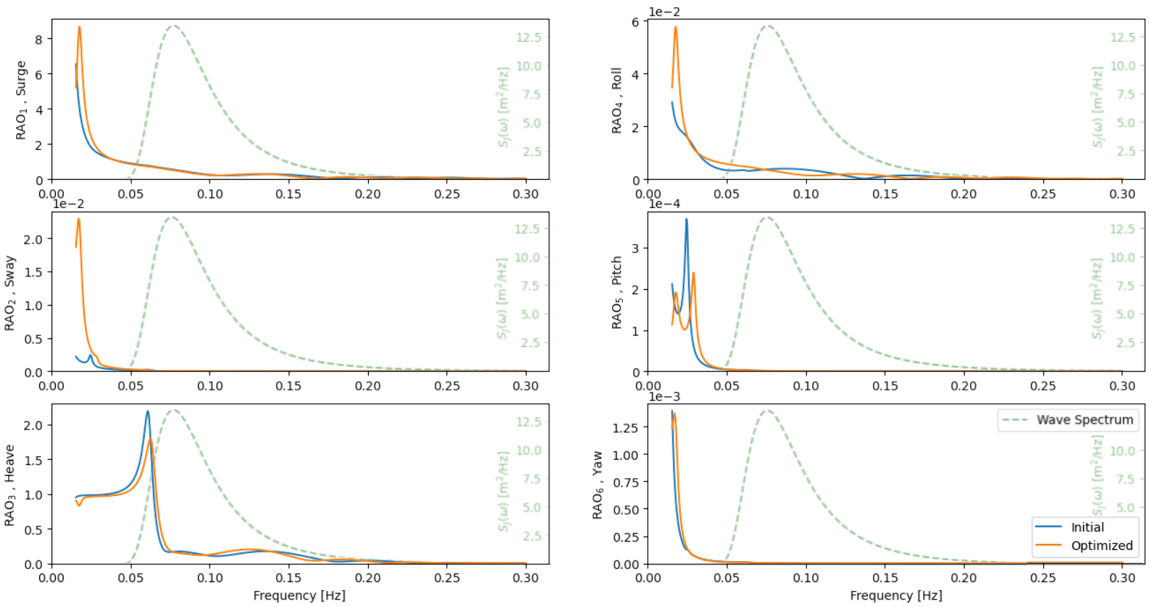


Figure 4.5: RAO of initial (blue) and optimized (orange) 15 MW OC4 designs. The wave spectrum for the operational case 1 show as well.

Though many of the RAO peaks are at low enough frequencies to be outside the realm of the wave spectrum, they are still very relevant for other parameters of the FOWT. The controller bandwidth for onshore WTs are set below the first natural frequency of the tower (lowest natural frequency of the system) to avoid controller-induced instabilities, however in floating the lowest natural frequency for pitch is an order of magnitude below that [82]. As seen in Fig. 4.5, the pitch peaks at a low frequency meaning that the controller bandwidth needs to be further reduced. This poses a problem as simply detuning the generator speed controller such that the generator speed only tracks signals lower than the platform natural frequency, would result in the generator regularly exceeding its over-speed limit in normal operation [82]. Alternative controllers are therefore necessary for FOWTs.

In App. A.8, the response and wave spectra are shown together with the RAO in order to visualize the relation between them.

Response Spectrum

The OC4 semi-submersible FOWT responses have been estimated with RAFT. In Fig. 4.6, the response spectra for surge, heave, and pitch are showed as well as the nacelle acceleration and tower bending moment. The preliminary and optimized designs are shown together for max thrust (left) and extreme storm conditions (right). In the last graph, the corresponding wave spectrum is shown.

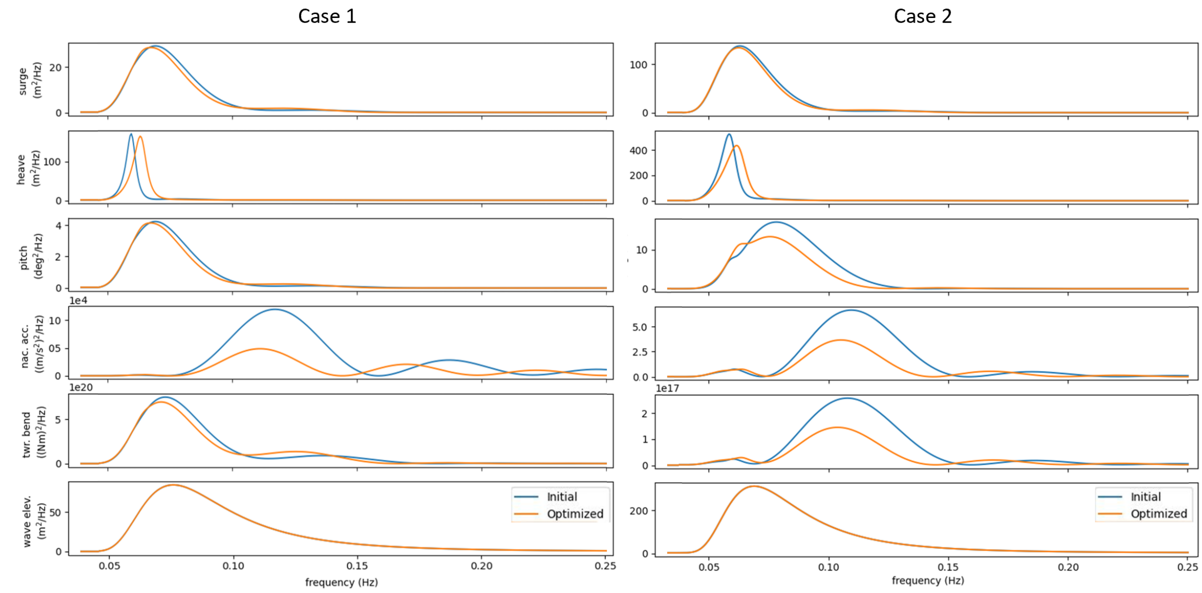


Figure 4.6: Response spectra for initial (blue) and optimized (orange) 15 MW OC4 designs during case 1 (left) and case 2 (right).

For both cases, the response spectra in surge and pitch mainly follows the wave spectrum, while the optimized design shows slightly lower response. This can be explained from the fact that both surge and pitch have their peaks well before the wave spectrum. The heave response still peaks around the natural frequency which is now higher and closer to the wave spectrum. However, the maximum value is still slightly reduced as a consequence of the reduced RAO_3 maximum as seen in Fig. 4.5. Furthermore, the nacelle acceleration response is significantly lower for the optimized design.

For the 50 year extreme case, the response spectrum of the nacelle acceleration is significantly higher, but still reduced for the optimized design.

Looking at Eq. 3.15, one might question how the response of nacelle acceleration is significantly decreased when both the surge and pitch PSD are almost unchanged. This can be explained by the phases of the two RAOs being more opposing and thereby cancelling each other out. This is a welcoming feature of the design. For an elaborated explanation, see App. A.9.

The tower bending moment is relevant for evaluating the load on the tower, which essentially influences how strong the tower needs to be and its required mass. The load is greatest when the turbine is operating, as the thrust is by far the largest contributor. A small reduction in tower bending moment is seen which may be due to the reduction in nacelle acceleration.

4.3 Comparison with Volturn US-S

The Volturn US-S reference semi-submersible floating platform is the result of a collaboration between the University of Maine (UMaine) and the U.S. Department of Energy. This reference platform was designed to support the IEA 15 MW Wind Turbine, and therefore provides details on a floating semi-submersible design developed especially for this turbine size. The dimensions of the Volturn US-S semi-submersible platform can be found in App. C.1 [31]. A comparison of the system motions of the optimized OC4 and Volturn US-S has been carried out. The two platforms have been compared for load case 1. The results are found in Tab. 4.7, where it is noted that the structure mass of the Optimized OC4 is 11.5 % higher than that of the Volturn US-S, while the latter compensates with a concrete ballast mass. The optimized OC4 has lower standard deviations in all DoF and a lower nacelle acceleration.

Table 4.7: Key design parameter and system motion (operating, load case 1) comparison of optimized 15 MW OC4 and Volturn US-S semi-submersible platforms.

Parameter	Symbol	Optimized OC4	Volturn US-S	Unit	Dev. [%]
Radius from center to OC	r_{MO}	57.84	51.75	[m]	11.8
Diameter of UC	d_{UC}	10.72	12.50	[m]	-14.2
Mass of steel structure	M_{sub}	4374	3922	[ton]	11.5
Ballast mass, water	$M_{bal,w}$	6224	11300	[ton]	-44.9
Ballast mass, concrete	$M_{bal,c}$	-	2536	[ton]	-
Hydrostatic stiffness in pitch	K_{55}	2172	3131	[MN/m]	-30.6
Maximum nacelle acceleration	a_{nac}	0.0152	0.0156	[m/s^2]	-2.6
Standard deviation in surge	σ_1	0.915	0.995	[m]	-8.0
Standard deviation in heave	σ_3	1.141	1.250	[m]	-8.7
Standard deviation in pitch	σ_5	0.349	0.381	[°]	-8.4

The response spectra of the two 15 MW platforms are shown in Fig. 4.7 for operational load case 1.

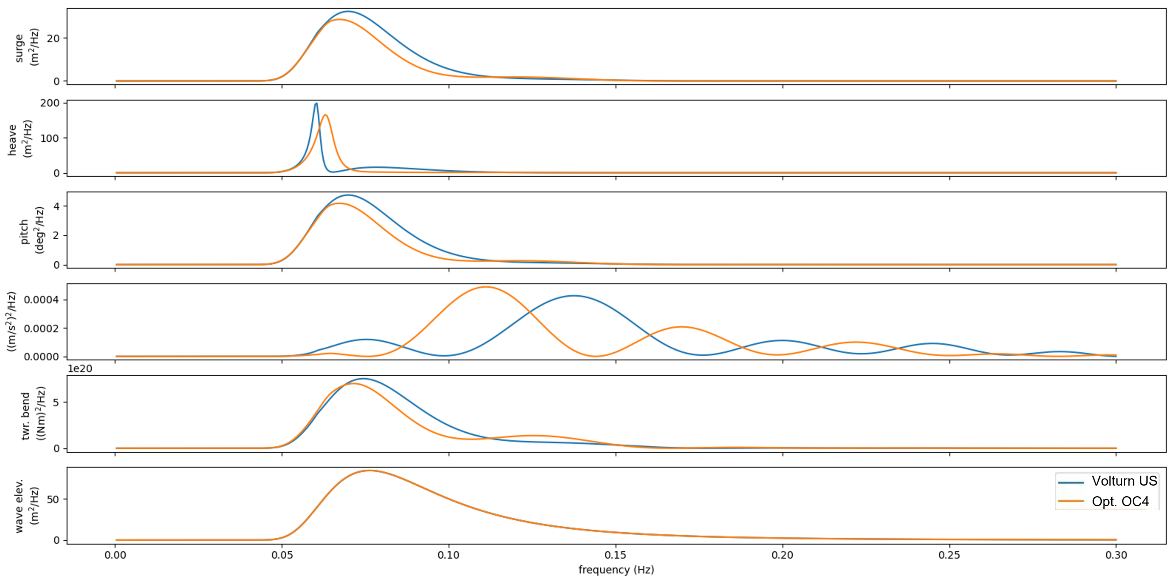


Figure 4.7: Response spectra for optimized OC4 (orange) and Volturn US-S (blue) 15 MW semi-submersible platforms.

In general, the behaviour of the two platforms are similar. One considerable difference is what frequencies influence the nacelle acceleration. The optimized OC4 has a higher peak than the Volturn US-S but a lower spectral moment. The shift in heave peaks can again be explained by the heavier Volturn US-S platform.

4.4 Sensitivity Analysis

To deal with uncertainties, a sensitivity analysis is performed on certain relevant parameters in the optimization model. The analysis has been performed on both design specification inputs in the model and on the constraints. The impact of these changes on the optimization model output has been analyzed and the results are discussed.

4.4.1 Maximum Static Pitch Angle

The restriction in maximum pitch angle is included both for stability reasons as well as for the wind turbine to maintain an upright position for optimal performance. This is a design decision and a larger angle might result in a lighter structure but a reduction in the power output. On the other hand, a smaller angle might result in a better power production but a more costly design of the structure. The initial maximum static pitch angle is set to 10° and is defined in the constraint for hydrostatic stability in Eq. 3.3. The optimization has been conducted with the maximum allowable pitch angle ranging between 6-14° in steps of 1°. In Fig. 4.8, the corresponding impact on the mass, design variables, and nacelle acceleration is shown for case 1 with the deviation in reference to the original optimized design.

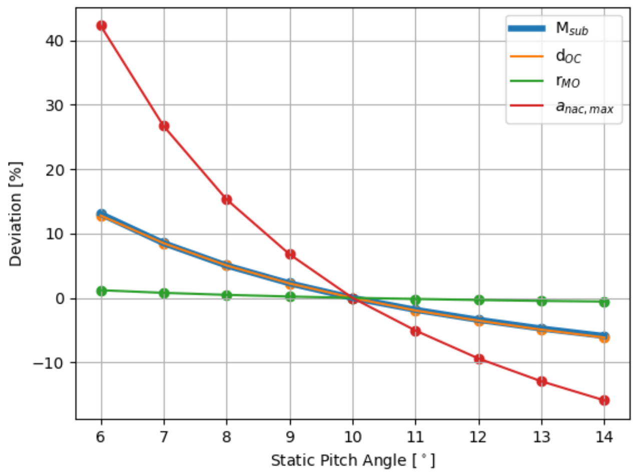


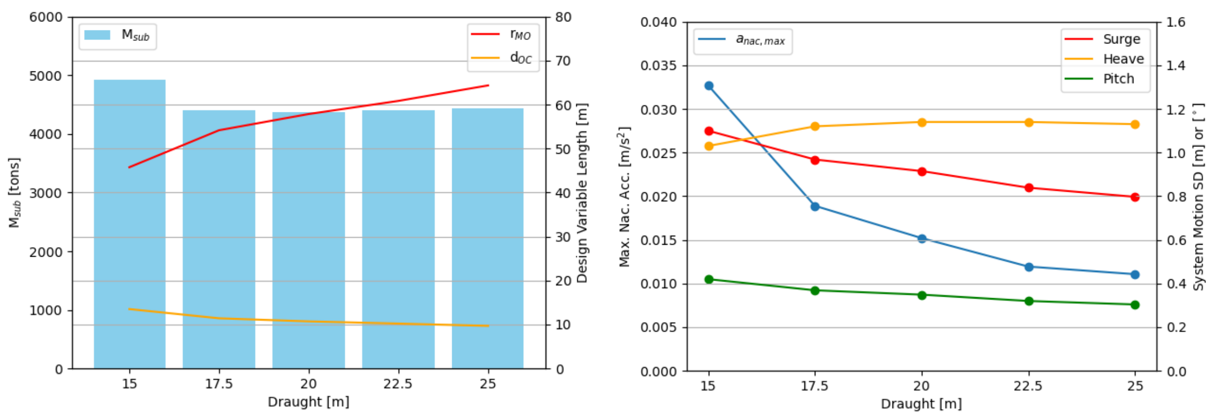
Figure 4.8: Key parameter change when subject to different maximum static pitch angles, $\theta_{\max}$.

A non-linear relationship between the mass of the structure and the maximum static pitch, $\theta_{\max}$, is seen. For a $\theta_{\max}$ of 6°, the mass of the design will increase with 12.5% compared to a design with $\theta_{\max}$ at 10°. On the other hand, allowing the platform to pitch up to a maximum of 14°, will only result in a platform mass reduction of 5.5%. r_{MO} remains around the optimized value of 57.84 m independent of the value of the maximum static pitch angle specified in the constraint, while d_{UC} varies, impacting the total mass of the structure. When looking at the relative changes in nacelle acceleration, the FOWT performs significantly worse for the designs where the static pitch angle is restricted to small values. However, the actual values are still very low as the maximum value is 0.02 m/s^2.

4.4.2 Draught

As mentioned, one of the reasons that semi-submersible platforms are favored by industry is the low draught which allows for easy assembly. Literature stated that platform developers try to keep the draught close to 20 m as this is considered the best compromise between stability and logistics. In the optimization model of this project, the draught of the floater was maintained at 20 m. A sensitivity analysis has been performed on the optimized design when changing the draught from 15-25 m in a 2.5 m interval. The height of the BCs has been kept constant at 6 m. The new designs are shown in Fig. 4.9a, where it is seen that the lightest structure is found for a draught of 20 m, which was the originally chosen value, although the structure with a draught of 22.5 m is only 0.25 % heavier. The standard deviations of system motions and nacelle acceleration of the new structures are compared in Fig. 4.9b. Increasing the draught will increase the added mass in surge as the cross-sectional area in this direction becomes larger, resulting in a lower SD. In heave, draught does not have the same impact on added mass and the changes are related to d_{UC} changes. The nacelle acceleration decrease as the draught increase and the d_{UC} decrease.

Keeping this observation in mind, while considering the fact that the structures with deeper draught are less than 1 % heavier than the 20 m design introduces the question of which draught level is the most appropriate compromise between mass and stability. This could be investigated further with multi-objective optimization. As the nacelle acceleration is also much more severe for extreme weather conditions, it would be interesting to investigate the reduction benefits of a deeper draught in these conditions.



(a) Changes in mass and design variables of optimized design when varying draught.

(b) System motion SD and max. nacelle acc. during operation (case 1) subject to varying draught.

Figure 4.9: Sensitivity to changes in draught

4.4.3 Thickness of Pontoons

In the upscaling of the OC4 from 5 MW to 15 MW, the thickness of the pontoons and cross-braces was scaled as to maintain the same internal stress. The details behind this approach is found in App. A.7. The thickness is constant in the optimization model, making Euler's critical load dependent on just the length of the member. The thickness impacts the limits of the structural constraints on both yielding and buckling in the optimization model, and a sensitivity analysis has been performed on the impact of this parameter on the optimized design. The optimization of the initial upscaled design with a thickness of 2.5 cm is seen in the middle, while one thinner and one thicker design has been optimized as well.

In Fig. 4.10 the mass and design variables are shown for the different pontoon thicknesses. The figure clarifies that increasing the wall thickness from 1.75 cm to 2.5 cm was a good decision as

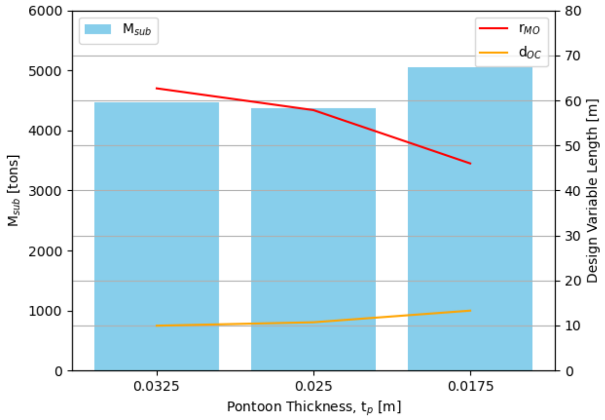


Figure 4.10: Changes in structure mass and design variables of optimized design with different thickness of pontoons and cross-braces.

the thinner walls a restricted in their ability increase the length of pontoons, which are mainly defined from r_{MO}. To ensure stability of the platform, d_{UC} will have to be larger, making the design significantly heavier. On the other hand, making the pontoons and cross-braces thicker allows for r_{MO} to be larger, but the added weight of these thicker members resulted in a structure which was still 2.2 % heavier than the initial optimized design. Furthermore, it should be noted that in the optimization with the thinner pontoons, yielding was the tight constraint, while for the other two designs, buckling was the defining factor. For all thicknesses in this range, the transition from yielding to buckling happens when the beam length surpasses 50.1 ± 0.1 m.

4.4.4 Ballast Levels

The OC4 semi-submersible is ballasted with water in the BCs and UCs. The ballast levels are adjusted in RAFT to get heave mean offset of 0±0.1 m. The ballast is adjusted with only a small amount in each iteration, and the levels in each column is merely adjusted so that the whole structure will reach the aforementioned heave mean offset of 0±0.1 m, while it is not optimized to reach the "optimal" distribution between the two columns, regarding floater stability. Specifying the default levels in each column close to an expected or desired optimal value will impact the range at which the levels are being adjusted, and where the final level will be found.

The distribution of the ballast between the two columns impact the z-coordinate of CoG which in particular impacts the hydrostatic stiffness in pitch. While the optimization model performs an optimization on the floater geometry, the ballast levels are not in fact optimized. From the preliminary to the optimized design, z_{CoG} changed from 0.529 m to 7.127 m, which impacted the contribution of the CoB-CoG relative position to K_{55} (see Eq. 2.18). The change in z_{CoG} was mainly due to the lighter floater structure, but the distribution of ballast also contributes.

In this section, the default ballast levels in the design have been varied with the purpose of making the optimization model able to find an optimal floater design with a lower z_{CoG}. By manually setting the default levels in the UCs to a lower level, the levels in the BCs were automatically adjusted to higher levels to ensure heave equilibrium. The effect of manually decreasing the default value in UCs on the optimization model output is shown in Fig. 4.11. Here it is seen that by adjusting the default ballast level in the UC, an optimized structure can take very different ballast level distributions in the two columns.

The preliminary design has a default value of 7.77 m in the UC, resulting in an optimized design with adjusted values of 3.79 m in the BCs and 7.65 m in the UCs. The adjustment ensured that the mean heave offset was within 0±0.1 m. As the BCs have an inner height of 5.88 m, there was potential for higher ballast levels in these, which would lower the overall z_{CoG}. By setting

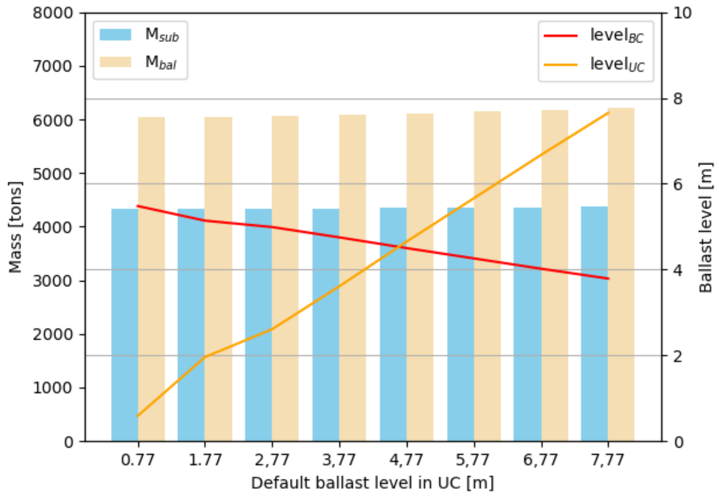


Figure 4.11: Changes in mass to ballast level adjustments.

the default value to 0.77 m in the UC, the optimization model was able to find a design with adjusted ballast levels at 5.48 m in the BCs and 0.59 m in the UCs. The new z_{CoG} was at 6.67 m. This gave a new optimized structure mass of 4334 tons, meaning a further optimization by 0.9 % as both r_{MO} and d_{UC} were slightly reduced to 57.79 m and 10.62 m respectively. The ballast mass was also reduced by 2.9 %.

4.4.5 Approach for Structural Constraints

In the process of deciding on a structural integrity criteria of the structure, different perspectives were considered. The first scenario, (a), represents a true static case for calm seas. Scenarios; (b), offset column submerged in water and (c), offset column above water, both represent extreme cases and with the imposed assumption of being static. As mentioned, the initial optimization model use case (a) for the structural constraint with a safety factor of 3 for both yielding and buckling.

In this section, the impact of using the two other scenarios, (b) and (c), in the optimization model, has been investigated. The safety factor was reduced to 1 for both yielding and buckling, in order to investigate the extreme case without any corrections. The bending criteria was also included with a safety factor of 1. The model was able to find an optimized design with a structure mass of 4118 tons, which is a reduction of 5.8 % compared to the optimized design with scenario (a) and safety factors of 3. This reduction in mass was achieved by increasing r_{MO} to 63.7 m and decreasing d_{UC} to 9.76 m.

In Fig. 4.12, the iteration tracker reveals that although the initial design does not comply with scenario (c), the model is able to find a design that does comply with the constraint by reducing the diameter and weight of offset columns and thereby reducing forces in the pontoons.

It can be seen how as r_{MO} increase, the bending constraints are reduced more rapidly than yielding, as it is proportional to the length of the pontoon. The design load is however lower as both Y-pontoons share the load, reducing the effect.

The results illustrates the models sensitivity to how the constraints are defined, and the importance of finding values that most accurately represent the reality. The many uncertainties connected to this also clarifies the importance of a more thorough analysis before a final design is fixed.

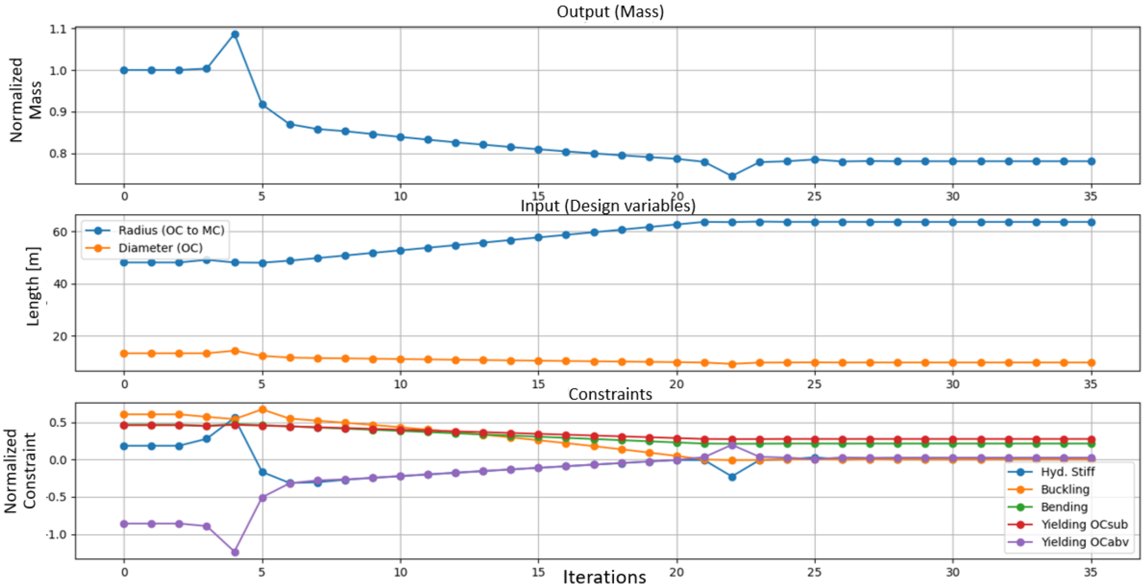


Figure 4.12: Iteration tracker for optimization with cases (b) and (c) on structural constraint.

4.4.6 Sea States

The dynamic analysis of the FOWT design uses case-specific wind and wave data. The default values as defined in Tab. 3.1, but it is also relevant to investigate the sensitivity of the platform design to a larger variety of wave loads. Environmental conditions are in principle site-specific, however, this study aims to evaluate the design with a general approach. The optimized design has been tested for eight different periodic sea states, representing general mild to extreme sea states [33]. The FOWT is first tested during operation, where a wind speed of 10.6 m/s is used for all cases, and the irregular sea state is included based on JONSWAP wave spectra with significant wave heights, H_s, and wave peak periods, T_p. The details for the sea states are specified in Tab. 4.8, and the wave spectra for sea state 4-8 are illustrated in Fig. 2.9.

Table 4.8: Specifications of different sea states.

Sea State	1	2	3	4	5	6	7	8
H_s [m]	0.09	0.67	1.4	2.44	3.66	5.49	9.14	15.24
T_p [s]	2.0	4.8	6.5	8.1	9.7	11.3	13.6	17.0

The system motion standard deviations in surge, heave, and pitch directions, and the maximum nacelle acceleration, at the different sea states are illustrated in Fig. 4.13. The FOWT shows exponentially increasing standard deviations along with more extreme sea states, while the maximum nacelle acceleration is increasing linearly and evenly for sea state 8, the value is still below 0.02 m/s^2. The SD of surge, heave, and pitch generally follows the exponential progress of the sea states, but heave more drastically than the rest. This might be explained by the natural period of the platform in heave being 15.9 s, which is between the wave period of sea state 7 and 8.

In Fig. 4.14 the RAOs for sea state 5-8 is shown in surge, heave, and pitch respectively. The wave period, T_p, of each sea state is also visualized with a vertical line. The graph shows that in heave, the RAO peak is strongly correlated with the natural frequency in heave which, as mentioned, is between T_p of the two most extreme sea states (7 and 8).

In pitch, the RAO peaks also have a strong correlation with the natural frequency, but in both

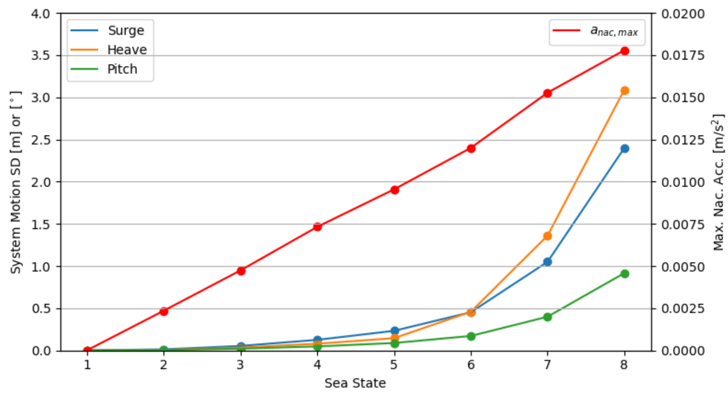


Figure 4.13: System motion SD and nacelle acceleration at different sea states during operation with constant wind speed of 10.6 m/s.

surge and pitch, the RAO peaks are far from the T_p of the specified sea states and also the general wave spectrum of sea waves. The fact that the peaks of RAO in heave have strong correlations with the natural frequency clarifies the importance of considering the sea state properties of the chosen site for the FOWT.

The graph also reveals that the largest peaks are experienced at the less extreme sea states. This can be explained by the RAOs dependency on wave amplitude, as expressed in Eq. 2.24.

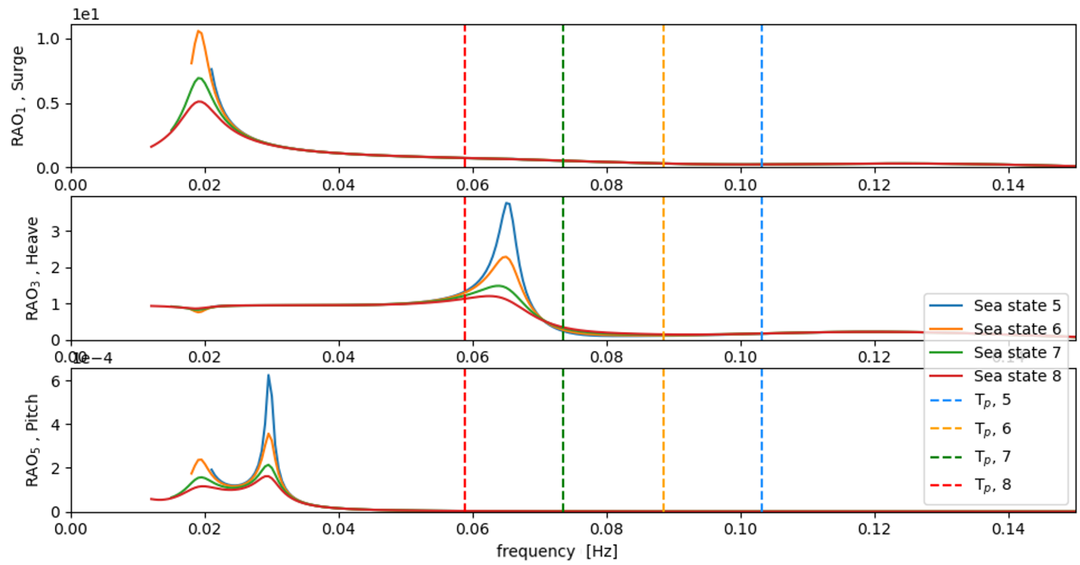


Figure 4.14: RAOs and T_p for different sea states during operation with constant wind speed of 10.6 m/s.

In Fig. 4.15 is seen the response spectra during operation under sea state 5-8. As the peaks of the RAOs in surge and pitch are far from the wave spectra, the surge and pitch responses follow the wave spectrum of the respective sea states, while the heave response seems to also be affected by the RAO peak around natural frequency.

Normally the use of RAOs to describe the behavior of a floating structures in waves assumes a linear relationship between the waves and motions of the floating structure. In practice, this means that if the wave amplitude is increased by a factor, then the amplitude of the structure motions is increased by the same factor, which, particular in large waves, are not always valid [56]. This is also the case for the RAOs produced by RAFT but with the difference that the RAOs are also coupled to the excitation force as of Eq. 3.11. It can be seen from Fig. 4.14 that as the sea state increase the peaks of the RAOs decrease, while the responses still increase as of

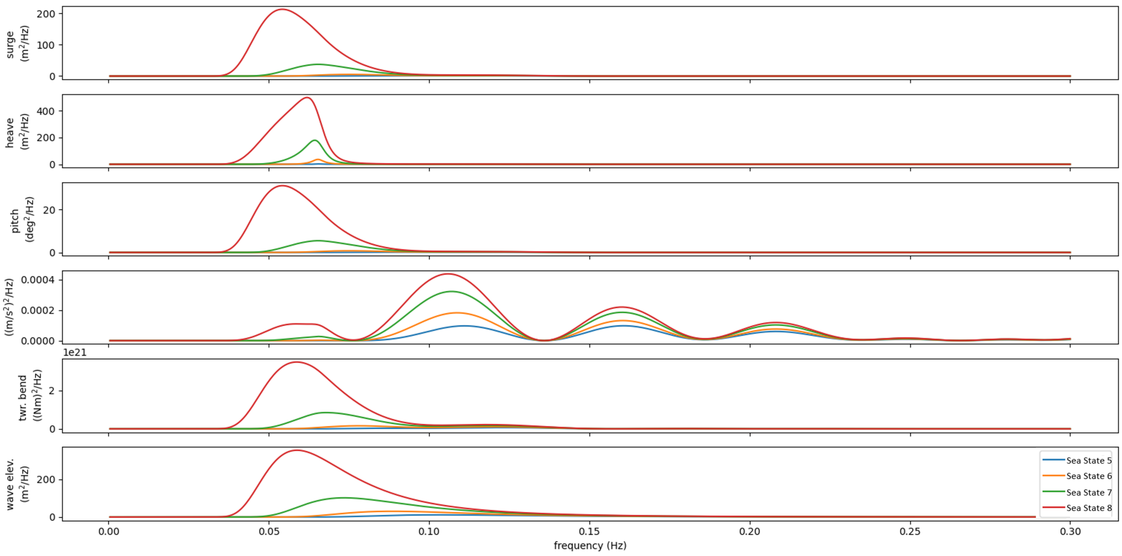


Figure 4.15: Response spectra for sea state 5-8 during operation with constant wind speed of 10.6 m/s.

Fig. 4.15.

The results of this sensitivity analysis show the importance of considering the sea state properties of the specific site when deciding for the installation of a FOWT. It might be necessary that restoring properties must be adjusted to achieve RAOs whose peaks do not coincide with the peak of the spectrum which is describing the likely sea state at that site.

Wind Speed Effects

As mentioned previously, the effect of rotor aerodynamics on the dynamic response is modeled using the derivative of thrust with respect to wind speed. This means that the wind will have a significant impact on the damping of the FOWT. This has been investigated in Fig. 4.16 where RAOs are shown for wind speeds of 8, 10.6, 14, and 17 m/s in the same sea state (case 1).

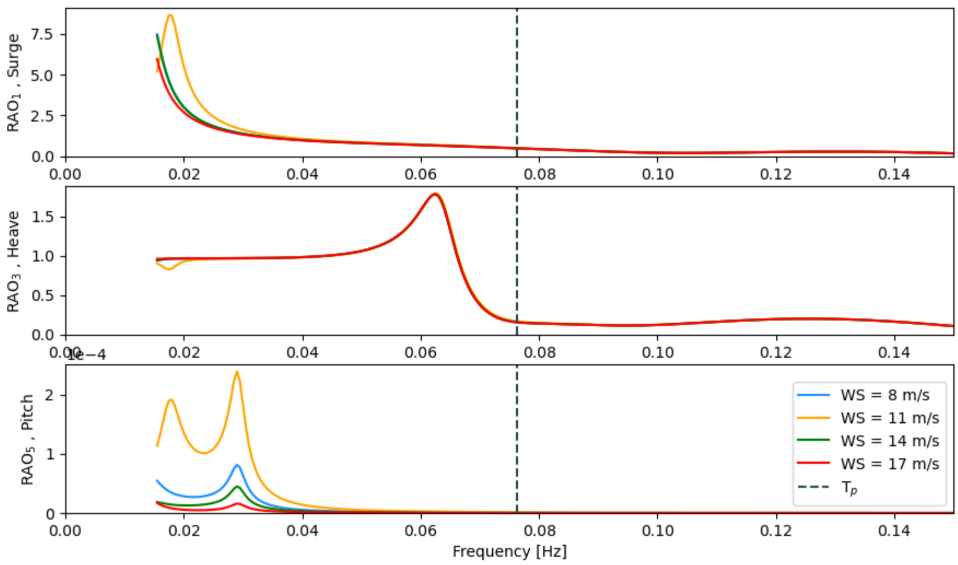


Figure 4.16: RAOs for different wind speeds and constant wave loads (case 1) during operation.

Especially the RAO in pitch is large for 10.6 m/s which is the rated wind speed and hence corresponds to the largest thrust force. Also for surge, it can be seen how the operation at rated wind speed results in a peak at a much higher frequency than the rest. The wind speed has almost no effect on the RAO in heave.

4.5 Final Discussions

The upscaling was done with consideration to existing semi-submersible reference platforms, and the scaling of the platform was done in relation to the increase in rotor diameter, as it has a direct and quantifiable impact on the loads, mass and stability of the whole system. The main purpose of the upscaling was to find a reasonable reference point for the optimization model, which was representing of a stable platform, without being over conservative. It is believed that this was done successfully, but since the upscaling was based on r_{MO} and d_{UC} which were also the design parameters for the optimization, the upscaling was quite forgiving in the way that the relation between the parameters would be corrected anyway.

As the choices made during upscaling impacted the initial reference for the optimized design, it is limited what the comparison between the two designs can be used for. It should be noted that a similar optimization was carried out for the original 5 MW OC4, with an adjusted maximum turbine thrust force, where the same tendencies were seen. For the same reason the optimized model was also compared with the Volturn US-S, a more mature reference model than the OC4 Semi-sub with a grid-connected 1:8 scale model in operation since 2013 and a full scale FOWT on the way. The similarity in responses between the optimized 15 MW OC4 system and the Volturn US-S provided comfort in the validation of the hydrodynamic reliability of the optimized OC4 system.

Many factors are important to consider when optimizing for the cost of a FOWT, such as material selection, manufacturing options, installation and operation. The only financial aspect which has been considered in this optimization is the reduction in amount of steel used for the substructure, with consideration to site and design specifications. Due to the immaturity of the technology, especially of larger capacity, there is limited experience meaning that cost models are subject to great uncertainties. In all this uncertainty, an optimization model based on a very tangible parameter, like the steel mass of the floater, can be an important tool in lowering the LCOE for FOWTs and support the commercialization of the industry in its early stage when floater design options are abundant.

The reliability of the model and the results are highly influenced by the criteria set in the optimization. As the structural constraints are simplified and ultimately insufficient, the geometry of the optimized design should not be considered final. For the implementation of other structures in the optimization model, it would, as a start, be sufficient to update the safety factors and decide between the relevant presented structural constraints. In order for the results to become more reliable, which would be necessary if optimization was to be used in the later stages of the design process, it is suggested to incorporate FEM in the structural evaluation of the floater. This is quite a substantial step-up in the computational capacity required of the model. For now, it is important to recognize the optimization model as a tool for the early-phase design development, where it can provide guidance on general design trends for a semi-submersible when subject to different customized constraints and considerations.

Some of the choices that were made in the upscaling; such as keeping the vertical dimensions unchanged and the definition of the wall thickness of pontoons, were constant in the optimization model. If other choices had been made, this would have resulted in a different optimal design. This underlines the importance of the sensitivity analysis. It would be valuable to investigate more parameters, like the specific dimensions of the BCs and different wall thicknesses. The inclusion of additional parameters as part of the design variables, such as wall thickness and vertical dimensions, would also be an interesting trajectory, as the optimization would have more options in the design development. This was compensated for with sensitivity analysis to some extent. With the capabilities of RAFT it is also possible to extend the scope of the optimization to include different mooring and anchoring configurations.

The model is a valuable tool to investigate design trends and tendencies, and since the model is able to extract fast dynamic results, it is an effective way to investigate relations between parameters and their effect on the floater mass and dynamic behavior. The dynamic are analyzed in RAFT with a frequency domain analysis, which is superior to the time domain in term of computational speed. Ideally, the process should be accompanied by a validation with a time domain analysis. This could e.g. be conducted in OpenFAST. RAFT has certain limitation as its hydrodynamic calculations are based on strip theory. OpenFAST evaluates the hydrodynamics with potential flow theory, which is generally more reliable. From the OpenFAST results, relevant coefficients and parameters may want to be adjusted and the optimization could be repeated in a refined model with more reliable results [51]. This would create a synergy between the more reliable output of the time domain analysis combined with the fast results showing general tendencies from the frequency domain analysis. It is here relevant to mention WEIS (Wind Energy with Integrated Servo-control). A framework that combines multiple NREL-developed tools to enable design optimization of floating offshore wind turbines [83]. This may prove to be a great tool for optimizing wind turbine systems utilizing the combination of frequency and time domain analysis.

A time domain validation of the results was attempted to be carried out in OpenFAST, but no results were obtained in time to be included in this report. The procedure is described in App. A.10.

Stiesdal has recently developed the Tetrasub foundation, which is one of the world's first fully industrialized floating offshore semi-submersible platforms. Although the platform is yet to be employed across sites, excused by the general immaturity of the industry, the company states that the tetra concept can be scaled to 20 MW rating without any difficulties, and the foundation weight range is expected to be 175-250 tons/MW depending on metocean conditions and turbine type [84]. For a 15 MW WT, this would require an expected platform mass of 2625-3750 tons.

This is considerable less than the 4374 tons that was found to be the optimized steel mass of the OC4. This speaks to the limits of the general OC4 design concept. As the optimization model is limited within the same design concept, it is important to identify a promising design concept before optimization of the specific dimensions are carried out. This can be an iterative process, where tendencies in the optimization results may help to clarify what aspect of an overall design is important. One example of this is the size of the BCs. If the optimization shows a tendency of a large diameter of the BCs to be beneficial for heave stabilization, while the ballast levels in the same BC has little effect, this would imply consent to adjust the height of the base columns or experiment with implementing heave plates instead.

5 Conclusion

In this work the 5 MW OC4 reference floating wind platform was upscaled to fit the 15 MW IEA WT, and an optimization model was defined to minimize the steel mass of the platform while maintaining structural integrity and hydrostatic stability. The optimized design was then evaluated from a hydrodynamic analysis. Finally, a sensitivity analysis was carried out in order to investigate the impact of certain design parameters and the general trends and tendencies for the optimization.

A literature study was conducted in order to investigate the theoretical and practical scaling trends utilized in upscaling of floating wind platforms. It was discovered that scaling laws were not strictly followed in practice, but rather followed a general trend relating the rotor diameter of the WT to the product of two key parameters; the distance between the center of the main column to the offset columns, r_{MO}, and the diameter of the upper offset column d_{UC}. It was concluded to follow this upscaling trend, while keeping the vertical dimensions constant.

An optimization problem was then formulated to minimize the steel mass of the OC4 platform with the same r_{MO} and d_{UC} as design parameters. The platform was constrained to maintain a maximum static pitch angle of 10° by regulating the hydrostatic stiffness which was calculated using RAFT. Constraints for the structural integrity of the platform were defined for resistance against yielding and buckling respectively. The constraints were based on a statics analysis of the pontoons which were deemed the weakest link. The structural constraints was chosen not to represent actual strong structural integrity, as to compensate for the limited strength of the original OC4 design. More conservative structural constraints were tested and included as options in the model.

The optimization model achieved a considerable reduction in the mass of the substructure by increasing r_{MO} and decreasing d_{UC}. It was noted that d_{UC} had the greatest impact on the substructure mass and hydrostatic stability while r_{MO} had a great impact on the structural integrity of the pontoons.

It was decided not to include hydrodynamic constraints in the optimization as this would make the optimization site specific. Instead, the dynamic results, calculated with RAFT, were used in the evaluation of the final design. Nevertheless, the optimized solution showed a reduction of the SD of system motions in surge and pitch as well as maximum nacelle acceleration both for operation at rated wind speed and for the 50 year extreme weather case. In heave, the SD increased in both cases due to the reduction in d_{UC}, which resulted in a reduced heave plate effect and a change in natural frequency due to a lower mass.

A sensitivity analysis was carried out in order to investigate the trend and tendencies of the optimization, where the following was revealed:

- Restricting the maximum allowable static pitch angle to 6° increased the structure mass by 12.5% (compared to 10°) while extending it to 14° reduced the structure mass by 5.5%.
- Increasing the draught from with 2.5-5 m reduced the reduce the SD of systems motions in surge and pitch as well as the nacelle acceleration while only increasing the substructure

mass with less than 1%.

- Redistributing the ballast mass to be placed lower in the semisubmersible structure reduced the structure mass up to 0.9% as this created a less negative contribution from the relative position of CoG and CoB to the hydrostatic stiffness.
- Increasing the initial wall thickness of the pontoons allowed the model to find an improved design with a larger r_{MO} and a smaller d_{UC}. Increasing the pontoons to much would instead lead to the pontoons being to heavy leading to a heavier overall substructure.
- The dynamic properties of the optimized structure was investigated in different sea states, and most notably it was observed that the peaks of RAO in heave have strong correlations with the natural frequency. This is relevant as the coupling of wave period with natural frequency strongly impacted the SD of system motion in heave. This clarified the importance of considering the sea state properties of the specific site when deciding for the installation of a FOWT.

The optimization model demonstrated the potential of serving as a valuable tool in the design process for future semi-submersible platforms for FOWTs.

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Appendices

A Supplementing Figures and Calculations

A.1 Upscaling defined from design variables

The design of the whole OC4 design must be regulating just the two design variables: r_{OM} and d_{UC}. Geometrically this was solved by defining each endpoint of all the pontoons from the design variables, see Fig. A.1.1. As the design variables does not affect the vertical position (z-coordinate) of the end points these are not represented here. For a semi-submersible with three outer columns $\nu = 360°/3 = 180°$. All elements of the design is defined as lines with a diameter and thickness. The elements are just defined for one offset column and then copied in a circular pattern of three. Also the relocation of the fair-lead location is following this procedure. It is assumed that the the diameter of the base column is two times that of the upper column.

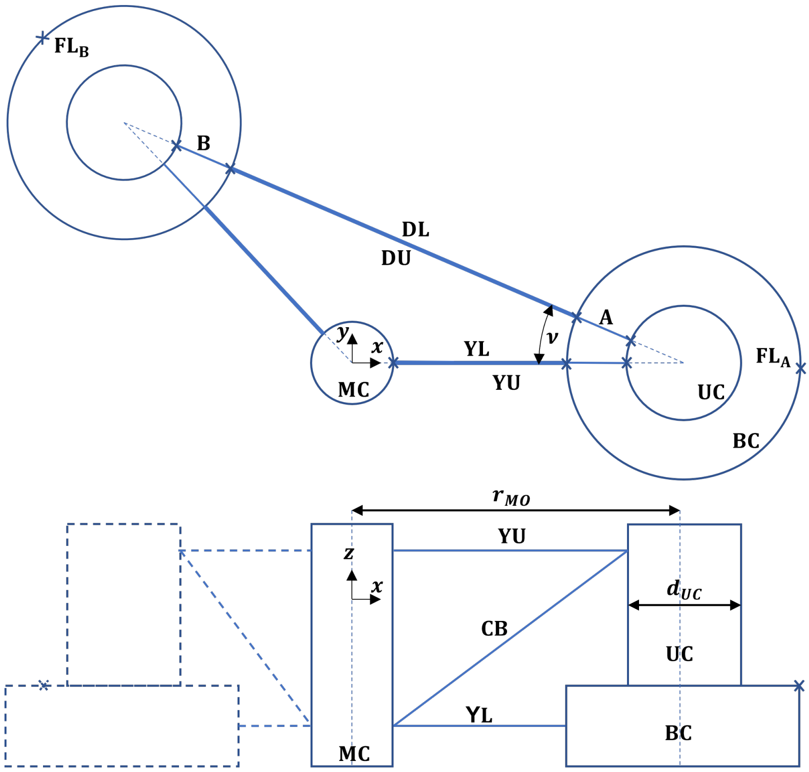

Figure A.1.1: Geometrical illustration of elements of the OC4. Dimensions not to scale.

For the delta upper pontoon (DL) the endpoints are given by:

$$DL_A = \begin{pmatrix} r_{MO} - \frac{d_{UC}}{2}\cos(\nu) \\ \frac{d_{UC}}{2}\sin(\nu) \end{pmatrix} \qquad DL_B = \begin{pmatrix} -r_{MO}\sin(\nu) + \frac{d_{UC}}{2}\cos(\nu) \\ r_{MO}\cos(\nu) - \frac{d_{UC}}{2}\sin(\nu) \end{pmatrix} \tag{A.1}$$

For the delta lower pontoon (DU) the endpoints are given by:

$$DU_A = \begin{pmatrix} r_{MO} - d_{UC}\cos(\nu) \\ d_{UC}\sin(\nu) \end{pmatrix} \qquad DU_B = \begin{pmatrix} -r_{MO}\sin(\nu) + \cos(\nu)d_{UC} \\ r_{MO}\cos(\nu) - d_{UC}\sin(\nu) \end{pmatrix} \tag{A.2}$$

For the upper and lower pontoon (YU and YL) and the cross brace (CB), only the x coordinate at the end of the offset column changes:

$$YU_{OC} = \begin{pmatrix} r_{MO} - \frac{d_{UC}}{2} \\ \text{uchanged} \end{pmatrix} \qquad YL_{OC} = \begin{pmatrix} r_{MO} - d_{UC} \\ \text{unchanged} \end{pmatrix} \qquad CB_{OC} = \begin{pmatrix} r_{MO} - \frac{d_{UC}}{2} \\ \text{unchanged} \end{pmatrix} \tag{A.3}$$

Contrary to the OC4 elements, the fair-lead points are defined by the mooring system and must be defined for each offset column individually. Their coordinate differs from the one of the OC4 system by being mirrored around the y-axis.

$$FL_A = \begin{pmatrix} r_{MO} - d_{UC} \\ 0 \end{pmatrix} \qquad FL_B = \begin{pmatrix} \sin(\nu)(d_{UC} + r_{MO}) \\ \cos(\nu)(d_{UC} + r_{MO}) \end{pmatrix} \qquad FL_C = \begin{pmatrix} \sin(\nu)(d_{UC} + r_{MO}) \\ -\cos(\nu)(d_{UC} + r_{MO}) \end{pmatrix} \tag{A.4}$$

A.2 Initial Stability of the OC4 Semisubmersible

In Fig. A.2.2 the OC4 is shown in a situation where an external force is acting on the hub of the turbine, resulting in a pitch of the floater. It can be seen why hydrostatic stability in pitch is a relevant parameter for determining the stability of the floater. Notice how the CoB is below the CoG making the hydrostatic contribution from the offset of these two points negative.

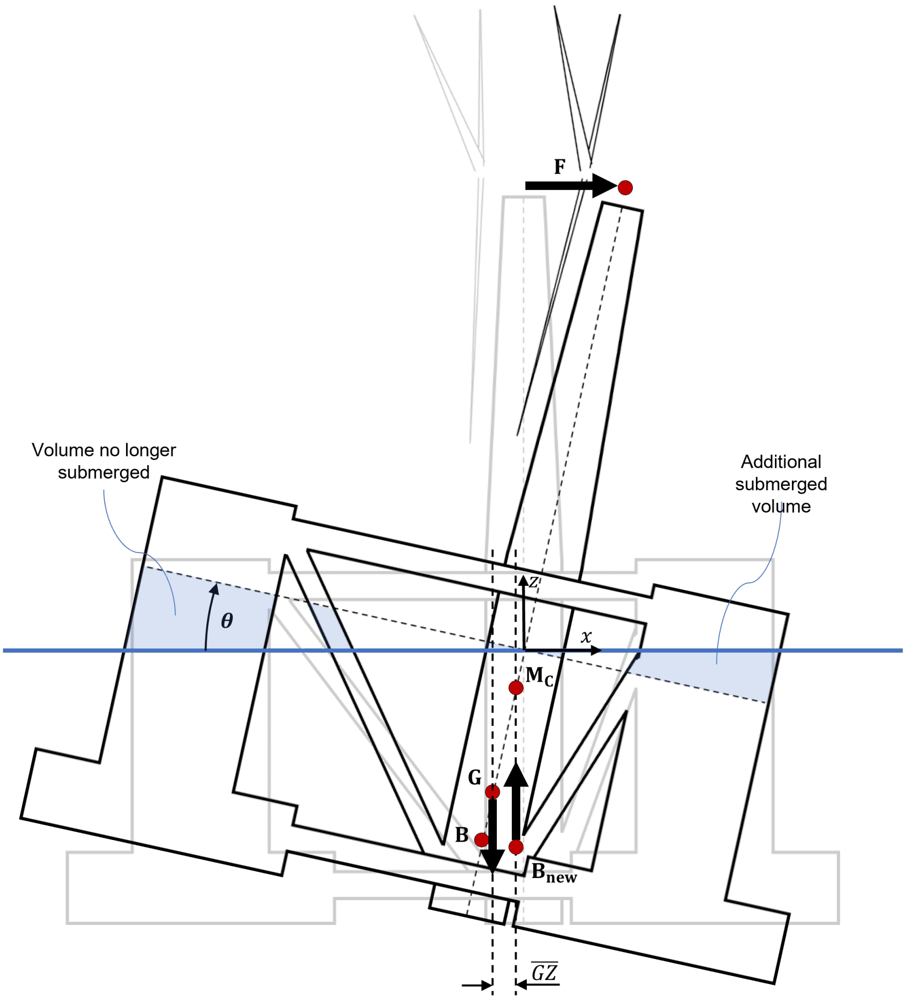

Figure A.2.2: Floating body diagram of an FOWT.

A.3 Redefined Mooring

In order to make the program able to take different water depths and floater designs while still keeping the same mooring configuration, it was necessary to make the geometry of the mooring system dependent on just the design variables and water depth.

The original OC4 5 MW setup was used as a standard for future mooring setups by defining two ratios to be maintained during upscaling and optimisation.

The slag-ratio, s_r, is an expression for how slag the mooring line is. It is defined by the relation between the straight line of the two mooring points, l_Δ and the actual mooring line length, l_m, see Eq. A.5:

$$s_r = \frac{l_m}{l_\Delta} = \frac{l_m}{\sqrt{(d_w - z_c)^2 + (r_a - r_c)^2}} \tag{A.5}$$

The straight line is found using the Pythagorean theorem where d_w is the water depth, z_c the draught of the cleats (fairleads), r_c the cleat radius, and r_a the footprint radius of the anchors. For the standard setup of the OC4 5 MW, $s_r = 1.0212$.

The footprint ratio, f_r, is an expression of the relation between the effective water depth and the effective mooring radius, see Eq. A.6:

$$f_r = \frac{r_a - r_c}{d_w - z_c} \tag{A.6}$$

Typically, f_r is about 4-8 times the water depth, which in turn will ensure a vertical and reduced load on the anchors. For the standard setup of the OC4 5 MW $f_r = 4.2835$.

In situations where the OC4 is to be investigated with minimal influence from the mooring system, a very slag cateneray setup was used of: $s_r = 1.5$ and $f_r = 4.2835$.

The two ratios are kept constant when upscaling and testing new designs. From these, a new expression for the length of the mooring line can be seen in Eq. A.7:

$$l_m = s_r\sqrt{(d_w - z_c)^2 + (r_a - r_c)^2} \tag{A.7}$$

The anchor points will change as well. The radius of these are defined from Eq. A.8 and their exact placements are found with the same approach as explained for the fair-leads in App. A.1.

$$r_a = f_r(d_w - z_c) + r_c \tag{A.8}$$

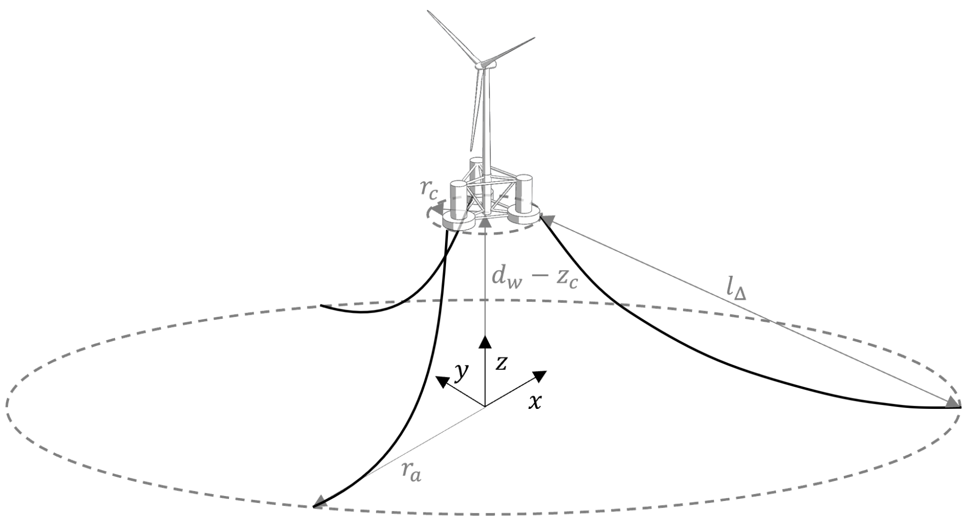

Figure A.3.3: Standard catenary mooring systems for the OC4 structure.

A.3.1 Low Mooring Stiffness

On some occasions, it was of interest to run simulations where the effect of the mooring system was limited. RAFT is failing when a mooring line becomes so slag that it lays on top of itself, meaning that the mooring line must always be longer than the sum of the horizontal and vertical distance from anchoring point to fair-lead point respectively.

In order to obtain a very slag configuration within these boundaries it was decided to define the length of the active mooring line differently than the two passive mooring lines. From Fig. A.3.5, the different lines can be seen. A position of the FOWT discreetly between anchor point 1 and 3 was assumed. Line 2 is then defined from the Pythagorean theorem as seen in A.9

$$l_2 = \sqrt{((d_w - z_c)^2 + (r_a(1 + sin(\nu) - z_c)^2))} \tag{A.9}$$

The lengths of mooring lines 1 and 3 were defined as sum of the horizontal distance from anchoring point to fair-lead point, and similar vertical distance:

$$l_1 = l_2 = cos(\nu)r_a + d_w - z_c \tag{A.10}$$

From this there is but to decide a footprint ratio to find the desired anchor radius for the given water depth: $r_a = d_w \cdot f_r$. The geometries are not exactly true, but they are within the right margins to give the desired result without RAFT failing.

In Fig. A.3.5, a comparison between the the original OC4 15 MW upscaled with mooring defined as in [blue] and mooring defined as described above is illustrated.

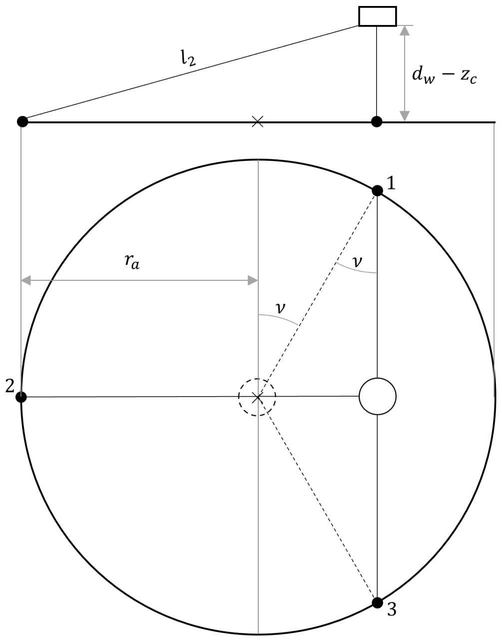

Figure A.3.4: Geometry for low mooring stiffness.

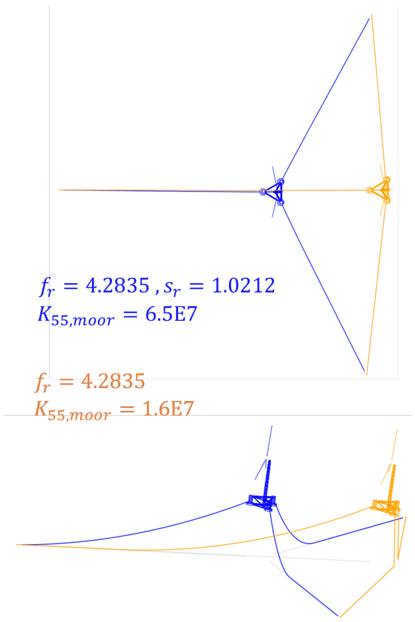

Figure A.3.5: Mooring configurations.

A.4 Analytical Hydrostatic Stiffness of the 5 MW OC4 Compared with Computational (RAFT) Results

In the following, the hydrostatic stiffness of the OC4 5 MW semi-submersible will be investigated analytically and later compared with the computational results using RAFT.

For the analytical results, the values presented in table A.2 were used.

The hydrostatic stiffness is given by Eq. 2.18:

$$K_{55} = \underbrace{\rho g I_{yy}}_{\text{waterplane}} + \underbrace{\rho g V z_B - m_{tot} g z_G}_{\text{B-G relative position}} + \underbrace{K_{moor,55}}_{\text{mooring}} \tag{A.11}$$

The **waterplane** contribution can be estimated using the approximation for the second moment of area from 2.20:

$$K_{55,wp} = \rho g I_{yy} \approx \rho g \frac{3\pi}{8}\left(r_{MO}^2 d_{UC}^2 + \frac{d_{UC}^4}{8}\right) = 1.45 \cdot 10^9 \text{Nm} \tag{A.12}$$

For the **B-G relation**, it must be estimated how much of the mooring line is resting on the seabed. With L being the straight line distance from anchors to fairleads: $L = \sqrt{h^2 + R_{Anch}^2} = 861$ m, it will be assumed that 2/3 of he mooring line is hanging above the seafloor. This estimate comes from the relation between L and L_0. It means that the mooring weight acting on the floater can be found by the weight of the hanging mooring line subtracted by the buoyancy from its water displacement:

$$w_{moor} = m_{moor} g - \rho_w g V_{moor} = (3\frac{2}{3} P_{moor} L_0) g - \rho_w g (3\frac{2}{3} A_{moor} L_0) = 1.89 \cdot 10^6 \frac{\text{kg m}}{\text{s}^2} \tag{A.13}$$

where $A_{moor} = \pi(D_{moor}/2)^2$ is the cross-sectional area of the mooring line. The buoyancy force can be found from the total mass of the system times g, as this is what will reach an equilibrium under Archimedes' principle, while the gravitation force on the floater is not including the weight of the mooring lines:

$$F_b = \rho_w g V = m_{tot} g = 1.32 \cdot 10^8 \frac{\text{kg m}}{\text{s}^2} \quad , \quad F_g = m_{tot} g - w_{moor} = 1.30 \cdot 10^8 \frac{\text{kg m}}{\text{s}^2} \tag{A.14}$$

The total CoG can be estimated from a weighted average of CoG of the nacelle, rotor, tower, and floater:

$$z_{tot} = \frac{(m_{rotor} + m_{nacelle}) z_{hub} + m_{tower} z_{G,tower} + m_{floater} z_{G,floater}}{m_{tot} - m_{moor}} = -9.89\text{m} \tag{A.15}$$

Now the buoyancy contribution to the hydrostatic stiffness can be found as:

$$K_{55,BG} = F_B z_{B,tot} - F_G z_{G,tot} = -4.49\text{Nm} \tag{A.16}$$

Finally the **mooring** stiffness will be investigated. The analytical estimation of the mooring stiffness is based on [85], where the stiffness in pitch is given by:

$$K_{55,moor} = \frac{4 T D_{fair}^2}{L} + 2 K_I R_{fair}^2 + 4 T D_{fair} \tag{A.17}$$

where T is the line pretension caused by platform excess buoyancy, and $K_I = EA_{moor}/L_0$. Assuming that $T = w_{moor}$, the mooring stiffness is found to be $3.97 \cdot 10^9$ Nm with the second element contribution far outweighing the rest. This result is far beyond what could be expected for this type of centenary mooring system whose primary purpose is station keeping rather than hydrostatic stability. In [85] similarly large deviations are seen. It is still included here in order to show that the modeling of the mooring system can be difficult to predict, but will not be used in the analytical calculation for the combined hydrostatic stiffness.

In table A.1, the results are compared with the RAFT calculations. An acceptably low deviation between the two are seen on all parameters except for the mooring.

Symbol	**Description**	**Analytical**	**RAFT**	**Dev. (%)**
$K_{55,wp}$	Waterplane	1.45E+09	1.46E+09	1%
$K_{55,BG}$	B-G relation	-4.49E+08	-4,81E+08	7%
	$F_B z_B$	-1.74E+09	-1.84E+09	6%
	$F_G z_G$	-1.29E+09	-1,36E+09	5%
$K_{55,moor}$	Mooring	(3.97E+09)	8.72E+07	(-4454)%
K_{55}	Total (analytical w.o. mooring)	1.00E+09	1,07E+09	6%
K_{55}	Total (both w.o. mooring)	1.00E+09	9.79E+08	2%

Table A.1: Values in [kg·m^2/s^2·rad] used for analytical and RAFT results of OC4 5 MW floater.

Table A.2: Values used for Analytical results of OC4 5MW floater.

Symbol	**Description**	**Value**	**Unit**
Density of water.	ρ_w	1025	[kg/m^3]
Gravitational acceleration constant.	g	9.81	[m/s^2]
Dimensional properties of OC4 [33]			
Diameter of upper offset column.	d_{UC}	12	[m]
Radius from center of main to offset column.	r_{MO}	28.86	[m]
Vertical coordinate of tower CoG from SWL.	$z_{G,tower}$	43.4	[m]
Vertical coordinate of floater CoG from SWL.	$z_{G,floater}$	-13.46	[m]
Overall (integrated) tower mass.	m_{tower}	$2.497 \cdot 10^5$	[kg]
Vertical coordinate of total CoB from SWL.	$z_{B,tot}$	-13.15	[m]
Total structure mass (incl. ballast and mooring)	m_{tot}	$1.3473 \cdot 10^7$	[kg]
Mooring properties of OC4 [33]			
Number of mooring lines.	n_{moor}	3	[-]
Mooring line diameter.	d_{moor}	0.0766	[m]
Unstretched mooring line length.	L_0	835.5	[m]
Radius to fairleads from platform centerline.	R_{fair}	40.868	[m]
Radius to anchors from platform centerline.	R_{Anch}	837.6	[m]
Depth to fairleads below SWL.	D_{fair}	14	[m]
Depth to anchors below SWL.	h	200	[m]
Young's modulus for steel.	E	210	[GPa]
Equivalent mooring line mass density.	P_{moor}	113.35	[kg/m]
NREL 5 MW wind turbine [86]			
Rotor mass.	m_{rotor}	$1.1 \cdot 10^5$	[kg]
Nacelle mass.	$m_{nacelle}$	$2.4 \cdot 10^5$	[kg]
Tower mass.	m_{tower}	$3.475 \cdot 10^5$	[kg]
Hub height above SWL.	z_{hub}	90	[m]

A.5 Static analysis of the OC4 structure

A.5.1 Truss Structure

Here the OC4 structure is analysed statically. This is done in three different cases. The first is under the assumption of still water. A realistic and common case, where the motion of the structure is in equilibrium. The two other cases are more extreme assumptions, investigated in order to get an understanding of the loads the structure may be subject to. This is done by assuming a wave situation resulting in the OC to be fully submerged and MC to be fully out of the water and vice versa.

To repeat, the assumptions are as given in sec 3.4.1:

- An isolated OC-MC link will be investigated.
- The delta pontoons will be disregarded.
- The pontoons and cross braces will be assumed to act like a truss structure meaning that they only carry axial loads (no torque or transverse loads).
- All three offset columns contribute equally to carry the weight on the main column.
- The weight and buoyancy of the pontoons can be disregarded.
- The force in the upper and lower pontoon carries the same horizontal load, $F_{YL} = F_{UL}$, as as both columns are regarded as rigid structures.

Still Water

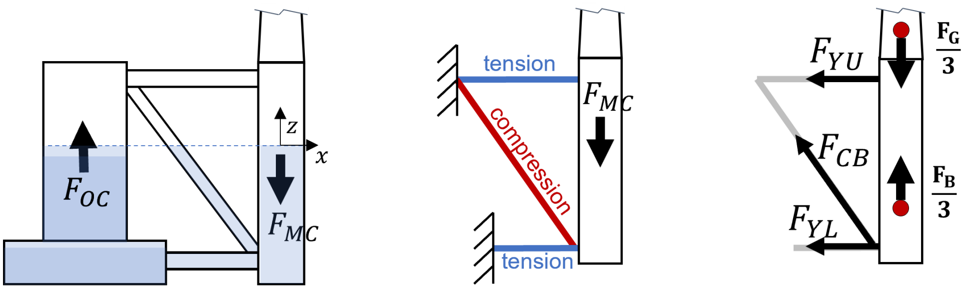

Figure A.5.6: Still water statics.

It is assumed that the motion of the OC4 is in equilibrium for still water. On the left in Fig. A.5.8, the isolated system is illustrated. The net buoyancy-gravitational force acting on the offset column is positive, while it for the main column is negative. As the isolated system is in equilibrium, it is given that $F_{OC} = F_{MC}/3$.

In the middle of Fig. A.5.8, the eternal forces of the truss structure is shown by assuming the offset column to be fixed. On the right side of Fig. A.5.8, a cut is made through the truss structure and the axial forces in each pontoon is shown. It is beneficial to analyse the load from the main column side rather than the offset column side, as this does now vary during optimization.

The gravitational forces and buoyancy force of the main column and what it is carrying is given by Eq. A.18:

$$F_{G,MC} = (m_{RNA} + m_{tower} + m_{MC})g \qquad F_{B,MC} = D_{MC}\pi r_{MC}^2 g\rho_w \tag{A.18}$$

where D_{MC} is the draught of the MC. The resulting force acting on the MC is given by the sum divided by three as seen in Eq. A.19

$$F_{Res} = \frac{F_{G,MC} - F_{B,MC}}{3} = \frac{(m_{RNA} + m_{tower} + m_{MC})g - D_{MC}\pi r_{MC}^2 g \rho_w}{3} \tag{A.19}$$

The vertical and horizontal force equilibriums are described in A.20:

$$\uparrow: 0 = -F_{RES} + F_{CB}\frac{h_{CB}}{l_{CB}} \qquad \rightarrow: 0 = -F_{YU} - F_{YL} - F_{CB}\frac{l_{YU}}{l_{CB}} \tag{A.20}$$

From a geometrical consideration the final forces in the trusses can be found to beA.21:

$$F_{CB} = F_{RES}\frac{l_{CB}}{h_{CB}} \qquad F_{YU} = F_{YL} = -F_{RES}\frac{l_{CB}}{2l_{YU}} \tag{A.21}$$

Wave Extreme at Offset Column

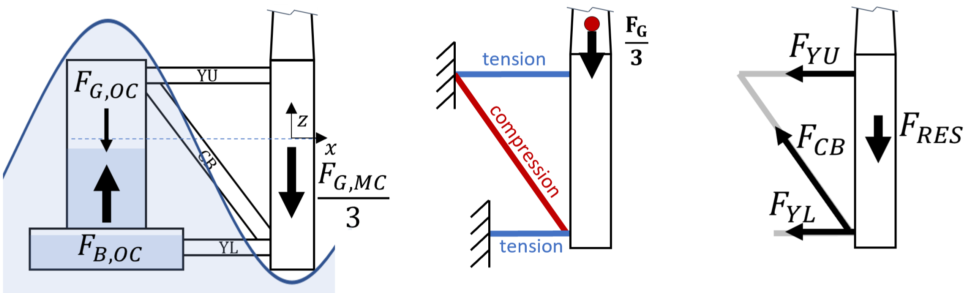

Figure A.5.7: Statics for wave extreme at offset column.

This case is not actually a static case, but in order to evaluate the extreme conditions the floater may experience and what loads are present in such a case, it will be regarded as such. In this case, a wave is expected to influence the structure in such a way that the entire offset column is submerged while the entire main column is above water. As the loads on the main column are greater than the loads on the offset column, $-F_{G,MC}/3 > F_{G,OC} + F_{B,OC}$, this side will now be isolated to find the reaction forces in the truss structure, meaning $F_{RES} = F_{G,MC}/3$. The pontoon forces are again given by A.21.

$$F_{RES,B} = \frac{(m_{RNA} + m_{tower} + m_{MC})g}{3} \tag{A.22}$$

Wave Extreme at Main Column

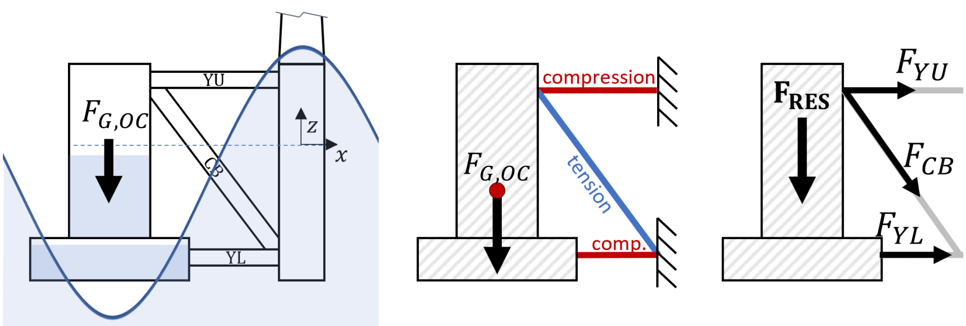

Figure A.5.8: Statics for wave extreme at main column.

This extreme case has waves influencing the structure in such a way that the offset column can be regarded as completely above water while the main column is fully submerged. As the loads on the offset column is greater than the loads on the main column, this side will now be isolated to find the reaction forces in the truss structure. The weight of the offset column can be found from Eq. A.23:

$$F_{G,OC} = (m_{UC} + m_{BC} + m_{ballast} + m_{caps})g \tag{A.23}$$

with $m_{ballast}$ being the mass of the ballast water, m_{caps} the combined weight of the caps, and m_{UC} and m_{BC} the steel weight of upper and lower part of the the offset column respectively. These are given in Eq. A.24 - A.26:

$$m_{iC} = \pi\rho_s(r_{iC}^2 - (r_{iC} - t_{iC})^2)h_{iC} \tag{A.24}$$

$$m_{caps} = t_{caps}\rho_s(r_{UC}^2 + 2r_{BC}^2) \tag{A.25}$$

$$m_{ballast} = \pi\rho_w((r_{UC} - t_{UC})^2 h_{b,UC} + (r_{BC} - t_{BC})^2 h_{b,BC}) \tag{A.26}$$

where h_b is the height of the ballast water in the given column. As RAFT can calculate the total ballast mass, m_{bal}, the ballast mass in the column, can instead be calculated as: $m_{bal,OC} = m_{bal}/3$. The resulting force influencing the truss structure is equal to the weight of the offset column; $F_{RES} = F_{G,OC}$. The pontoon forces are again given by A.21. As $t_{UC} = t_{BC} = t_{caps}$, $r_{BC} = 2r_{UC}$ and $h_{BC} + h_{UC} = h_{OC}$ the following is obtained:

$$F_{RES} = \left(\pi\left(r_{UC}^2 - (r_{UC} - t)^2\right)h_{UC}\rho_s + \pi\left(4r_{UC}^2 - (2r_{UC} - t)^2\right)h_{BC}\rho_s + 9t\rho_s {r_{UC}}^2 + m_{bal,OC}\right)g \tag{A.27}$$

A.5.2 Bending

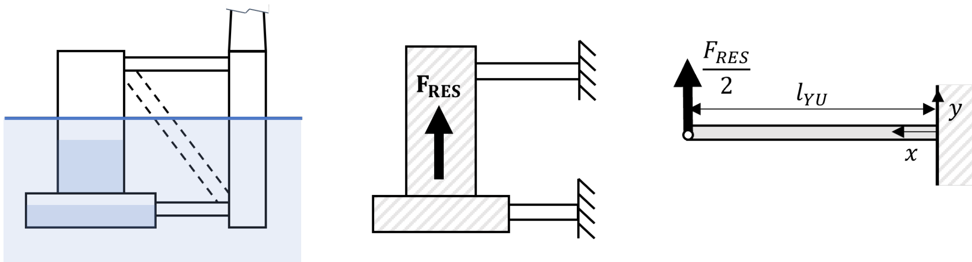

Figure A.5.9: Still water statics with cross beam excluded.

In an attempt to include bending in the model, a new structural analysis was conducted under the following assumptions:

- An isolated OC-MC link will be investigated with the cross beam missing.
- The delta pontoons will be disregarded.
- All three offset columns contribute equally to carry the weight on the main column.
- The weight and buoyancy of the pontoons can be disregarded.
- The force in the upper and lower pontoon carries the same vertical loads, as both columns are regarded as rigid structures: $F_{RES} = 2T_{YU} = 2T_{YL}$
- The critical load will be adjusted using the same correction as presented in ANSYS analysis of the OC4 5 MW [33].

The resulting force is the same as for the still water case, Eq. A.19.

As the upper pontoon is longer, it will experience a greater internal stress. From Eq. 2.38 this is found to occur at the end of where the cantilever is fixed. From this, a critial load, for which yielding will occur, can be defined as shown in Eq. A.28:

$$P_{cr} = \frac{\sigma_y I_{YU}}{l_{YU} r_{YU}} \tag{A.28}$$

where σ_y is the yielding strength of steel, I_{YU} is the cross sectional moment of inertia of the YU pontoon given by Eq. 2.35 and l_{YU}, and r_{YU} is the pontoon length and radius respectively.

In the definition of the OC4 an ANSYS analysis is carried out for the full scale 5 MW substructure [33]. It was found that the free-free bending modes of the platform were fairly low, most likely because the pontoon and cross members were reasonably sized for a model-scale semi-submersible but too narrow at full scale. Here it was determined to increase the bending stiffness of the members until the lowest free-free bending mode was greater than 3 Hz. This was accomplished by increasing the Young's modulus of steel for all members from a typical value of 2.10E+2 GPa to 2.10E+4 GPa, an increase of two orders of magnitude. In Tab. 3.4 of [33], the recommended property to use for the effective bending stiffness, EI, is given for the different members. For the upper pontoon $EI_{A,YU} = 5.720\text{E}{+}11$ Nm^2.

The maximum slope, ϕ_{max}, of a cantilever beam with an point load at the end (as of Fig. 2.12) is given by Eq. A.30 [63]:

$$\phi_{max} = \frac{Pl^2}{2EI} \tag{A.29}$$

The relation between the correction present in ANSYS as of [33] and in our model can be obtained as a measure of critical load by setting Eq. A.30 for respectively the ANSYS model and actual dimensions equal to each other. Let situation **A** represent the ANSYS model with corrected effective bending stiffness, $EI_{A,YU}$, and situation **B** represent the real values with a Young's modulus of steel of $E_s = 2.10$ GPa and I_{an} defined by Eq. 2.35. The equality can then be reformulated to find the relation between the critical loads which correspond to the correction factor, c_b, then can be used to account for the difference:

$$\phi_{max,A} = \phi_{max,B} \qquad \Leftrightarrow \qquad \frac{P_A l^2}{2EI_{A,YU}} = \frac{P_B l^2}{2E_s I_{an}} \qquad \Leftrightarrow \qquad \frac{P_A}{P_B} = \frac{EI_{A,YU}}{E_s I_{an}} = c_b \tag{A.30}$$

Using the relevant values for the 5 MW OC4, we get the relation between the critical loads to be $c_b = 12.3$. This correction is multiplied to the critical bending load in the model.

A.5.3 Linear Correction Factor for Extreme Cases

Cases (b) and (c) from Fig. 3.2 are very extreme cases. The closer the main column and offset column are to each other, the more unlikely it is to happen. In order to account for this, a factor, f_p, correcting for the probability of the case to happen can be implemented as a function of the distance between the offset and main column, r_{MO}. This has been done using linear interpolation between two points. The first is for the original 5 MW OC4 design, and the other is when r_{MO} has increased to double the original distance:

$$f_p(r_{MO,org}) = 0.1 \qquad \& \qquad f_p(2r_{MO,org}) = 1 \qquad \Rightarrow \qquad f_p(r_{MO}) = 0.9\frac{r_{MO}}{r_{MO,org}} - 0.8 \tag{A.31}$$

A.6 Dynamic analysis of the OC4 structure

In reality, the FOWT is in motion when affected by the environment, making the system dynamic. Looking at the same isolated OC-MC link as for the static analysis, the dynamic forces can be estimated by finding acceleration of the MC and OC. By estimating the maximum difference in acceleration between the two columns, the force that the truss structure is subject to can be estimated.

The local RAO in heave of MC and OC respectively can be found as described in Eq. A.32 and A.33. For the OC, the RAO in pitch is also relevant as this is offset from the origin with a distance in the x-direction of x_{OC} meaning that rotation here will contribute to vertical movement.

$$RAO_{a_{OC}}(\omega) = -\omega^2(RAO_3(\omega) + RAO_5(\omega)x_{OC}) \tag{A.32}$$

$$RAO_{a_{MC}}(\omega) = -\omega^2(RAO_3(\omega)) \tag{A.33}$$

The difference in movement between the two columns is then given by Eq. A.34:

$$RAO_{a_{O-M}}(\omega) = -\omega^2(RAO_5(\omega)x_{OC}) \tag{A.34}$$

The RMS of the response σ_r can be found as the 0-th order moment from Eqs. 2.27 and 2.25:

$$\sigma_{a_{O-M}}(\omega) = \sqrt{\int |RAO_{a_{O-M}}(\omega)|^2 S(\omega) d\omega} \tag{A.35}$$

From the `helpers` function `getVelocity`, the accelerations in the top node of the MC and OC are found respectively. The acceleration in translation of the OC can be seen plotted in the frequency spectrum in Fig. A.6.10 top. Heave is then further investigated by looking at acceleration of the MC, OC, and the difference between the two at the bottom of Fig. A.6.10.

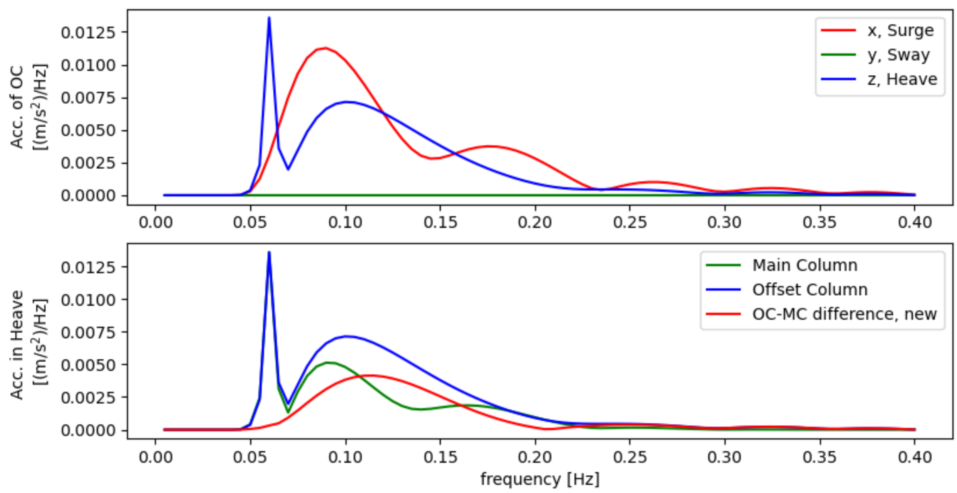

Figure A.6.10: Acceleration of MC and OC.

Regardless of whether `getVelocity` or Eq. A.34 are used, the RMS is found to be $\sigma_{a_{O-M}} = 9E-4$. From this, the maximum can be estimated. One way to do this is with a "three-sigma" approach, 2.28. As the mean is zero it is given by Eq. A.37

$$\max(a_{O-M}) = 3\sigma_{a_{O-M}} = 2.7 \cdot 10^{-3}\text{m/s}^2 \tag{A.36}$$

If instead, we assume that the distribution of the acceleration follows the distribution of wave heights making it Rayleigh distributed, Eq. is used, which yields the following:

$$\text{for} \quad p(\max(a_{O-M})) = 99.7\% \qquad \Rightarrow \qquad \max(a_{O-M}) = 9.3 \cdot 10^{-3}\text{m/s}^2 \tag{A.37}$$

A.7 Redefine Pontoon Thickness for Upscaled OC4

The thickness of the pontoons for the upscaled UC4 was defined in order to maintain the same internal stress under the same circumstances. The cross brace was used for investigation, as this pontoon experience the greatest load under the assumptions in App. A.5. The stress is defined by Eq. 2.32 and as it is to stay constant during upscaling,

$$\frac{N_5}{A_5} = \frac{N_{15}}{A_{15}} \tag{A.38}$$

where N is the normal force in the cross brace and A is the area, which for an annular is given by A.39, where the value of the radius, r, is maintained:

$$A_5 = \pi(r^2 - (r - t_{p,5})^2) \qquad\qquad A_{15} = \pi(r^2 - (r - t_{p,15})^2) \tag{A.39}$$

Using RAFT, the normal force in CB was obtained for the original OC4 and the upscaled OC4 using the original pontoon thickness. Under the static case for wave extreme at MC, it was found that: $N_5 = 4.6 \cdot 10^7$ N and $N_{15} = 6.3 \cdot 10^7$ N. Combining this with Eqs. A.39 and A.38 it is found that $t_{p,15} = 0.0242$ m which is an increase of 38 %. The thickness is chosen to be rounded up to $t_{p,15} = 0.025$ m.

A.8 Combination of RAO, Wave- and Response Spectra of 15MW Optimized OC4

In Fig. A.8.11, the RAO, Wave- and Response Spectra are shown together for the 15 MW Optimized OC4 as generated by RAFT from 0.1 to 0.25 Hz. Notice how the RAO acts as a filter on the wave spectrum to create the response spectrum.

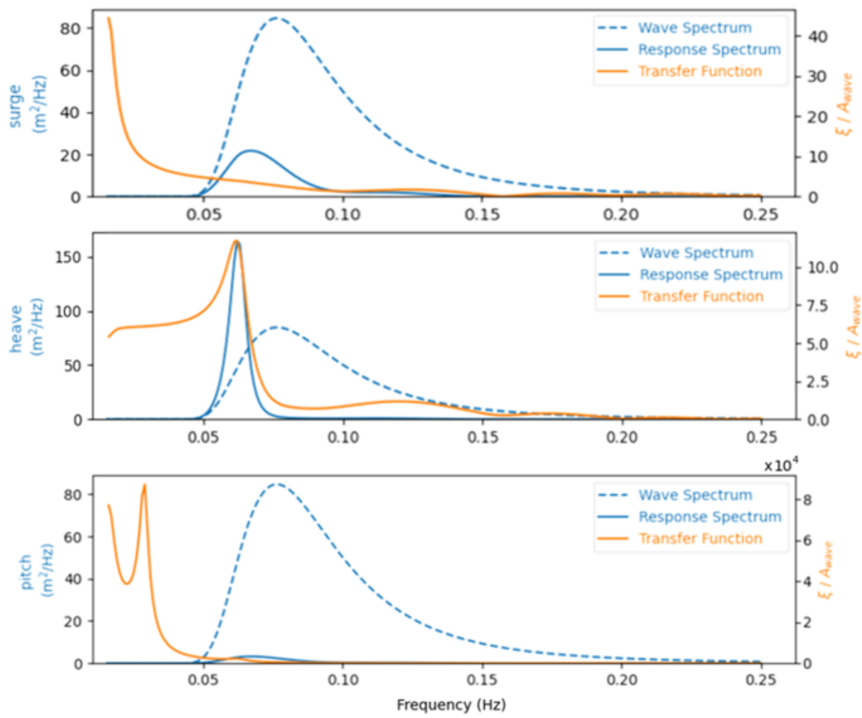

Figure A.8.11: Wave and response spectra along with transfer functions (RAOs) of the optimized 15 MW OC4 semi-submersible at max thrust operating conditions with wind speed of 11 m/s, wave height of 8.5 m, and wave period of 13.1 s.

A.9 Nacelle Acceleration Relative to Surge and Pitch

As mentioned in section 2.6.2, RAOs are graphs of the resulting non-dimensional amplitudes plotted as a function of encounter frequency. They are complex-valued and consist of amplitude functions and phase angles belonging - physically - to set of real numbers.

The phase angles, δ_i, give the phase relationship between the motion and the wave. A positive value means that the maximum positive motion occurs δ_i/ω_e seconds before the maximum wave depression is experienced at the origin of the coordinate system. Negative values imply that the motion lags the wave depression at the origin [56].

In Fig. A.9.12, the phase angles in surge and pitch are shown for both the upscaled and optimized FOWT at *case 1* conditions. Notice how the phases are opposite each other between 0.025 and 0.75 Hz resulting in them canceling each other out. The result of the phase cancellation can be seen in the third graph, where the phases of the nacelle acceleration are plotted. In the mentioned interval, the effective phase angle of the nacelle is 0, which in turn results in the PSD of the nacelle being zero. This, combined with the difference in amplitudes, is what generates the significant difference in nacelle acceleration from a marginal difference in pitch and surge PSD.

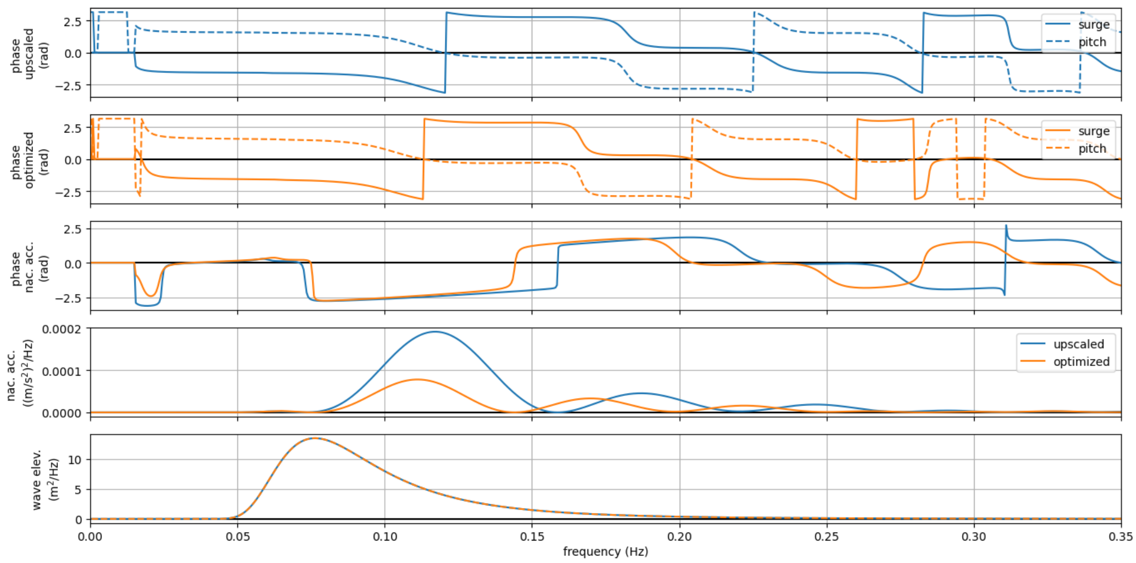

Figure A.9.12: Upscaled versus optimized 15MW OC4 semi-submersible at max thrust operating conditions with wind speed of 11 m/s, wave height of 8.5 m, and wave period of 13.1 s.

A.10 OpenFAST Validation

An attempt to validate the frequency domain results from RAFT with a time domain analysis in OpenFAST was made. This is a brief summery of that attempt.

With the installation of OpenFAST follows some test examples of different turbines in different settings. One of these is a simulation of the 5 MW OC4 semisubmersible with the NREL offshore 5-MW baseline wind turbine. Another is a simulation of the 15 MW Umaine VolturnUS-S FOWT with the IEA 15 MW reference wind turbine. Both of these were run successfully. The time histories (including response for all 6 DoF, the blade pitch, rotor speed, wave elevation and wind speed) were stored and later transformed to frequency domain. This was done using with Fast Fourier Transformation (FFT) using the MatLab library extension Wave Analysis for Fatigue and Oceanography (WAFO)[87]. The results for the VolturnUS-S can be seen for a 500 s simulation in Fig. A.10.13. In order to get a reasonable representation of the FOWTs response in the frequency domain, the FOWT movement was simulated for 1h and 10 minutes, where the first 10 minutes were not included in the results, as to make sure that the system was fully developed. The transformation from time domain to frequency domain is shown in Fig. A.10.14 for heave. 4 different smoothness are used.

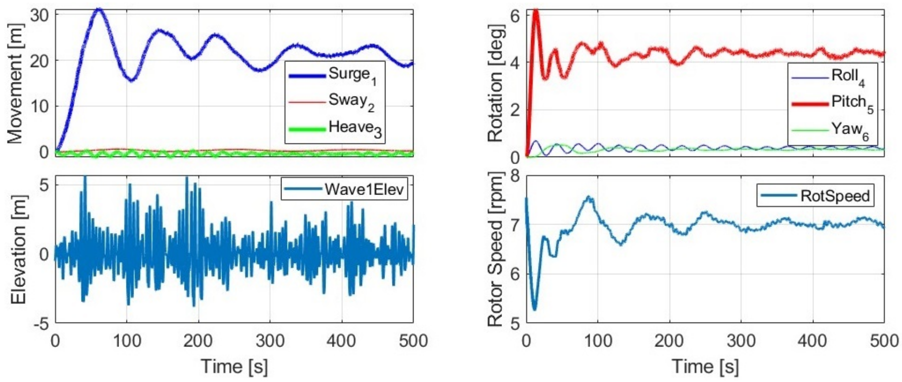

Figure A.10.13: OpenFAST simulation of the 15 MW Umaine VolturnUS-S FOWT with a wind speed of 10 m/s in a sea state 6 ($H_s = 5.5$ and $T_P = 5.5$).

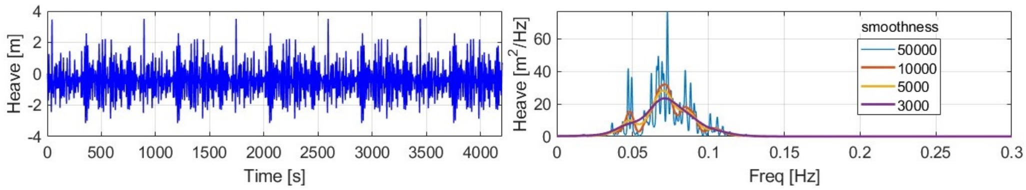

Figure A.10.14: Heave response and transformation to frequency response of OpenFAST simulation of the 15 MW Umaine VolturnUS-S FOWT with a wind speed of 11 m/s in a JONSWAP sea of $H_s = 8.5$ and $T_P = 13.1$.

In order to make a model of the optimized 15 MW OC4 FOWT the setup for the VolturnUS-S system was taken as the starting point. All semisubmersible defining sections were then changed to fit the definition of the optimized 15 MW OC4 structure. This was done by using the 5 MW OC4 example as a starting point to redefine all dimensions and properties to fit that of the optimized 15 MW OC4 structure. RAFT's calcBEM function was used to generate a mesh for the platform and run a BEM analysis on it using pyHAMS in order to write adjusted .1 and .3 output files suitable for use with OpenFAST. The geometry of the structure was controlled by converting the point-definition into Solid Works.

Unfortunately the construction of an OpenFAST model with a IEA 15 MW WT fitted on the optimized 15MW OC4 platform was unsuccessful. Results were obtained and analyzed, but proved unreliable as the heave would find its mean at a water depth of around 20 m and the pitch would be negative for low wind speeds. A lot of trouble shooting was conducted in cooperation with NREL personnel, but the problems were not solved in time for reliable results to take part in this thesis.

B Programs, Installation, and Codes

B.1 RAFT Setup Guide

This is a guide to help getting started with RAFT.

B.1.1 General hints and comments

- RAFT is still in the development phase, so it is important to regularly check the Github repository (`https://github.com/WISDEM/RAFT`) – you can then update your WEIS folders using git pull.
- Following a modular philosophy, RAFT can be used independently for various frequency-domain modelling needs or it can be used as an integrated part of WEIS through its OpenMDAO wrapper. In this guide it will be installed to be used independently.

B.1.2 RAFT installation

1. Install **Anaconda** (`https://www.anaconda.com/`) if not already installed.
2. Set up and activate the Anaconda environment from a prompt (**Anaconda3 Power Shell** on Windows or Terminal.app on Mac:)

```
conda config --add channels conda-forge
conda create -y --name raft-env python=3.9
conda activate raft-env
```

3. As a prerequisite for running RAFT it is also necessary to import and install **MoorPy**, **PyHAMS**, and **WISDEM** (alternatively **CCBlade**). The necessary libraries must first be installed in the relevant new environment, and the repository for each tool must be downloaded/cloned.

```
conda install -y cython git jsonschema make matplotlib nlopt numpy openmdao
    openpyxl pandas pip pyside2 pytest python-benedict pyyaml ruamel_yaml
    scipy setuptools simpy sortedcontainers statsmodels swig

conda install -y pyoptsparse                # (Linux only)
conda install -y gfortran pyoptsparse       # (Mac only)
conda install -y m2w64-toolchain libpython  # (Windows only)

pip install marmot-agents

git clone https://github.com/WISDEM/WISDEM.git  # (WISDEM repository)
```

```
git clone https://github.com/NREL/MoorPy.git  # (MoorPy repository)
git clone https://github.com/WISDEM/pyHAMS.git   # (PyHAMS repository)
git clone https://github.com/WISDEM/RAFT.git   # (RAFT repository)
```

NOTICE: If running on a (non-private) DTU PC, the line "pip install (...)" must be substituted by "`python -m pip install -user (...)`".

4. Now each tool can be installed by running the **setup.py** program located inside each individual tools folder. In order to directly use the examples in the repository and look at the code when necessary, we recommend all users install the tools in developer / editable mode. Access each folder individually, set up the program, and leave the folder before moving on to the next step.

```
cd WISDEM
python setup.py develop

cd ..
cd MoorPy
python setup.py develop

cd ..
cd PyHams
python setup.py develop

cd ..
cd RAFT
python setup.py develop
```

After each installation the last line should state a successful finished installation, e.g.: "`Finished processing dependencies for RAFT==x.x`" for the RAFT installation.

Now it should all be set up. Check that there a no red errors.

Each time you want to run RAFT, remember to always activate the raft environment:

`conda activate raft-env`

B.1.3 Running RAFT from Spyder

If you experience problems with running RAFT from Spyder, rather than Prompt, this may be due to the Python interpreter not having access to the raft-env. To address this, open Spyder and go to **ToolsPreferencesPython interpreter**. The python interpreter is probably set as Default. Instead select "**Use the following Python interpreter**:" Now select the location of the raft-env that you created. Something like:

```
C:/Users/YOURUSERNAME/.conda/envs/raft-env/python.exe
```

Select "**Apply**", then "**OK**". Open **Anaconda3 Power Shell**, activate raft-env and install spyder kernls:

```
conda activate raft-env
pip install spyder-kernels==0.*
```

Restart Spyder and the problem should be solved. To get the plots as pop-up-windows run `%matplotlib qt` in the console.

B.1.4 Useful links and contacts

At this point, it is also worth mentioning other useful resources:

RAFT documentation: `https://openraft.readthedocs.io/en/latest/index.html`

Source code (Github repository): `https://github.com/WISDEM/RAFT`

Examples: `https://github.com/WISDEM/RAFT/tree/master/examples`

B.2 Optimization Code

```
import os, sys
import yaml
import raft
import copy
import math
import numpy as np
import pandas as pd
#import openmdao as om
import matplotlib.pyplot as plt # To get figures in new window: %matplotlib qt
from openmdao.api import ExplicitComponent, IndepVarComp, Group, Problem,
ScipyOptimizeDriver, SqliteRecorder, CaseReader
from raft.helpers  import getVelocity , getRMS
#from openmdao.test_suite.components.sellar_feature import SellarDerivatives
# python -m pip install --user openmdao[notebooks]

##############################################################################
#########################    Initialize Design    ############################

# open the design YAML file and parse it into a dictionary for passing to raft
with open('OC4semisub_IEA15MW_case_RAFT.yaml') as file:
    design1 = yaml.load(file, Loader=yaml.FullLoader)

with open('OC4semi.yaml') as file:
    design5MW = yaml.load(file, Loader=yaml.FullLoader)

# Create the RAFT model (will set up all model objects based on the design dict)
model1 = raft.Model(design1)

# Evaluate the system properties and equilibrium position before loads are applied
model1.analyzeUnloaded()

# Compute natural frequencie
model1.solveEigen()

# Simule the different load cases
model1.analyzeCases(display=1)

# Returns the resutls dictionary
results1 = model1.calcOutputs()


##############################################################################
###########################    Setup values    ###############################

# Choose what Constraints to include (1 = include , 0 = exclude):
Con_Stiff   = 1     # Hydrostatic stiffness
```

```
Con_Bend    = 1     # Bending of Y_upper pontoon
Con_Buck    = 1     # Buckling of Y_upper pontoon
Con_Yield   = 1     # Yielding of Cross Brace
Con_Pitch   = 0     # Pitch (dynamic - case specific) (not used but optional)

# Choose what static cases will be investigated
still = 0
OCsub = 1
OCabv = 1

# Choose mooring type
mooring_type = 'OC4 5MW equivalent'
# Options:  'low mooring effect' and 'OC4 5MW equivalent'

# The design variables
#Input variable: Distance from offset to main column [m]
R = design1['platform']['members'][1]['rA'][0]
#Input variable: Diameter of offset column [m]
d_off = design1['platform']['members'][1]['d'][2]
# Wall-thickness length ratio for all pontoons [-]
ratio_tl = 0.011
# Boundaries of design variables, +- of default.
x = R-1   # for R [m]
y = d_off-1   # for d_off [m]

# Design criteria
# Maximum thrust force [N], (2.3 MN for 15 MW IEA and 0.7 or 0.8 MN for 5 MW NREL)
F_T_max = 2300000
# Theta max, [degrees] , (10 degree is recomended)
Theta_maximum = 10

# Safety Factors
s_buckling = 1
s_bending = 1
c_bending = 12.3
s_yielding = 1

    # Linear SF to correct for wave extreme probability
SF_1 = 1  # The probability at l_YU = l_YU_5MW_org
SF_2 = 1  # The probability at l_YU = 2*l_YU_5MW_org



################################################################################
############################    Functions    ##################################

#Function to redefine the design dictionary from new inputs
def Design_New(R_radius, d_offset):
    #Creating a copy of the design dictionary, while keeping the origional as is
    design_new = copy.deepcopy(design1)
    nu = 30 * np.pi/180                 # [rad] angle between DL and CB

    #Redefine the input values to design dictionary
    # 1 - offset_column
    design_new['platform']['members'][1]['d'][0] = 2*d_offset # Diameter of lower OC [m]
    design_new['platform']['members'][1]['d'][1] = 2*d_offset # Diameter of lower OC [m]
    design_new['platform']['members'][1]['d'][2] = d_offset   # Diameter of upper OC [m]
    design_new['platform']['members'][1]['d'][3] = d_offset   # Diameter of upper OC [m]
    design_new['platform']['members'][1]['rA'][0] = R_radius  # Dist. center of MC to OC [m]
    design_new['platform']['members'][1]['rB'][0] = R_radius  # Dist. center of MC to OC [m]
    # 2 - delta_upper_pontoon
    design_new['platform']['members'][2]['rA'][0] = R_radius - math.cos(nu)*d_offset/2
    design_new['platform']['members'][2]['rA'][1] = math.sin(nu)*d_offset/2
```

```
    design_new['platform']['members'][2]['rB'][0] = - math.sin(nu)*R_radius
                                                    + math.cos(nu)*d_offset/2
    design_new['platform']['members'][2]['rB'][1] = math.cos(nu)*R_radius
                                                    - math.sin(nu)*d_offset/2
    # 3 - delta_lower_pontoon
    design_new['platform']['members'][3]['rA'][0] = R_radius - math.cos(nu)*(d_offset*2)/2
    design_new['platform']['members'][3]['rA'][1] = math.sin(nu)*(d_offset*2)/2
    design_new['platform']['members'][3]['rB'][0] = - math.sin(nu)*R_radius
                                                    + math.cos(nu)*(d_offset*2)/2
    design_new['platform']['members'][3]['rB'][1] = math.cos(nu)*R_radius
                                                    - math.sin(nu)*(d_offset*2)/2
    # 4 - Y_upper_pontoon
    design_new['platform']['members'][4]['rB'][0] = R_radius - d_offset/2
    # 5 - Y_lower_pontoon
    design_new['platform']['members'][5]['rB'][0] = R_radius - (d_offset*2)/2
    # 6 - cross_brace
    design_new['platform']['members'][6]['rB'][0] = R_radius - d_offset/2
    # Anchor-line to vessel points
    design_new['mooring']['points'][3]['location'][0] =  math.sin(nu)*((d_offset*2)/2
                                                        + R_radius)    # FL_B
    design_new['mooring']['points'][3]['location'][1] =  math.cos(nu)
                                                        *((d_offset*2)/2 + R_radius)
# FL_B
    design_new['mooring']['points'][4]['location'][0] = -((d_offset*2)/2 + R_radius)
# FL_A
    design_new['mooring']['points'][4]['location'][1] =  0
# FL_A
    design_new['mooring']['points'][5]['location'][0] =  math.sin(nu)
                                                        *((d_offset*2)/2 + R_radius)
# FL_C
    design_new['mooring']['points'][5]['location'][1] = -math.cos(nu)
                                                        *((d_offset*2)/2 + R_radius)
# FL_C
    # Tools for changing anchor positions and mooring line length:
    d_w =  design_new['mooring']['water_depth']                   # water depth
    z_c = -design_new['mooring']['points'][3]['location'][2]
    r_c = -design_new['mooring']['points'][4]['location'][0]      # radius of cleats

    if mooring_type == 'OC4 5MW equivalent':
        # water depth
        d_w_5MW =  design5MW['mooring']['water_depth']
        # draught of cleats
        z_c_5MW = -design5MW['mooring']['points'][3]['location'][2]
        # radius of anchors (footprint)
        r_f_5MW = -design5MW['mooring']['points'][1]['location'][0]
        # radius of cleats
        r_c_5MW = -design5MW['mooring']['points'][4]['location'][0]
        # length of mooring lines
        l_m_5MW =  design5MW['mooring']['lines'][0]['length']
        # direct line length from anchor to cleat
        l_d_5MW =  math.sqrt((d_w_5MW-z_c_5MW)**2 + (r_f_5MW-r_c_5MW)**2)

        s_r = l_m_5MW/l_d_5MW                                 # slag ratio
        f_r = (r_f_5MW - r_c_5MW)/(d_w_5MW-z_c_5MW)           # footprint to height ratio

        r_a = f_r*(d_w-z_c) + r_c                             # radius of anchors

        l_m = s_r * math.sqrt((d_w-z_c)**2 + (r_a - r_c)**2)  # length of mooring line

        l_1 , l_2 , l_3 = l_m , l_m , l_m

    elif mooring_type == 'low mooring effect':
        r_a = 4.2835 * d_w
```

```
        l_2 = math.sqrt((d_w-z_c)**2 + (r_a*(math.sin(nu) + 1)-z_c)**2)
        l_1 = math.cos(nu)*r_a + d_w - z_c
        l_3 = l_2
    else:
        print('choose mooring type')
        sys.exit()


    # Anchor points in x and z coordinate
    design_new['mooring']['points'][0]['location'][0] =  math.sin(nu)*r_a
    design_new['mooring']['points'][0]['location'][1] =  math.cos(nu)*r_a
    design_new['mooring']['points'][1]['location'][0] = -r_a
    design_new['mooring']['points'][1]['location'][1] =  0
    design_new['mooring']['points'][2]['location'][0] =  math.sin(nu)*r_a
    design_new['mooring']['points'][2]['location'][1] = -math.cos(nu)*r_a
    # Mooring line length
    design_new['mooring']['lines'][0]['length'] = l_1
    design_new['mooring']['lines'][1]['length'] = l_2
    design_new['mooring']['lines'][2]['length'] = l_3
    # Water depth correction in z-axis
    design_new['mooring']['points'][0]['location'][2]=-d_w
    design_new['mooring']['points'][1]['location'][2]=-d_w
    design_new['mooring']['points'][2]['location'][2]=-d_w
    return design_new

#Function to compute the RAFT results from the given design
def RAFTY(designRUN):
    # RAFT run
    with HiddenPrints():
        modelRUN = raft.Model(designRUN)
        modelRUN.analyzeUnloaded()

        if Con_Pitch == 1:
            modelRUN.solveEigen()
            modelRUN.analyzeCases(display=1)

    resultsRUN = modelRUN.calcOutputs()

    return resultsRUN

#Function to define minimum stiffness required
def Stiff_Min(designRUN,resultsRUN):
    # Hub height above waterline, [m]
    Z_hub   = designRUN['turbine']['Zhub']
    # Vertical fairlead location (point of rotation) [m]
    Z_fair = designRUN['platform']['members'][1]['stations'][2]

    stiff_min = (F_T_max*(Z_hub-Z_fair))/(np.sin(Theta_maximum*(np.pi/180)))
    return stiff_min


# Function to define the forces in YU and CB for given design
def Struc_Des(design,results):
    g = 9.81  # Gravity acceleration
    # Gravitational Force on MC
    m_RNA = design['turbine']['mRNA']
    m_tower = results['properties']['tower mass']
    A_MC_annular = np.pi*((design['platform']['members'][0]['d'][0]/2)**2 -
                 ((design['platform']['members'][0]['d'][0]/2) -
                 design['platform']['members'][0]['t'])**2)
    V_MC_annular = A_MC_annular*(design['platform']['members'][0]['rB'][2]
                    -design['platform']['members'][0]['rA'][2])
    m_MC = V_MC_annular*design['platform']['members'][0]['rho_shell']
```

```
    F_G_MC = (m_MC+m_RNA+m_tower) * g

    # Bouyancy Force on MC
    A_MC_cylinder = np.pi*(design['platform']['members'][0]['d'][0]/2)**2
    V_MC_cylinder = A_MC_cylinder*abs(design['platform']['members'][0]['rA'][2])
    F_B_MC = V_MC_cylinder * g * design['site']['rho_water']

    # Gravitational Force on OC
    m_ballast = results['properties']['ballast mass'][0]/3
    t = design['platform']['members'][1]['t']
    h_UC = design['platform']['members'][1]['cap_stations'][3]
            -design['platform']['members'][1]['cap_stations'][2]
    h_BC = design['platform']['members'][1]['cap_stations'][1]
            +design['platform']['members'][1]['cap_stations'][0]
    r_UC = design['platform']['members'][1]['d'][2]/2
    r_BC = design['platform']['members'][1]['d'][0]/2
    m_UC = h_UC*math.pi*(r_UC**2-(r_UC-t)**2)*design['platform']['members'][1]['rho_shell']
    m_BC = h_BC*math.pi*(r_BC**2-(r_BC-t)**2)*design['platform']['members'][1]['rho_shell']
    m_caps = design['platform']['members'][1]['cap_t'][0]
            *math.pi*(r_UC**2+2*r_BC**2)*design['platform']['members'][1]['rho_shell']
    F_G_OC = (m_ballast + m_UC + m_BC + m_caps)*g

    # Length of YU
    l_YU = math.dist(design['platform']['members'][4]['rB'],
                    design['platform']['members'][4]['rA'])
    # Length of CB
    l_CB = math.dist(design['platform']['members'][6]['rB'],
                    design['platform']['members'][6]['rA'])
    # Height of CB
    h_CB = design['platform']['members'][6]['rB'][2]
            -design['platform']['members'][6]['rA'][2]

    # Resulting forces for different cases:
    F_RES_sw = (F_B_MC-F_G_MC)/3    # still water     - F_RES on MC
    F_RES_mc = (-F_G_MC)/3          # OC submerged    - F_RES on MC
    F_RES_oc = F_G_OC               # OC above water  - F_RES on OC

    # Calculation of linear probability correction factor s_linear
    R_new = design['platform']['members'][1]['rB'][0]
    R_5MW = design5MW['platform']['members'][1]['rA'][0]
    a = (SF_2-SF_1)/(2-1)
    b = SF_1 - a*1
    s_linear = a*R_new/R_5MW +b

    F_RES = np.zeros(3)
    if still == 1:
        F_RES[0] = F_RES_sw
    if OCsub == 1:
        F_RES[1] = F_RES_mc*s_linear
    if OCabv == 1:
        F_RES[2] = F_RES_oc*s_linear

    N_CB = F_RES* (l_CB/h_CB)       # Normal-Force in CB
    N_YU = -(F_RES*(l_YU/h_CB))/2   # Normal-Force in YU
    T_YU = F_RES_sw/2               # Transverse load on YU (for bending)

    return N_CB , N_YU , T_YU

#Function to define critical force for yielding in CM and YL
def Yield_Crit(design):
    # Yield Strength of Steel, structural ASTM A36 steel [Pa]
    sigma_y = 250E6
    # Cros sectional area of CM and YL (they are the same)
```

```
    A = math.pi*((design['platform']['members'][4]['d'][0]/2)**2
                 -(design['platform']['members'][4]['d'][0]
                 /2-design['platform']['members'][4]['t'])**2)
    # Critical yielding force in tension and compression
    F_yield = sigma_y * A
    return F_yield

#Function to define critical force for buckling
def Buck_Crit(design):
    # Youngs Modulus of Steel
    E = 210E9
    # Length of Y_upper_pontoon
    l_YU = design['platform']['members'][4]['rB'][0]
            - design['platform']['members'][4]['rA'][0]
    # Length of Cross Brace
    l_CB = math.dist(design['platform']['members'][6]['rB'],
                     design['platform']['members'][6]['rA'])
    # radii of Y_upper_pontoon
    r_out = design['platform']['members'][4]['d'][0]/2
    r_in = r_out - design['platform']['members'][4]['t']
    # Second moments of area the YU annulus
    I = np.pi/4*(r_out**4 - r_in**4)
    # Critical buckling force
    F_buck_YU = np.pi**2*E*I/l_YU**2
    F_buck_CB = np.pi**2*E*I/l_CB**2

    return F_buck_CB , F_buck_YU

#Function to define critical force for bending in YU
def Bend_Crit(design):
    # Yield Strength of Steel, structural ASTM A36 steel [Pa]
    sigma_y = 250E6
    # Length of Y_upper_pontoon
    l_YU = design['platform']['members'][4]['rB'][0]
            - design['platform']['members'][4]['rA'][0]
    # radii of Y_upper_pontoon
    r_YU_out = design['platform']['members'][4]['d'][0]/2
    r_YU_in = r_YU_out - design['platform']['members'][4]['t']
    # Second moments of area the YU annulus
    I_YU = np.pi/4*(r_YU_out**4 - r_YU_in**4)
    # Critical bending force
    F_bend = sigma_y * I_YU / (l_YU * r_YU_out)
    F_bend = F_bend * c_bending
    return F_bend


#Function to plot Response spectrum of optimized results along original results
def plotResponsesModel(design1,results1,results2):
    # Plots the response spectrum for both results
    # [Hz] lowest frequency to consider, also the frequency bin width
    min_freq     = design1['settings']['min_freq']
    # [Hz] highest frequency to consider
    max_freq     = design1['settings']['max_freq']
    TwoPi        = 2.0*np.pi
    # angular frequencies to analyze (rad/s)
    w = np.arange(min_freq, max_freq+0.5*min_freq, min_freq) *2*np.pi

    nModel = 2
    nplots = 6

    A = np.zeros((nModel, nplots, len(np.transpose(results1['case_metrics']['surge_PSD']))))

    A[0,0,:] = results1['case_metrics']['surge_PSD']
```

```
    A[0,1,:] = results1['case_metrics']['heave_PSD']
    A[0,2,:] = results1['case_metrics']['pitch_PSD']
    A[0,3,:] = results1['case_metrics']['AxRNA_PSD']
    A[0,4,:] = results1['case_metrics']['Mbase_PSD']
    A[0,5,:] = results1['case_metrics']['wave_PSD']

    A[1,0,:] = results2['case_metrics']['surge_PSD'][0]
    A[1,1,:] = results2['case_metrics']['heave_PSD'][0]
    A[1,2,:] = results2['case_metrics']['pitch_PSD'][0]
    A[1,3,:] = results2['case_metrics']['AxRNA_PSD'][0]
    A[1,4,:] = results2['case_metrics']['Mbase_PSD'][0]
    A[1,5,:] = results2['case_metrics']['wave_PSD'][0]

    fig, ax = plt.subplots(6, 1, sharex=True)

    for iModel in range(nModel):
        ax[0].plot(w/TwoPi, TwoPi*A[iModel,0,:])  # surge
        ax[1].plot(w/TwoPi, TwoPi*A[iModel,1,:])  # heave
        ax[2].plot(w/TwoPi, TwoPi*A[iModel,2,:])  # pitch [deg]
        ax[3].plot(w/TwoPi, TwoPi*A[iModel,3,:])  # nacelle acceleration
        ax[4].plot(w/TwoPi, TwoPi*A[iModel,4,:])  # tower base bending moment (FAST's kN-m)
        ax[5].plot(w/TwoPi, TwoPi*A[iModel,5,:], label=f'case {iModel+1}')  # wave spectrum

    ax[0].set_ylabel('surge \n'+r'(m$^2$/Hz)')
    ax[1].set_ylabel('heave \n'+r'(m$^2$/Hz)')
    ax[2].set_ylabel('pitch \n'+r'(deg$^2$/Hz)')
    ax[3].set_ylabel('nac. acc. \n'+r'((m/s$^2$)$^2$/Hz)')
    ax[4].set_ylabel('twr. bend \n'+r'((Nm)$^2$/Hz)')
    ax[5].set_ylabel('wave elev.\n'+r'(m$^2$/Hz)')
    ax[-1].set_xlabel('frequency (Hz)')
    ax[-1].legend()
    fig.suptitle('RAFT power spectral densities')


################################################################################
#############################    Classes    ###################################

class Mass_C(ExplicitComponent):
    def initialize(self):
        self.options.declare('mass_org', default = RAFTY(design1)['properties']['shell mass'])

    #Define inputs and outputs for the component
    def setup(self):
        self.add_input('R_radius')    # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset')    # Input variable: Diameter of offset column [m]

        self.add_output('mass_semi')  # Output variable: Mass of the semisub [kg]

        #Use finite differences to compute derivatives
        self.declare_partials('*','*', method='fd')

    def compute(self, inputs, outputs):
        #Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        #Compute mass of the structure
        mass_NEW = RAFTY(Design_New(R_radius, d_offset))['properties']['shell mass']
        mass_org = RAFTY(design1)['properties']['shell mass']
        mass_semi = mass_NEW/mass_org

        # The starting objective value
```

```
        print( ' - - - - - - - - - - - - - - - - - - - - - - - - - - -
                        - - - - - - - - - - - - - - - - - - -')
        print(   '[D =' , "%.2g" % d_offset[0] , 'm] ',
                 '[R =' , "%.4g" % R_radius[0] , 'm] ',
                 '[M ='     , "%.3g" % mass_semi   , '] ', end=' ')

        #Assign the mass of the semisub to the output variable
        outputs['mass_semi'] = mass_semi


class Stiff_Zero(ExplicitComponent):
    def setup(self):
        self.add_input('R_radius') # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset') # Input variable: Diameter of offset column [m]

        self.add_output('stiff_zero') # Output var: Stiff. diff. (actual vs min), K [N/m]

        self.declare_partials('*','*', method='fd')

    def compute(self, inputs, outputs):
        #Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        stiff_des = RAFTY(Design_New(R_radius, d_offset))
                            ['properties']['C support structure'][4,4]

        stiff_min = Stiff_Min(Design_New(R_radius, d_offset),
                                RAFTY(Design_New(R_radius, d_offset)))

        stiff_zero = (stiff_des - stiff_min)/stiff_min # Normalized stiffness constraint

        print(   '[stiff_z =' , "%.3g" % stiff_zero , '] ', end=' ')

        #Assign the stiffness of the semisub to the output variable
        outputs['stiff_zero'] = stiff_zero


class Yield_Zero(ExplicitComponent):
    def initialize(self):
        self.options.declare('s_yield', default = s_yielding)

    def setup(self):
        self.add_input('R_radius') # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset') # Input variable: Diameter of offset column [m]

        if still == 1:
            self.add_output('yield_zero_still')  # Output variable: Yielding (case a)
        if OCsub == 1:
            self.add_output('yield_zero_OCsub')  # Output variable: Yielding (case b)
        if OCabv == 1:
            self.add_output('yield_zero_OCabv')  # Output variable: Yielding (case c)

        self.declare_partials('*','*', method='fd')

    def compute(self, inputs, outputs):
        # Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        #Unpack options
        s_yield = self.options['s_yield']
```

```
        # Normal force in Cross Brace is always greater than YU pontoon
        N_CB = Struc_Des(Design_New(R_radius, d_offset),
                            RAFTY(Design_New(R_radius, d_offset)))[0]

        F_yield = Yield_Crit(Design_New(R_radius, d_offset))

        yield_zero = (F_yield-abs(max(N_CB, key = abs))*s_yield)/F_yield

        if still == 1:
            yield_zero_still = (F_yield-abs(N_CB[0])*s_yield)/F_yield     # Normalized
        if OCsub == 1:
            yield_zero_OCsub = (F_yield-abs(N_CB[1])*s_yield)/F_yield     # Normalized
        if OCabv == 1:
            yield_zero_OCabv = (F_yield-abs(N_CB[2])*s_yield)/F_yield     # Normalized

        n = np.argmax(N_CB)

        print(   f'[yield_{n} =' , "%.3g" % yield_zero , ']', end=' ')

        if still == 1:
            outputs['yield_zero_still'] = yield_zero_still
        if OCsub == 1:
            outputs['yield_zero_OCsub'] = yield_zero_OCsub
        if OCabv == 1:
            outputs['yield_zero_OCabv'] = yield_zero_OCabv


class Buck_Zero(ExplicitComponent):
    def initialize(self):
        #Safety factor for buckling
        self.options.declare('s_buck', default = s_buckling)

    def setup(self):
        self.add_input('R_radius')   # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset')   # Input variable: Diameter of offset column [m]

        self.add_output('buck_zero')

        self.declare_partials('*','*', method='fd')

    def compute(self, inputs, outputs):
        # Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        #Unpack options
        s_buck = self.options['s_buck']

        # Buckling only occurs in compression => Find largest negative force
        N_CB_b = min(Struc_Des(Design_New(R_radius, d_offset),
                        RAFTY(Design_New(R_radius, d_offset)))[0])

        N_YU_b = min(Struc_Des(Design_New(R_radius, d_offset),
                        RAFTY(Design_New(R_radius, d_offset)))[1])


        # Find critical values
        F_buck_CB = Buck_Crit(Design_New(R_radius, d_offset))[0]

        F_buck_YU = Buck_Crit(Design_New(R_radius, d_offset))[1]

        # Choose the one with largest negative force
```

```
        N = np.array([N_CB_b,N_YU_b])
        crit = np.array([F_buck_CB,F_buck_YU])
        index = np.argmin(N)

        buck_zero =  (crit[index]+N[index]*s_buck)/crit[index]
        print(   f'[buck_index = {index} ]' , end=' ')
        print(   '[buck =' , "%.3g" % buck_zero, "]", end=' ')

        # Control:
        buck_zero_CB = (F_buck_CB-abs(N_CB_b)*s_buck)/F_buck_CB
        buck_zero_YU = (F_buck_YU-abs(N_YU_b)*s_buck)/F_buck_YU
        #print(   '[buck_CB =' , "%.3g" % buck_zero_CB, "]", end=' ')
        #print(   '[buck_YU =' , "%.3g" % buck_zero_YU, "]", end=' ')


        outputs['buck_zero'] = buck_zero


class Bend_Zero(ExplicitComponent):
    def initialize(self):
        #Safety factor for buckling
        self.options.declare('s_bend', default = s_bending)

    def setup(self):
        self.add_input('R_radius') # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset') # Input variable: Diameter of offset column [m]

        self.add_output('bend_zero')

        self.declare_partials('*','*', method='fd')

    def compute(self, inputs, outputs):
        # Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        #Unpack options
        s_bend = self.options['s_bend']

        # Transverse force in YU pontoon
        T_YU = Struc_Des(Design_New(R_radius, d_offset),
                        RAFTY(Design_New(R_radius, d_offset)))[2]

        F_bend = Bend_Crit(Design_New(R_radius, d_offset))

        bend_zero = (F_bend-abs(T_YU)*s_bend)/F_bend     # Normalized

        print(   '[bend_z =' , "%.3g" % bend_zero , '] ', end=' ')

        outputs['bend_zero'] = bend_zero



class Pitch_M(ExplicitComponent):

    def setup(self):
        self.add_input('R_radius')  # Input variable: Distance from offset to MC [m]
        self.add_input('d_offset')  # Input variable: Diameter of offset column [m]

        self.add_output('pitch_max')  # Output variable: Maximum pitch angle (dynamic)

        self.declare_partials('*','*', method='fd')
```

```
    def compute(self, inputs, outputs):
        #Unpack inputs
        R_radius = inputs['R_radius']
        d_offset = inputs['d_offset']

        #Compute maximum pitch
        pitch_max = RAFTY(Design_New(R_radius, d_offset))['case_metrics']['pitch_max'][0]

        print(   '[pitch_max =' , "%.3g" % pitch_max , 'deg] ', end=' ')

        outputs['pitch_max'] = pitch_max

class HiddenPrints:
    def __enter__(self):
        self._original_stdout = sys.stdout
        sys.stdout = open(os.devnull, 'w')

    def __exit__(self, exc_type, exc_val, exc_tb):
        sys.stdout.close()
        sys.stdout = self._original_stdout


################################################################################
###################   Optimization Workflow    ################################

#Define a group of components apt to the semisub optimization
model = Group()

#Define the independent variables' component
indep_var = IndepVarComp()
indep_var.add_output('R_radius', val = R)
indep_var.add_output('d_offset', val = d_off)

#Add component with independent design variables
model.add_subsystem('indep_var', indep_var)

#Add component to compute the mass of the semisub
model.add_subsystem('Mass_C', Mass_C())

#Connect inputs and outputs
model.connect('indep_var.R_radius',['Mass_C.R_radius'])
model.connect('indep_var.d_offset',['Mass_C.d_offset'])

#Add component to compute the stiffness difference and connect in- and outputs
if Con_Stiff == 1:
    model.add_subsystem('Stiff_Zero', Stiff_Zero())
    model.connect('indep_var.R_radius',['Stiff_Zero.R_radius'])
    model.connect('indep_var.d_offset',['Stiff_Zero.d_offset'])

#Add component to compute the buckling difference and connect in- and outputs
if Con_Buck == 1:
    model.add_subsystem('Buck_Zero', Buck_Zero(s_buck = s_buckling))
    model.connect('indep_var.R_radius',['Buck_Zero.R_radius'])
    model.connect('indep_var.d_offset',['Buck_Zero.d_offset'])

#Add component to compute the bending difference and connect in- and outputs
if Con_Bend == 1:
    model.add_subsystem('Bend_Zero', Bend_Zero(s_bend = s_bending))
    model.connect('indep_var.R_radius',['Bend_Zero.R_radius'])
    model.connect('indep_var.d_offset',['Bend_Zero.d_offset'])

    #Add component to compute the yielding difference and connect in- and outputs
```

```
if Con_Yield == 1:
    model.add_subsystem('Yield_Zero', Yield_Zero(s_yield = s_yielding))
    model.connect('indep_var.R_radius',['Yield_Zero.R_radius'])
    model.connect('indep_var.d_offset',['Yield_Zero.d_offset'])

#Add component to compute the max incl angle and connect in- and outputs
if Con_Pitch == 1:
    model.add_subsystem('Pitch_M', Pitch_M())
    model.connect('indep_var.R_radius',['Pitch_M.R_radius'])
    model.connect('indep_var.d_offset',['Pitch_M.d_offset'])


#Add constraints to the model
if Con_Stiff == 1:
    model.add_constraint('Stiff_Zero.stiff_zero', lower = 0 )

if Con_Buck == 1:
    model.add_constraint('Buck_Zero.buck_zero', lower = 0 )

if Con_Bend == 1:
    model.add_constraint('Bend_Zero.bend_zero', lower = 0 )

if Con_Yield == 1:
    if still == 1:
        model.add_constraint('Yield_Zero.yield_zero_still', lower = 0 )
    if OCsub == 1:
        model.add_constraint('Yield_Zero.yield_zero_OCsub', lower = 0 )
    if OCabv == 1:
        model.add_constraint('Yield_Zero.yield_zero_OCabv', lower = 0 )

if Con_Pitch == 1:
    model.add_constraint('Pitch_M.pitch_max', upper = Theta_maximum )

#Specify design variables with boundaries
model.add_design_var('indep_var.R_radius', lower = R-x    , upper = R+x)
model.add_design_var('indep_var.d_offset', lower = d_off-y, upper = d_off+y)

#Specify objective function
model.add_objective('Mass_C.mass_semi')

#Create a problem from the model
sub_problem = Problem(model)

#set up the problem
sub_problem.setup()

#######################   Recorder   ##########################################

recorder = SqliteRecorder("cases.sql", record_viewer_data=False)

sub_problem.model.add_recorder(recorder)
sub_problem.driver.add_recorder(recorder)

sub_problem.driver.recording_options['record_desvars'] = True
sub_problem.driver.recording_options['record_outputs'] = True
sub_problem.driver.recording_options['includes'] = []
sub_problem.driver.recording_options['excludes'] = []

######################## Running optimization  ################################

# Minimum stiffness for initial design
print('         minimum   stiffness:  ', "%.3g" % Stiff_Min(design1,results1) , '[Nm]')
print('________________________________________________________________________________')
```

```
#Run the problem with the initial values

sub_problem.run_model(case_prefix='Model_Run1')

#Add a driver to the problem
sub_problem.driver = ScipyOptimizeDriver()
sub_problem.driver.options['optimizer'] = 'COBYLA'    # Type of driver
sub_problem.driver.options['tol'] = 1e-5              # Tolerance

#Set up the problem with driver
sub_problem.setup()

#Run the problem with driver
sub_problem.run_driver(case_prefix='Driver_Run1')

sub_problem.cleanup()

cr = CaseReader("cases.sql")

# all cases recorded by the root system
model_cases = cr.list_cases('root', recurse=True)

################################################################################
########################### Plot final results ################################

# Find the optimized design from the optimized R and D
design2 = Design_New(sub_problem['indep_var.R_radius'][0],
                        sub_problem['indep_var.d_offset'][0])

# Create the RAFT model (will set up all model objects based on the design dict)
model2 = raft.Model(design2)

# Evaluate the system properties and equilibrium position before loads are applied
model2.analyzeUnloaded()

# Compute natural frequencie
model1.solveEigen()     # for the initial
model2.solveEigen()     # for the new

# Simule the different load cases
model1.analyzeCases(display=1)    # for the initial
model2.analyzeCases(display=1)    # for the new

# Returns the resutls dictionary
results2 = model2.calcOutputs()

# Plot the power spectral densities from the load cases
plotResponsesModel(design1,results1,results2)

# Visualize the system in its most recently evaluated mean offset position
# plot the first RAFT model, and return the figure and axis handles of the plot
fig1, ax1 = model1.plot(hideGrid=True, color='blue')
# plot the second RAFT model, and pass it the axis to plot on
model2.plot(ax=ax1, color='orange')

plt.show()

################################################################################
############################# Final Print #####################################
#Print final results
print('----------------------------------------------------------------')
print('         Optimized Solution:              ')
print('R_radius:  ', np.round( sub_problem['indep_var.R_radius'][0],2) , '[m]      and
```

```
',
      'd_offset:  ', np.round( sub_problem['indep_var.d_offset'][0],2) , '[m]')
print('Mass:      ', np.round(results2['properties']['shell mass']/10**3 ,2),'[ton]' )
print('Hyd.Stiff: ', "%.3g" % results2['properties']['C support structure'][4,4],'[Nm]' )
print('---------------------------------------------------------------')
print('Natural Periods in 6 DOF', 1/results2['eigen']['frequencies'], '[sec]')
print('---------------------------------------------------------------')

################################################################################
######################## Case Recorder Plotting ###############################

# Create matrix of size: itertations x outputs
A = np.zeros((len(model_cases),len(cr.get_case(0).outputs)+1))

# Gather cases values in Matrix A
for i in range (len(model_cases)):
    A[i,0] = i
    A[i,1] = cr.get_case(i).outputs['Mass_C.mass_semi'][0]
    A[i,2] = cr.get_case(i).outputs['indep_var.R_radius'][0]
    A[i,3] = cr.get_case(i).outputs['indep_var.d_offset'][0]

    a = 3   # number to control the inputs into A

    if Con_Stiff == 1:
        a += 1
        a1 = a
        A[i,a] = cr.get_case(i).outputs['Stiff_Zero.stiff_zero'][0]
    if Con_Buck == 1:
        a += 1
        a2 = a
        A[i,a] = cr.get_case(i).outputs['Buck_Zero.buck_zero'][0]
    if Con_Bend == 1:
        a += 1
        a3 = a
        A[i,a] = cr.get_case(i).outputs['Bend_Zero.bend_zero'][0]
    if Con_Yield == 1:
        if still == 1:
            a += 1
            a4 = a
            A[i,a] = cr.get_case(i).outputs['Yield_Zero.yield_zero_still'][0]
        if OCsub == 1:
            a += 1
            a5 = a
            A[i,a] = cr.get_case(i).outputs['Yield_Zero.yield_zero_OCsub'][0]
        if OCabv == 1:
            a += 1
            a6 = a
            A[i,a] = cr.get_case(i).outputs['Yield_Zero.yield_zero_OCabv'][0]
    if Con_Pitch == 1:
        a += 1
        a7 = a
        A[i,a] = cr.get_case(i).outputs['Pitch_M.pitch_max'][0]


fig, ax = plt.subplots(3)
# 1st plot: Output (Mass)
ax[0].plot(A[:,0], A[:,1], "-o")
ax[0].set(title='Output (Mass)', ylabel="Mass [kg]")
ax[0].grid()

# 2nd plot: Descision Variables (Radius and Diameter)
ax[1].plot(A[:,0], A[:,2], "-o", label='Radius (OC to MC)')
ax[1].plot(A[:,0], A[:,3], "-o", label='Diameter (OC)')
```

```
ax[1].legend()
ax[1].grid()
ax[1].set(title='Input Dimensions',ylabel="length [m]")

# 3rd plot: Constraints (choose constraints used)
if Con_Stiff == 1:
    ax[2].plot(A[:,0], A[:,a1], "-o", label='Hyd. Stiff')
if Con_Buck == 1:
    ax[2].plot(A[:,0], A[:,a2], "-o", label='Buckling')
if Con_Bend == 1:
    ax[2].plot(A[:,0], A[:,a3], "-o", label='Bending')
if Con_Yield == 1:
    if still == 1:
        ax[2].plot(A[:,0], A[:,a4], "-o", label='Yielding still')
    if OCsub == 1:
         ax[2].plot(A[:,0], A[:,a5], "-o", label='Yielding OCsub')
    if OCabv == 1:
        ax[2].plot(A[:,0], A[:,a6], "-o", label='Yielding OCabv')
if Con_Pitch == 1:
    ax[2].plot(A[:,0], A[:,a7], "-o", label='Pitch')
ax[2].legend()
ax[2].grid()
ax[2].set(title='Constraints', xlabel="# Iteration",ylabel=" --- ")


plt.show()

################################################################################
######################## Store Case Recorder eXcel ############################

# Store Setup values
A1 = np.array([[ 'Con_Stiff' , Con_Stiff],[ 'Con_Bend' , Con_Bend], ['Con_Buck',Con_Buck],
               ['Con_Yield',Con_Yield],[ 'Con_Pitch', Con_Pitch],['R_max', R + x],
               ['d_off_min',d_off-y], ['Theta_maximum', Theta_maximum],
               ['s_buckling', s_buckling],
               ['s_bending',s_bending],[ 's_yielding', s_yielding]])

# Name the Columns for Outputs from Case Recorder
output_names = [None]*A.shape[1]
output_names[0:4] = ['iteration #','Output (Mass)' ,'Radius (OC to MC)', 'Diameter (OC)' ]
a = 3   # number to control the inputs into A
if Con_Stiff == 1:
    output_names[a1] = 'Hyd. Stiff'
if Con_Buck == 1:
    output_names[a2] = 'Buckling'
if Con_Bend == 1:
    output_names[a3] = 'Bending'
if Con_Yield == 1:
    if still == 1:
        output_names[a4] = 'Yielding still'
    if OCsub == 1:
        output_names[a5] = 'Yielding OCsub'
    if OCabv == 1:
        output_names[a6] = 'Yielding OCabv'
if Con_Pitch == 1:
    output_names[a7] = 'Pitch'

# Define data frame
df1 = pd.DataFrame(A1).T
df2 = pd.DataFrame(A , columns=output_names).T   # Save as type: data frame

n = 3
```

```
with pd.ExcelWriter(f'output{n}.xlsx') as writer:
    df1.to_excel(writer, sheet_name=f'output{n}_info')
    df2.to_excel(writer, sheet_name=f'output{n}_val')
```

C Data Sheets

C.1 Dimensional Specifications of Volturn US-S

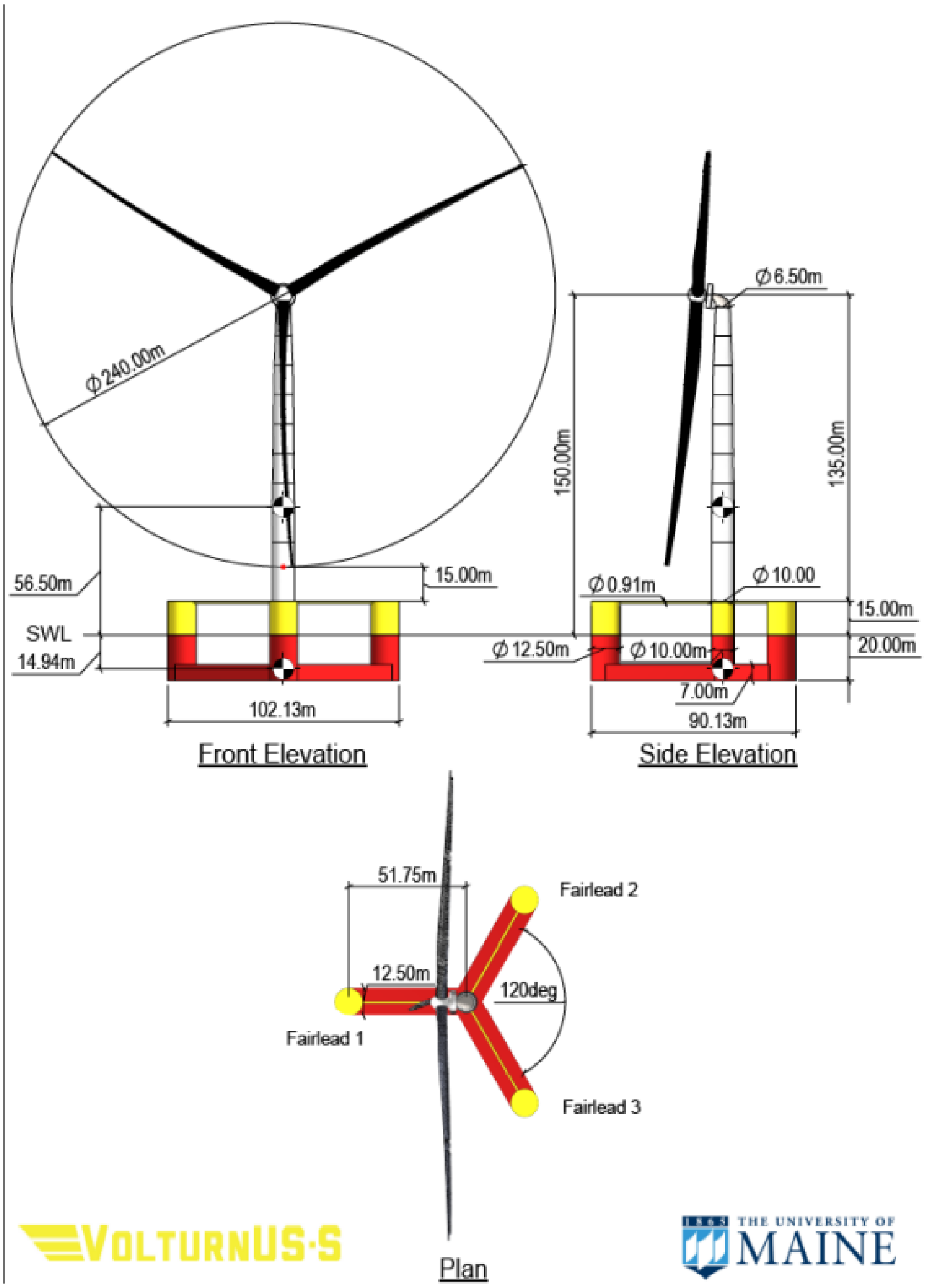

Figure 3. General arrangement. *Figure courtesy of the University of Maine*